



GE VERNOVA

ADMS

16.25.1

System Overview

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Change History

Date	Change Description
May 29, 2023	ADMS 3.16: • Updated references to "Storm Assist" to new product name "Outage Assist". • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Apr 1, 2024	ADMS 16.7.0: • Updated document format and styling.
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Contents

1. Introduction	5
2. A Tour of the ADMS Solution	9
2.1. ADMS Feature Summary	10
2.2. Distribution Network Object Model	14
2.3. Network View	15
2.3.1. Tracing the Network	17
2.3.2. Topology Processing	19
2.3.3. Temporary Network Modification	20
2.3.4. Viewports	21
2.3.5. Container Objects	23
2.4. Network Analysis	24
2.5. Distribution System Plan Generation Engine	26
2.6. Switching Operations	27
2.7. Network Outage Management	29
2.8. Distribution Operator Training Simulator	31
2.9. Historical Data	34
2.9.1. Archiving Overview	34
2.9.2. ADMS Archive	36
2.10. Read-Only Data Access	36
2.11. System Architecture	37
2.11.1. Main ADMS Servers	38
2.11.2. ADMS Client	39
2.12. Scada Server	40
3. ADMS Modeling Environment	42
3.1. Distribution Network Object Model	42
3.2. ADMS Modeling Tools	43
3.3. Substation Editor	46
3.4. Substation Builder	48
3.5. Substation File Builder Lite	49
3.6. Converter	49
3.7. Model Manager	51
4. Main ADMS Server	54
4.1. Loader Module	56
4.2. Dynamics Module	56
4.3. SCADA Module	56
4.4. Network Topology Processor Module	56
4.5. Query Module	57
4.6. Client Module	57
4.7. Distribution Network Analysis Functions Module	58
4.8. OMS Modules	58

4.9. Plugin Module.....	59
4.10. Plan Manager Module	60
5. ADMS Client.....	61
5.1. Network Topology Processor Module	62
5.2. Navigation	63
5.3. Configuration.....	63
5.4. Browser	64
6. ADMS Configuration Tools	65
6.1. Geographic Background Compiler	65
6.2. Navigation Builder	66
6.3. Net Dynamics Configuration Editor	67
6.4. Viewer Configuration Editor	68
6.5. Resource Editor	68
6.6. File Updater Utility	69
6.7. Grid Tabular Builder.....	70
6.8. DNAF Configuration Editor	71
6.9. Switching Order Client Configuration Editor	72
6.10. Switching Order Server Configuration Editor	73
6.11. OMS Client Configuration Editor	74
6.12. OMS Server Configuration Editor.....	75
6.13. Model Manager Configuration Editor	76
6.14. Coded Translation Editor.....	77
6.15. Snapshot Viewer.....	78
6.16. File Updater Configuration Editor	79
7. ADMS External Interfaces	81
7.1. Distribution Management Interface.....	81
7.2. Outage Management Interfaces	82
7.3. Outage Management Read-Only Interface.....	83
7.4. Call and Crew Interfaces	84
8. Other System Functions.....	86
8.1. High Availability	86
8.2. Authentication and Authorization.....	88
8.3. Centralized Alarm Management	90

1. Introduction

ADMS is GE's solution for the real-time management of electrical distribution networks.

ADMS is a suite of modular applications that combine to form a robust and scalable Advanced Distribution Management System(ADMS). ADMS meets the real-time network management needs of both large and small electric distribution utilities, providing a full range of high-quality network operation functions designed for seamless integration with other systems, such as Supervisory Control and Data Acquisition (SCADA) systems. In addition, ADMS provides programmatic interfaces to enable integration with third-party products and enterprise applications and to support read-only access to ADMS data.

The conceptual view of the components that make up an Integrated Distribution Management System is shown in below image.

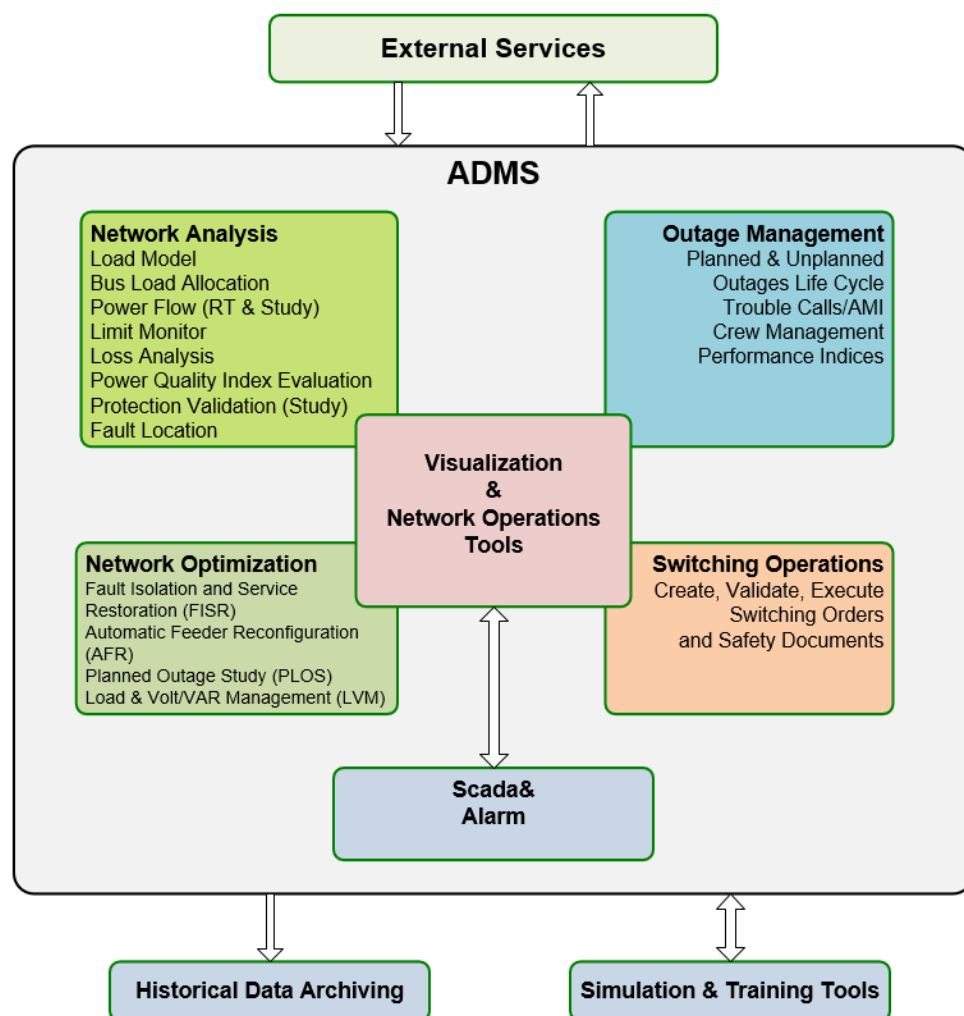


Figure 1. Conceptual View of ADMS

The key application modules are:

- **Network View:** Real-time geographic and schematic views of the distribution network.
- **Outage Management:** Management of unplanned outages, planned outages, and trouble calls, along with prediction of outage locations.
- **Switching Operations:** Electronic management of switching orders and safety documents.
- **Network Analysis:** Distribution power flow analysis, fault location, and bus load allocation.
- **Network Optimization:** Advanced analysis of network configurations to produce restoration plans, Volt-VAR management, and load balancing.
- **Interfaces:** Programmatic interfaces that support integration with surrounding enterprise systems.

As shown in the above image, ADMS also includes a Network Simulator for operator training and performance testing, and a Data Archive for historical data recording.

In addition to these application modules, real-time data and controls are integrated via the SCADA system.

In essence, the functionality offered by ADMS allows distribution operators and engineers to monitor, control, analyze, and plan the operation of the power distribution system. ADMS provides all the functionality required to view and manage the planned and unplanned events that occur on a distribution network.

The design of ADMS is based on a client-server architecture. As with many other similar systems, the client provides the user interface, while the server performs CPU intensive processing and interfaces with external systems.

Unlike thin clients, the ADMS smart client performs a key portion of the application processing in addition to providing complex and dynamic data presentation. This adds a number of benefits:

- Reduced server load, which allows for supporting many clients (for example, hundreds) that might be required under major storm recovery conditions.
- Lightweight data traffic between the client and server, providing acceptable performance on limited bandwidth connections to devices such as mobile workstations and temporary control centers.
- Some capability for the client to run autonomously and to accommodate short disconnections from the server by running in read-only mode.
- Segregated user environments, which allow for stand-alone operation for certain functions such as study-mode operations for planning staff.
- Improved performance by leveraging the CPU of the workstation to perform tasks locally. For example, display call-up is performed by the client and does not impact the server.

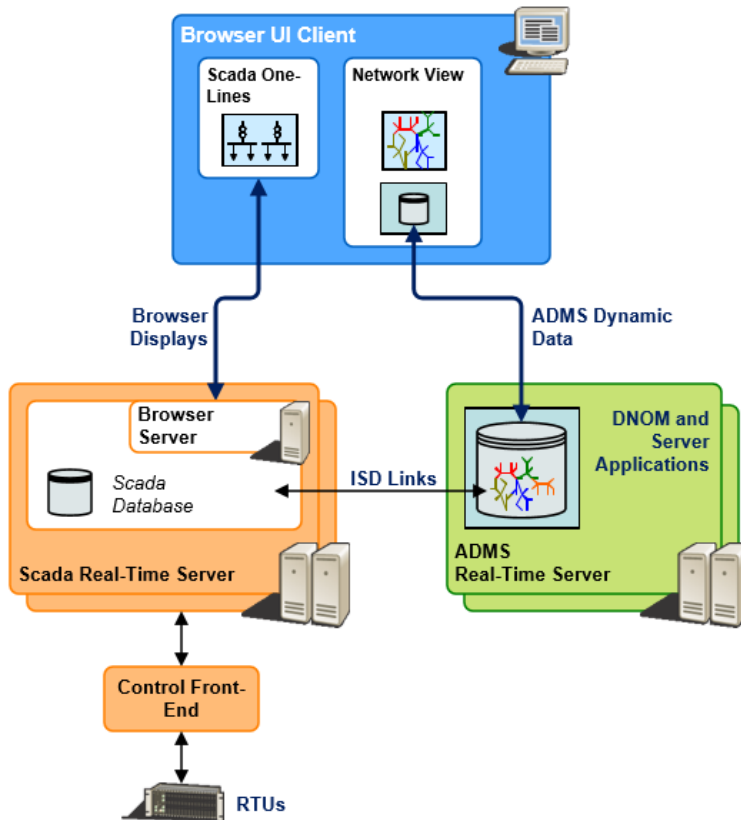


Figure 2. ADMS Architecture Overview

The client server architecture of ADMS is implemented as a set of building blocks that provide a complete operating, configuration, and support environment consisting of the following main components:

- ADMS servers, for real-time operations control
- ADMS clients provide both geographic and schematic representations of the distribution network combined with advanced functionality for operators and analysts
- ADMS configuration tools, for support engineers to configure the ADMS applications
- ADMS modeling tools, for distribution modeling engineers to model both the distribution network and the SCADA subsystem
- ADMS File Updater keeps the models, pictures (grid displays), and bitmaps available on the ADMS server and the ADMS client machines up to date
- ADMS web services and TCP interfaces to communicate with external systems and services

ADMS leverages some of the core foundation that is shared among related GE products. The core technology components used by ADMS include:

- Browser, a graphical user interface for control center operators
- Scada, for the control and monitoring of medium to large transmission and distribution networks

- Scada add-on components, including the Configuration Manager Proxy server (CFGPROXY) and the Permission Manager Proxy server (NETPERMSRV), respectively, which are responsible for interacting with the Habitat Configuration Manager (CFGMAN) and Permission server (PERMIT)
- **e-terra***modeler*, an environment for managing model changes in the Platform subsystems (up to the 2.5 release)
- Source, an environment for managing model changes in the Platform subsystems, including Scada
- Historian, a data management and reporting application for historical data and operational knowledge

In addition, the ADMS architecture allows a full-function simulator to be implemented by running the standard network server modules in an alternate mode. The performance and accuracy of the simulator means that it can be used for both training and system validation testing.

2. A Tour of the ADMS Solution

The aim of an Advanced Distribution Management System (ADMS) is to provide a single consistent operating environment that encompasses all the necessary functions for the efficient management of the distribution network. GE's solution for real-time distribution network management is ADMS.

ADMS is a complete real-time Integrated Distribution Management System, as shown in the below image. The integrated applications are specifically designed to support the large variety of activities related to distribution power system operations. The system will help distribution operators accomplish day-to-day routine tasks as well as handle critical emergency situations resulting from major outages.

The other utility systems that interact directly with ADMS to provide network model, customer, and metering information are shown in orange at the bottom of the below image.

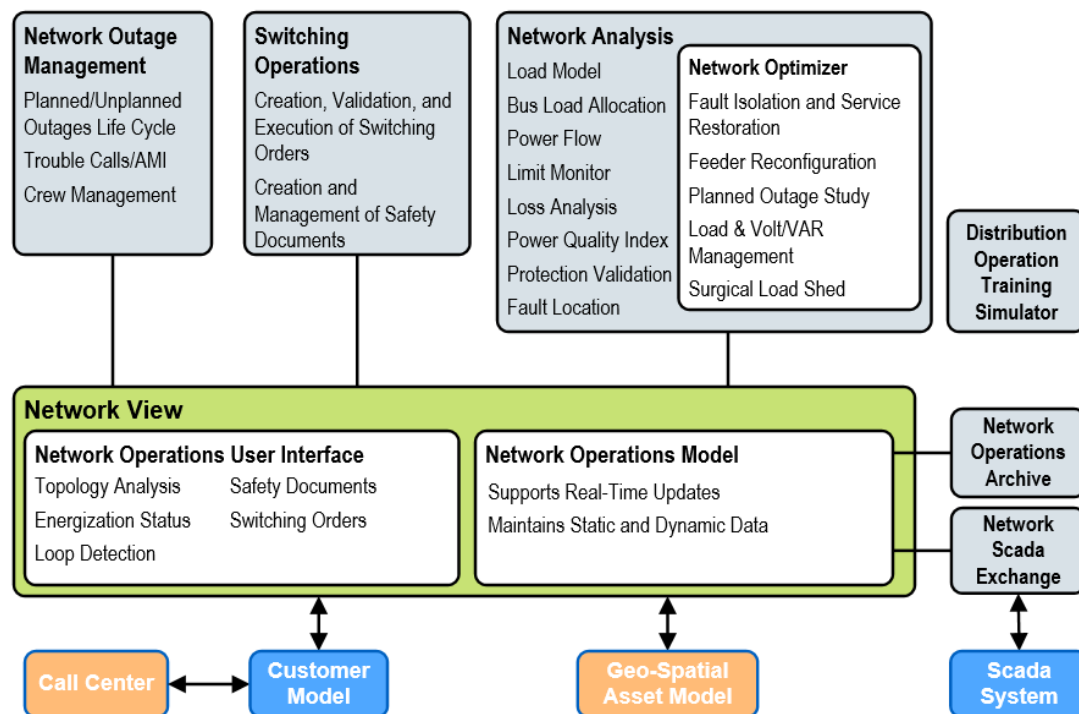


Figure 3. The Complete ADMS Suite of Applications

ADMS was developed to meet the real-time network management needs of both large and small electricity distribution utilities. These applications provide a full range of high-quality network operation tools, as well as adding value to existing utility systems such as the Geographic Information System (GIS), which is used as the source of the data model. This interface will be achieved by using adaptors based on the Common Information Model (CIM) data description language. Further development of this adaptor strategy in the future will allow ADMS to interoperate with other corporate systems, such as mobile resource management systems or planned work management systems, to provide a comprehensive solution that extends beyond the control room's boundaries.

SCADA integration is a key part of the ADMS concept, allowing the Distribution Network Object Model to be automatically updated with real-time data (status and measurements) from the field, and allowing the distribution operator to directly control network devices from the ADMS user interface.

The below image shows an example of how operation of a SCADA feeder breaker (center feeder bank in the schematic one-line display) automatically updates the energization status of the Network View. Notice the white de-energization halo on the corresponding feeder of the geographic display.

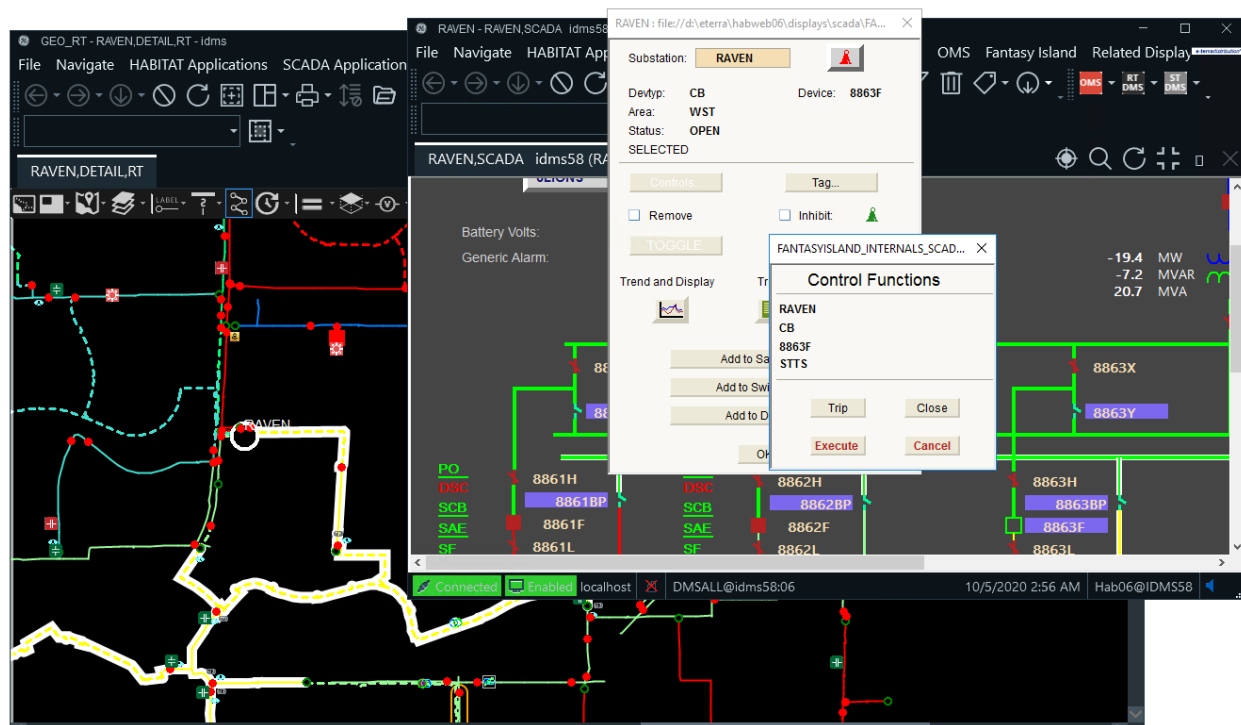


Figure 4. Feeder De-Energization Due to SCADA Operation of Feeder Breaker

2.1. ADMS Feature Summary

ADMS is a complete end-to-end real-time network management environment with the following features:

- **System Level Features**
 - High availability (hot standby and multi-host support)
 - Install packages
 - Citrix environment support
- **Distribution Network Object Model (DNOM)**

Real-time electrical network model with equipment, connectivity, electrical properties, load model, and static and dynamic data with updates.

- **Network Operations User Interface**

Geographic displays, schematic displays, customer displays, topology analysis, energization status, annotations, tagging, loop detection, operator notes/annotations, network tracing, temporary network modifications (cuts, jumpers), nested objects, tagging, mobile substations, landbase information, and future construction.

- **Scada Integration**

- Monitor the operating status of devices in a monitored station in real time
- SCADA control from DMS
- Tag support on SCADA and DMS objects
- Tag Summary Display and Processing List
- Intelligent Alarm Processing
- Alarm and Event Logging
- RTU Scanning
- Display Notes
- Orphaned Tags Utility
- Safety Documents (creation and maintenance of Safety Documents)

- **Switching Operations**

- Create, validate, and execute Switching Orders
- Save partially complete switch orders
- Track switch order status
- Search for and retrieve a Switching Order
- Daily Log
- Templates
- Switching Order client Editor
- Switching Order Simulation in Study Mode
- Create Safety Documents
- Switching Operations client Plug-in
- Switching Operations server
- Switching Operations Web server

- **Distribution Network Analysis Functions (DNAF)**

Real-time and study mode DNAF applications:

- Distribution Power Flow (DPF)
- Bus Load Allocation (BLA)
- Power Quality Index (PQI) Evaluation

- Loss Analysis (LA)
- Limit Monitor (LIM)
- Fault Location (FL)
- Fault Isolation and Service Restoration (FISR)
- Short Circuit Calculation (SCC)
- Protection Validation (PRV)
- Automatic Feeder Reconfiguration (AFR)
- Load and Volt/VAR Management (LVM)

Study mode applications:

- Planned Outage Switching (PLOS)
- Fault Isolation and Service Restoration Contingency Analysis (FIRCSA)

- **Network Outage Management**

Unplanned/Planned Outage Management:

- Outage extent analysis
- Outage lifecycle management
- Network performance indices
- Interruption notices (customer communication)
- Coordination with switching operations
- Outage history
- Maintenance planning
- Interfaces (including for AMR / AMI, for Customer Energization and Consumption)

Trouble Call Management:

- Planned work management
- Customer notification
- Resource management
- Mobile data system
- Vehicle location
- Call entry (call handling front end) IVR/TCS
- Interfaces (including for Postal)
- Call aggregation

Outage Prediction

Crew Management:

- Monitor and assign crews
- Crew callout
- Crew optimization
- Interfaces (including for the Workforce Management System)
- **Study Mode**
 - Simulated planned switching based upon actual network state
 - Run network view and network optimization standalone on clients
- **Network Simulator**
 - Distribution Operator Training Simulator (DOTS)
 - Integration with SCADA simulation
 - Outage Management System simulation (RoboCrew to simulate crew behavior, RoboDispatcher to assign crews and clear the incidents, and other features)
 - Event Scripter to simulate scenarios
 - Snapshot of real-time environment
- **Network Operations Archive** (Historian, ADMS Archive, and DNOM Database)
- **Application Programming Interfaces**
 - Distribution Management interface
 - Outage Management read-only interface
 - Call and Crew interfaces
- **Sample Reports**
- **Modeling** (Input, Tools, and Management)
 - Common Information Model (CIM) based XML schema for data sources including station internals, GIS ("station externals"), and Customer Information System (customer data and electrical address).
 - Model Export (exporting DNOM in CIM format)
 - Substation Editor
 - Substation Builder
 - Substation File Builder Lite
 - Annotation Importer-Exporter
 - DNOM Database Loader application and service
 - DNOM Unique ID Checker
 - DNOM Viewer
 - Model Manager Application and service

- Converter
- Customer List Compiler
- Geographic Background Compiler
- Navigation Builder
- Permit Builder
- File Updater (server and client)
- Crew Modeling
- Model Quality Stat Logger
- **Permission Management** (Permit server, Permission Management Plug-In)
- **Configuration Tools**
 - Viewer Configuration Editor
 - Net Dynamics Configuration Editor
 - Model Manager Configuration Editor
 - Coded Translation Editor
 - DNAF Configuration Editor
 - DNOM DB Loader Configuration Editor
 - OMS Client and OMS Server Configuration Editors
 - Resource Editor
 - Encryption Tool
 - Snapshot Viewer (view, import, export, and edit snapshots, journals, and dynamics)
 - Switching Order Client and Switching Order Server Configuration Editors
 - Grid Tabular Builder
 - File Updater Configuration Editor

ADMS is a comprehensive network management solution that spans the full extent of the electrical supply network and delivers advanced functionality.

2.2. Distribution Network Object Model

The Distribution Network Object Model (DNOM) is the foundation for all ADMS functionality. This network model supports all the operational and analytical functions that might need to be undertaken by both operational and network planning staff working either in the control centers or in the field. This means that there is no need to support separate analysis modeling and display definition efforts.

The DNOM is a phase-domain, memory-resident, object-oriented model. To support the performance and functional requirements of the ADMS, it has the following characteristics:

- Large model sizes (more than 25 million objects)
- Frequent updates (normally updated every day)
- High volume of switching activity (planned and unplanned)
- The ability to model a wide range of network configurations, including temporary modifications

The static data model of the DNOM is normally generated and maintained by extracting data from the utility's asset management system— usually a Geographic Information System (GIS). These data updates are performed on a per-station basis using the vendor-neutral Common Information Model (CIM) interface standards.

2.3. Network View

The Distribution Network Object Model (DNOM) and the Network Operations user interface combine as a stand-alone module known as "Network View", which forms the basis of the ADMS suite of applications.

Network View supports full real-time operations by maintaining both static and dynamic data, and it presents this information to the operator in a single consistent view of the network, as shown in the below image. The user interface reads the data to be displayed directly from the Distribution Network Object Model without the need for separately prepared display definitions.

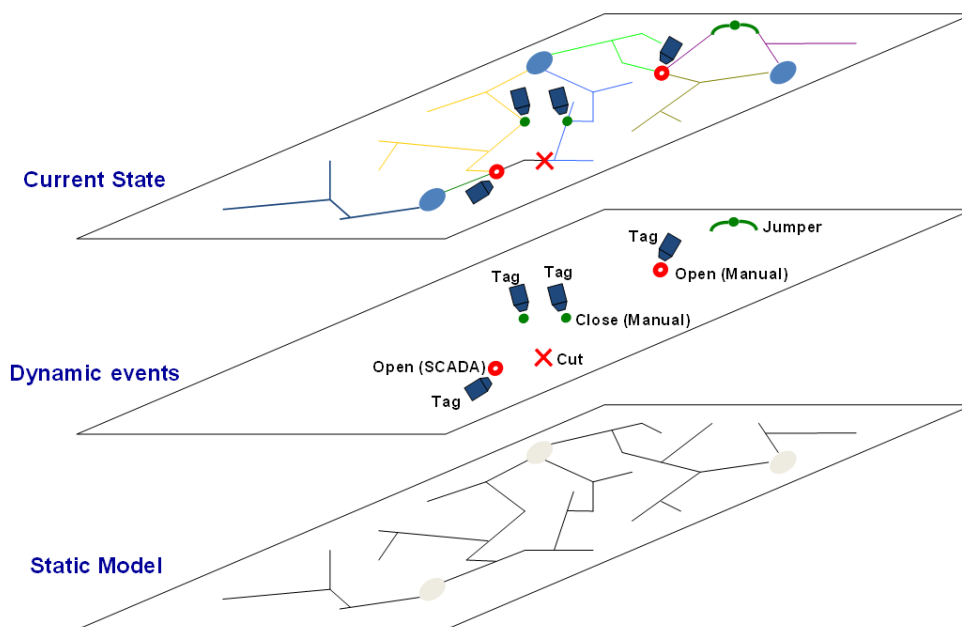


Figure 5. Network View Display Components

The Network View user interface is an environment that provides panning, zooming, and de-cluttering features. It uses a URL-based command structure to call up different types of displays and control layers, and it positions the views of the distribution network.

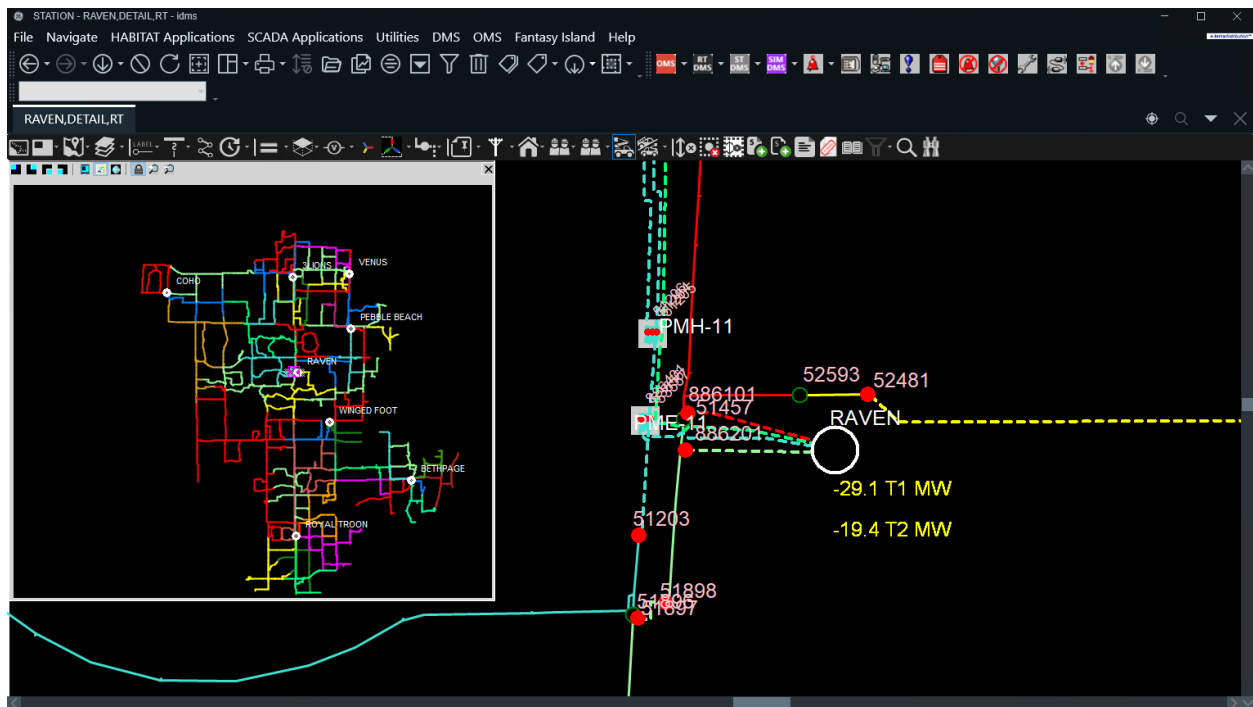


Figure 6. An Example of a Geographic Network View

Network View directly supports the following functions:

- Upstream and downstream network tracing (see [Network Downstream Trace Function \(Cyan Halo\)](#))
- Network and display navigation
- Device search
- Topology processing (see [Topology Processor Indicating De-Energized Line Due to Open Switch](#))
- Device operations (for example, opening and closing switches)
- Temporary network modifications (jumpers, cuts, grounds) (see [Temporary Network Modifications \(Cut and Jumper\)](#))
- Tabular displays of DNOM data

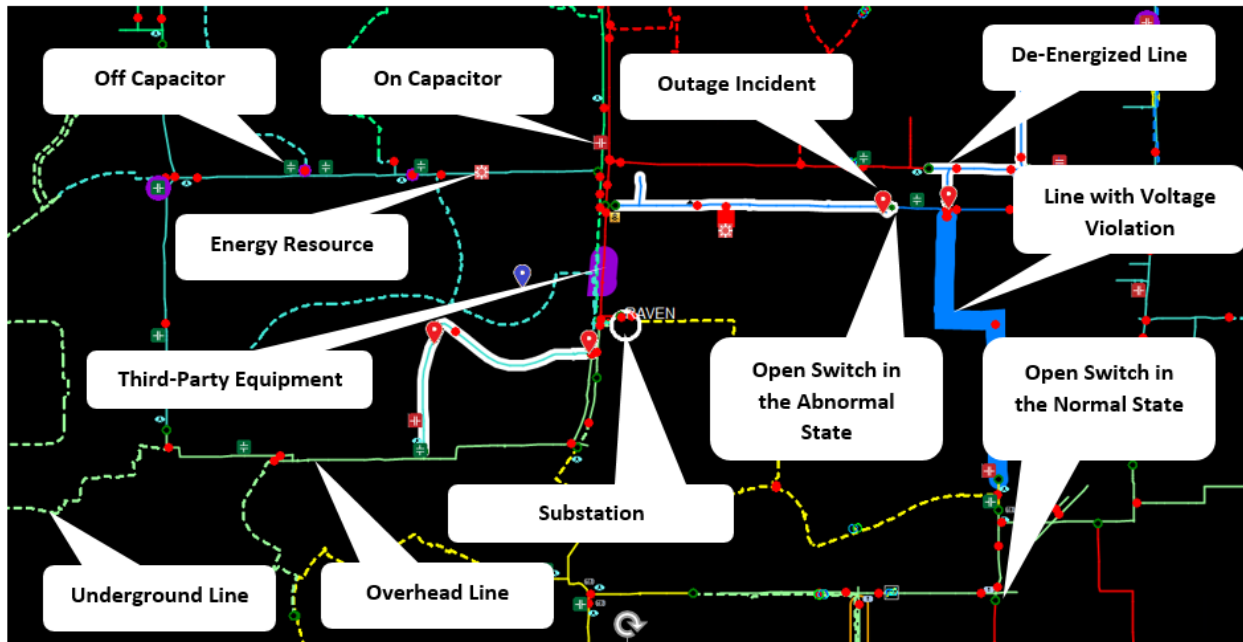


Figure 7. Typical Network Component Symbology

2.3.1. Tracing the Network

The topology processor performs network trace functions on the local client. These traces have no effect on the operations model or other clients viewing the network. The trace functions allow operators to easily identify energization sources or end points of a network section. Traces are indicated by a cyan halo applied to all the affected devices including laterals and customers that may not be visible on the current display. A trace is performed by selecting a point on the network and then using the toolbar buttons. Traces recognize the current dynamic state of the network including temporary modifications. Up to eight traces with different colors can exist simultaneously on a geographic display.

The trace functions available are:

- **Upstream Trace:** Traces from the selected point to the energization source – usually the feeder head.
- **Upstream Trace to SCADA:** Traces from the selected point to the first upstream SCADA controllable device. If no device is found, an informational message appears and the tracing is canceled.
- **Upstream Trace to Protective Device:** Traces from the selected point to the first upstream protective device such as a fuse, recloser, breaker, etc. If no protective device is found, an informational message appears and the tracing is canceled.
- **Upstream Trace to Switching Device:** Traces from the selected point to the first upstream switching device, including temporary switches such as cuts and jumpers. If no device is found, an informational message appears and the tracing is canceled.
- **Upstream Trace to Recloser/Breaker:** Draws a trace upstream from the selected point to the

next upstream recloser or breaker. If there is no upstream recloser or breaker, an info message appears and the tracing is canceled.

- **Downstream Trace:** Traces from the selected point out to all the downstream extents on the network. The system status message displays the number of customers and transformers included in the trace. If a downstream trace is performed on a de-energized section of network, the trace identifies all devices electrically connected to the selected point.
- **Downstream Trace to SCADA:** Traces downstream to the first SCADA controllable device. If no device is found, an informational message appears and the tracing is canceled.
- **Downstream Trace to Protective Device:** Traces downstream to the first protective device such as a fuse, recloser, or breaker. If no device is found, an informational message appears and the tracing is canceled.
- **Downstream Trace to Switching Device:** Traces downstream to the first switching device, including temporary switches such as cuts and jumpers. If no device is found, an informational message appears and the tracing is canceled.
- **Up and Downstream Trace:** Traces from the selected point upstream to the source, then out to all the downstream extents on the network. The system status message displays the number of customers and transformers included in the trace. If a downstream trace is performed on a de-energized section of network, the trace identifies all devices electrically connected to the selected point.
- **Extent Trace:** Identifies all the circuit elements connected to the selected network component. For de-energized components, this traces to the first open device.
- **Trace Between Marks:** Traces from the selected component out to any marked devices. If no marked devices are discovered, the trace defaults to an Extent Trace.
- **Path Trace:** Identifies all the circuit elements on the most direct path between the two selected points on the network, if electrically possible. It is important to note that feeders are normally modeled in such a way as to prevent traces passing through the feeder head and via the substation bus.
- **Trace Between:** Identifies all the circuit elements on the most direct path or connected to the path between the two selected points on the network, if electrically possible. It is important to note that feeders are normally modeled in such a way as to prevent traces passing through the feeder head and via the substation bus.
- **Switchable Section Trace:** For a selected line section, the switchable section trace identifies all the line sections, including laterals, between bounding feeder backbone switches. The system status message displays the number of customers and transformers within the bounded section. In this way, an operator can quickly identify the switches needed to isolate or energize a point in the network and obtain a count of the number of customers and/or transformers that will be affected.
- **Phase Tracing:** Upstream and downstream traces of individual phases may be performed using this function, allowing the operator to identify the path of a given phase through an unbalanced section of the network, such as an underground residential loop.
- **Trace to All Potential Isolating Devices:** Marks the switches that need to be tripped to isolate

the selected point on the line, and also draws the trace to the switches.

- **Trace to All Normally Open Switches:** Marks all normally open switches that are connected to the selected point on the line, and also draws the trace to the switches.
- **Clear Trace:** Clears all traces in all viewports.

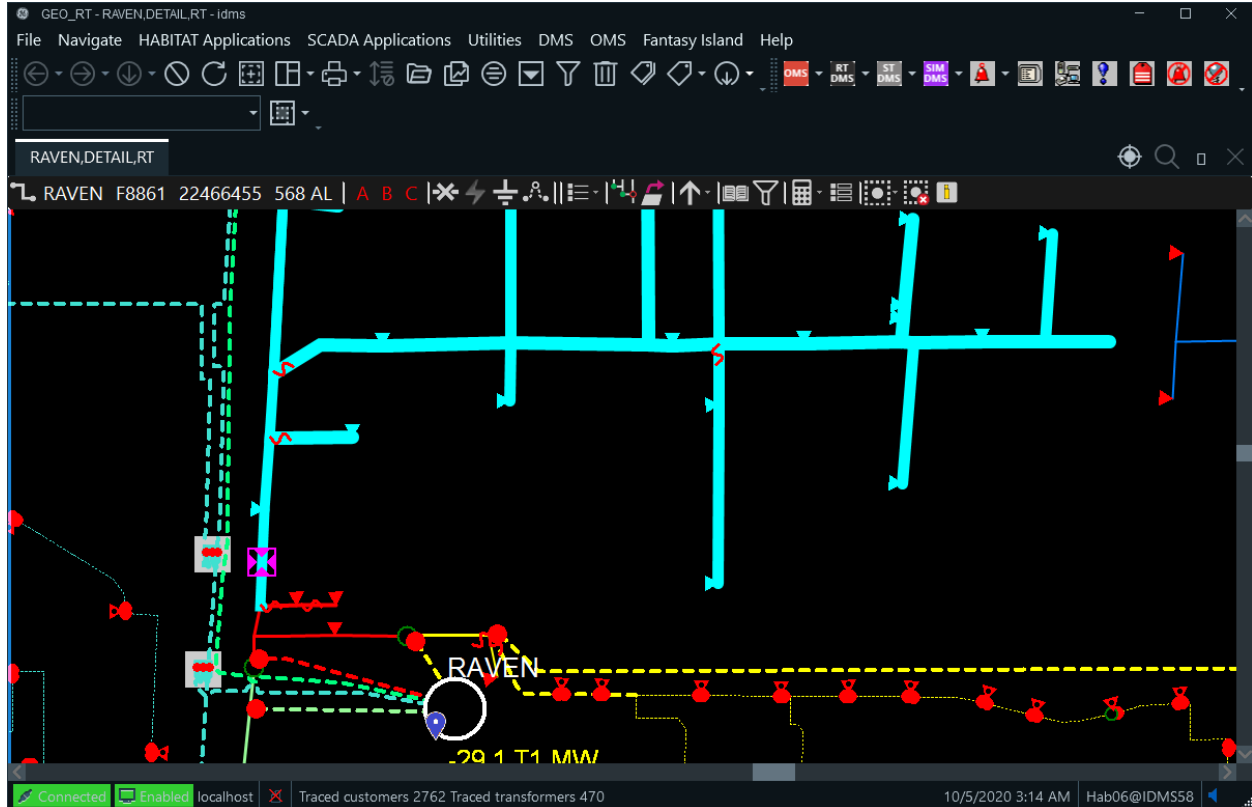


Figure 8. Network Downstream Trace Function (Cyan Halo)

2.3.2. Topology Processing

When there is a status change on the network, either via a SCADA update or manual switching operation, the topology processor automatically updates the network view to reflect the impact of the change on the network. For example, if a lateral fuse is opened, the lateral is shown as de-energized.

The Topology states shown on the display are:

- **Normal:** The lines are shown in the nominal feeder color.
- **De-energized:** The lines are shown with a white halo (configurable).
- **Abnormally energized:** The lines are shown in the color of the energizing feeder with white cross-hatching.
- **Network and Parallel Loop:** Any condition where a section of the network is energized from more than one point, either on the same feeder or on neighboring feeders, is considered a

loop. The main energization path is identified with a yellow halo (configurable). The line sections connected to the main energization loop are shown in a single color (typically magenta) with white cross hatching.

- **Powerflow Mismatch:** This refers to a multiphase line section where there are both energized and de-energized phases, or the direction of the power-flow in one of the phases is opposite the other phases. The affected line sections are identified with an orange (configurable) halo.
- **Grounded:** Network sections that are de-energized and have one or more grounds applied are identified with a green or green-yellow halo (configurable).
- **Faulted:** Likely locations of a fault (based on the fault indicator and feeder protection relay statuses, as well as the fault location application results) are identified by halos (configurable).
- **Dangerous Conditions:** The Dangerous Switching Condition decoration appears on open switches whenever closing a switch could result in an unsafe network condition, including dynamic phase mismatch, phase angle mismatch, and nominal voltage mismatch.

The topology processor runs locally on each client to ensure a fast response to network changes and to reduce the network traffic and server loads.

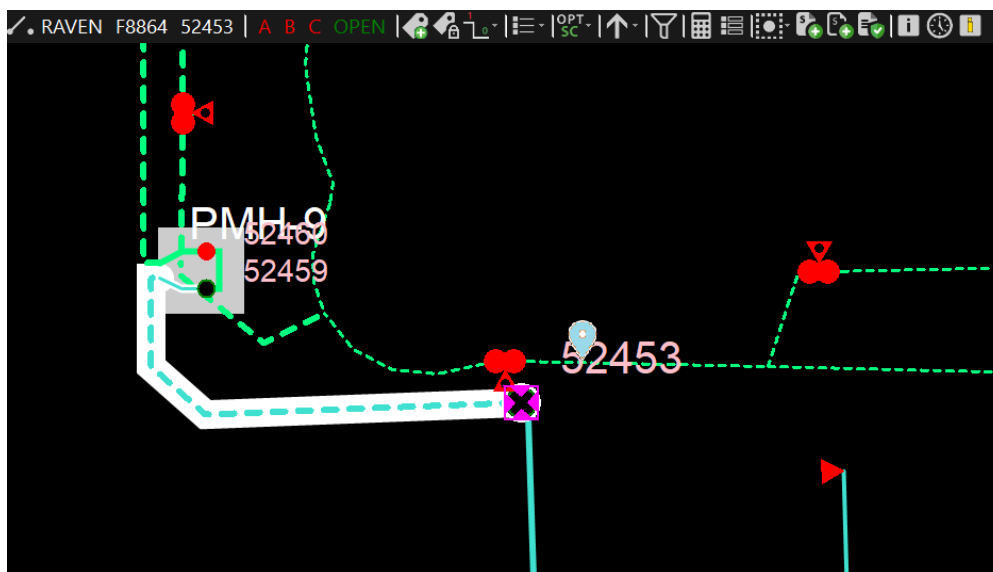


Figure 9. Topology Processor Indicating De-Energized Line Due to Open Switch

2.3.3. Temporary Network Modification

Temporary modifications are made to the distribution network to isolate faults, restore services, or allow maintenance work to be undertaken. These modifications are performed by field crews and must be reflected in the network operations model in order to maintain the correct network topology. ADMS provides tools to allow operators to apply cuts, jumpers, temporary switches, and grounds to any point in the network and enter a name/description for the modification. The Abnormal Summary display lists abnormal topologies and temporary devices (for example jumpers, faults, cuts, and grounds) that are currently applied.

Cuts are used to place a temporary break in any line segment in the network. The "cut" object is actually a temporary in-line switch and the status can be changed as required. The symbol can be selected to show either a cut or a temporary switch. Once placed, the cut or temporary switch can be opened, closed, and tagged in the same way as a standard disconnect switch. The topology of the network is automatically updated to reflect the changes caused by the placement of the cut.

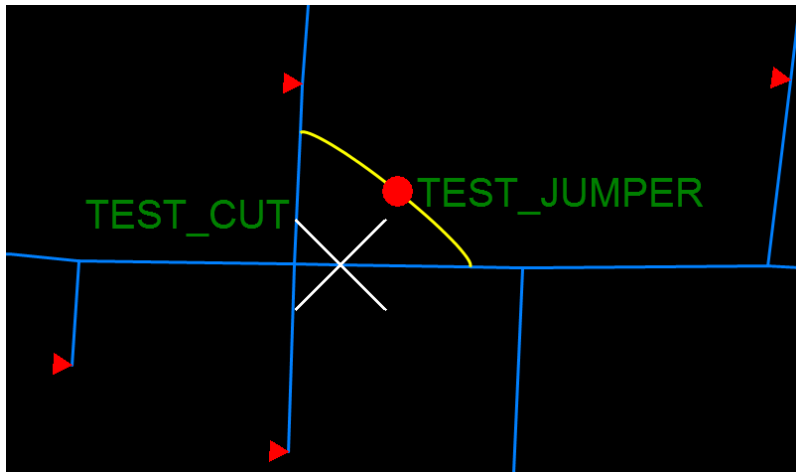


Figure 10. Temporary Network Modifications (Cut and Jumper)

Jumpers provide a temporary, switchable connection between two points in the network. The initial state of the jumper is set to open. Once the jumper has been placed, the switch symbol in the midpoint of the jumper may be selected and opened or closed in the same way as a standard disconnect switch. The topology of the network model is updated accordingly. Jumpers can be attached to a line, or to a distribution transformer. When attached to a distribution transformer, the jumper is electrically connected to the high side of the transformer.

Temporary grounds can only be placed, for obvious reasons, on de-energized line sections. These grounds represent the physical grounds placed on lines for safety purposes during maintenance and construction.

A temporary ground may be placed on one or more available phases. The mandatory description becomes the ground label and any information entered becomes the explanation.

2.3.4. Viewports

The ADMS client is part of the unified Browser interface. This provides consistency of operation and integration with other GE products.

Multiple independent viewports allow for multiple parts of the network to be readily monitored and operated simultaneously. Viewport configurations can be saved by operators as "rooms" so that standard or particular arrangements of displays can be quickly called up.

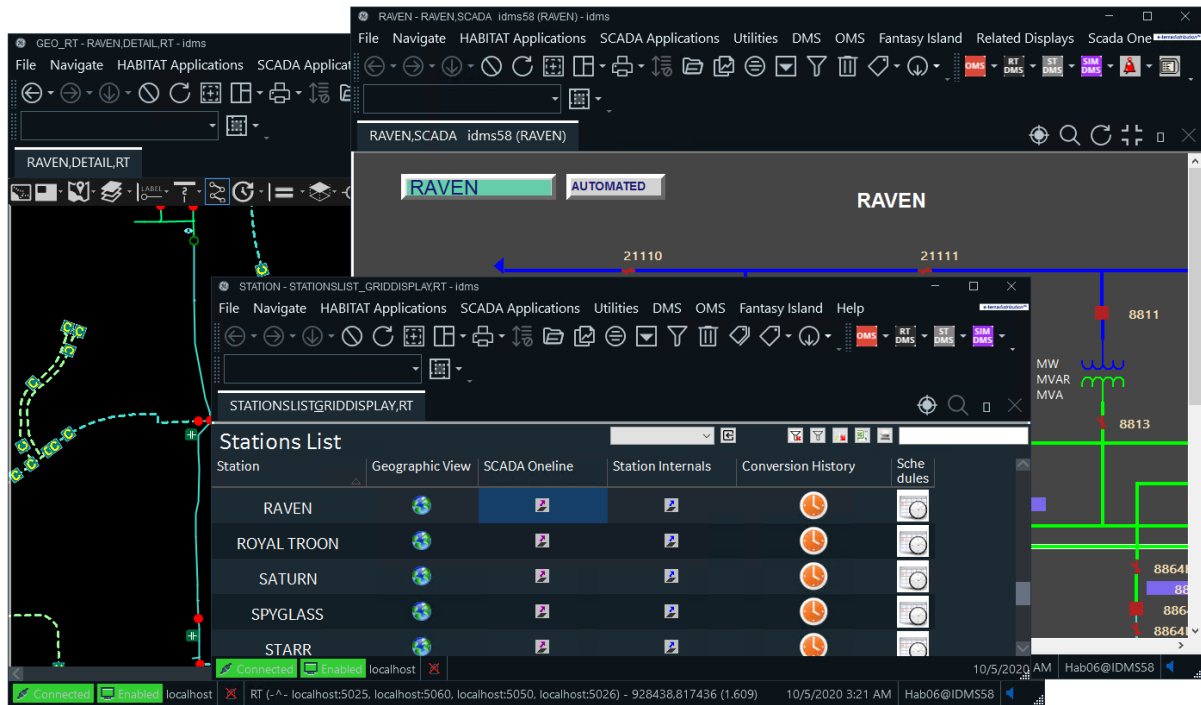


Figure 11. Browser Showing Multiple Views

The ADMS client supports a wide range of tools to allow the operator to effectively navigate and manage the network, such as:

- Context sensitive toolbars and menus
- Indicate phases for all line sections
- Display Layer controls
- Tabular displays
- Device operation directly from the network view displays



Figure 12. Elements of the Network View Display

2.3.5. Container Objects

Visualizing the current state of the network is the critical function of the network view. In cases where switches are contained within cabinets or vaults, a simple geographic view is not satisfactory for effectively operating the network. ADMS can present an internal view of cabinets and vaults in two ways:

- The internal switching of the cabinet can be extracted and presented as a small, automatically generated schematic within the overall geographic network view. The "explosion" of the cabinet/vault internals from the single symbol is based on the display zoom level.
- Separate "internal world" views of cabinets and vaults may be created in the GIS (model data source) or the Substation Editor tool. These displays may be called up in a separate window as a related display to the cabinet/vault object.

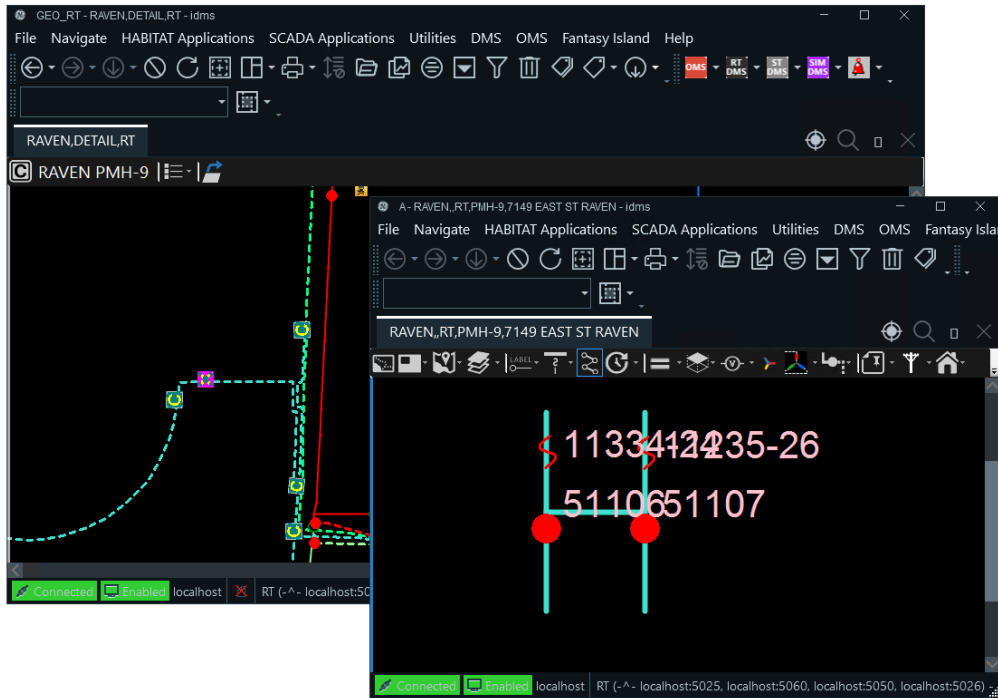


Figure 13. Internal World of a Switch Cabinet

2.4. Network Analysis

The Network Analysis module is a powerful set of tools based on a three-phase unbalanced distribution network power flow. Network Analysis allows users to study either the current state of the network or its state at some future time. Power flows can be solved for all types of network configurations, including meshed, radial, and unbalanced networks. Individual phase voltages are calculated at all nodes, and power flows are calculated for all feeder segments. Additional tools are provided to calculate system losses and loads at distribution feeder nodes. The limit monitor identifies feeder voltage violations and line flows that are outside of acceptable operating limits (see below image).

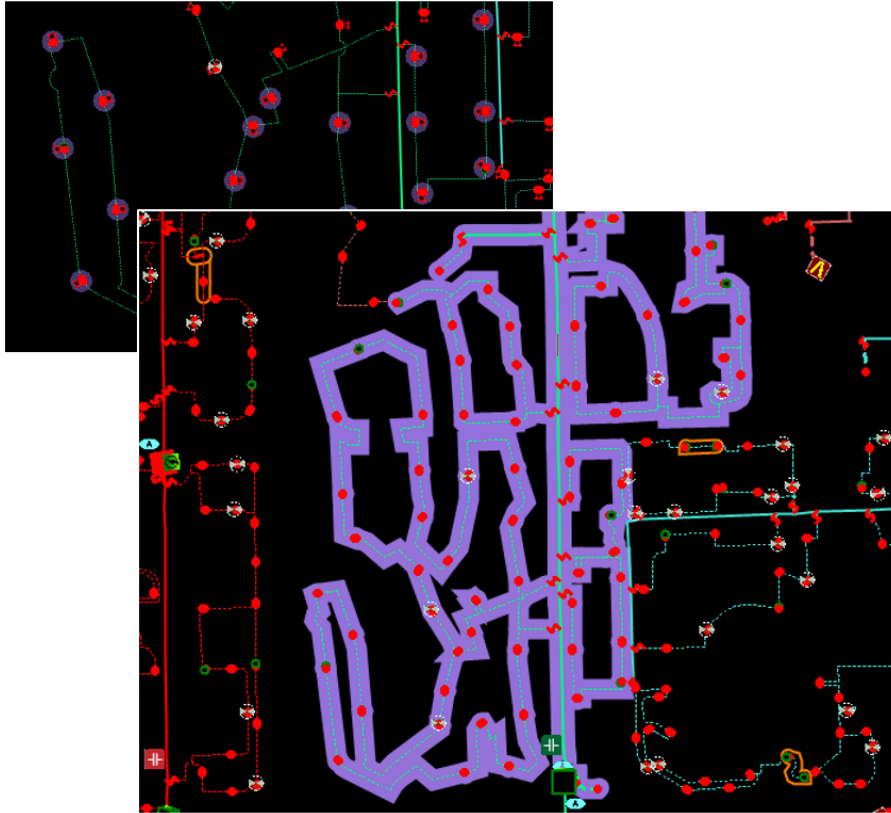


Figure 14. Power Flow Analysis Showing Voltage Violations

Network View is used as the input and display system. The Network Analysis functions are further enhanced by adding the Distribution System Plan Generation Engine (DSPGE) module, which provides network reconfiguration and load control functions.

ADMS includes the following network analysis functions:

- **Distribution Power Flow (DPF):** Finds the complex voltages at all nodes and the power flows through all line segments in the distribution network. Included within the Distribution Power Flow is the Bus Load Allocation function, which provides estimates for the loads (kW and kVAR) at each feeder node, and Limit Monitor (LM), which checks the DPF solution for line flows and voltages that are outside of defined operating limits.
- **Power Quality Index (PQI) Evaluation:** Determines the extent of feeder voltage violations, including voltage imbalance.
- **Loss Analysis (LA):** Calculates the technical losses for conductors, transformers, and feeders.
- **Protection Validation (PRV):** Calculates the maximum and minimum short circuit current for a phase-to-phase or phase-to-ground fault at selected points in the distribution network. The calculated fault currents are then used to verify that each segment of the feeder is properly protected.
- **Load Model and Estimation:** Estimates the loads based on the current time, day type, and season.
- **Fault Location:** Uses information from protection relays and fault detectors to determine the

most probable locations of a fault on a feeder.

The Network Analysis functions can be executed in both real-time and study modes. The real-time mode uses the current power system conditions (SCADA measurements and schedules) to evaluate the system. The functions are triggered to run periodically on pre-specified events or on operator demand. The trigger events include:

- Topology change: feeder main or lateral
- Sudden feeder load and substation bus voltage changes

In study mode, operational data is retrieved either from a savecase or a snapshot of the real-time system. Studies for some point in the future are estimated using load profile values. The load scaler is based on the study time and the day type. The forecast state of network switching is obtained by applying existing or planned switching orders to the network model.

2.5. Distribution System Plan Generation Engine

The Distribution System Plan Generation Engine (DSPGE) module is a set of advanced network reconfiguration and load management tools that augment the functionality of the Network Analysis system. These tools assist the dispatcher with operating the system in an optimal fashion during both normal operation and outage conditions.

The functionality provided by the Plan Generation Engine includes Fault Isolation and Service Restoration (FISR), Automatic Feeder Reconfiguration (AFR), Load and Volt/VAR management (LVM), Planned Outage Switching (PLOS), and Fault Isolation and Service Restoration – Contingency Analysis (FISRCA).

Output switching plans created by the Plan Generation Engine can be executed manually by the dispatcher, or fed to switching operations to be turned into a complete switching order. In addition, the Plan Generation Engine module provides a Closed-Loop functionality in which the solved switching plans can be automatically executed. This assumes the necessary SCADA measurements and controls are modeled in the integrated SCADA subsystem. The Closed-Loop mode is a key functionality for a self-healing distribution management system.

- The Load and Volt/VAR Management (LVM) function provides recommendations to improve the voltage quality and provide reactive support to the surrounding distribution system. By using analysis of a significant portion of the distribution network and coordinating the operating commands for all active regulating devices (including capacitors, tap changing transformers, and generators), LVM achieves considerable improvement over conventional load and Volt/VAR control techniques.
- The Fault Isolation and Service Restoration function (FISR) allows optimally restoring service to customers. In the case of a sustained outage, FISR creates a Switching Plan that isolates the faulted area and provides service restoration by using optimal feeder reconfigurations, including cascading reconfiguration to off-load one feeder in order to pick up part of the load of an adjacent feeder. FISR uses a problem formulation that defines one or more operational objectives, including minimizing un-served kW, minimizing customer minutes interrupted,

minimizing the number of switching operations, and minimizing voltage drop along the reconfigured feeders. The problem formulation also specifies the network and operational constraints, including segment overloads, voltage violations, and relay settings. FISR is triggered manually either by placing a fault symbol on the real-time network view or opening the DNAF Solution display.

- The Automatic Feeder Reconfiguration (AFR) function provides switching plans to optimally unload overloaded segments. The function can be used to reduce overloads on both substation transformers and feeder segments. The function takes into account peak load conditions within a user-configured time window.
- The Planned Outage Switching (PLOS) recommends switching plans to support planned outages. PLOS is only available in study mode. The user specifies the network device that is to be taken out of service and PLOS generates one or more possible switching plans. PLOS first transfers the affected load and then isolates the specified device to prevent the unnecessary dropping of load. The operator can prioritize the optimization of the switching plan by selecting an appropriate problem formulation.
- The Fault Isolation and Service Restoration– Contingency Analysis (FISRCA) application executes a FISR analysis for every feeder main switchable section in a given substation. FISRCA is only available in study mode. FISRCA packages a group of FISR executions, sequentially placing a fault on each switchable section, and then solving FISR to determine if a valid restoration plan can be found that restores power to all customers connected to the non-faulted portion of the network.

2.6. Switching Operations

The network switching operations consist of two main components: switching orders and safety documents.

Switching orders define the step-by-step procedures for taking a piece of the network out of service, restoring it, or making a change in network configuration. Switching orders can be a formally approved list of switching instructions or a step-by-step log of operator actions.

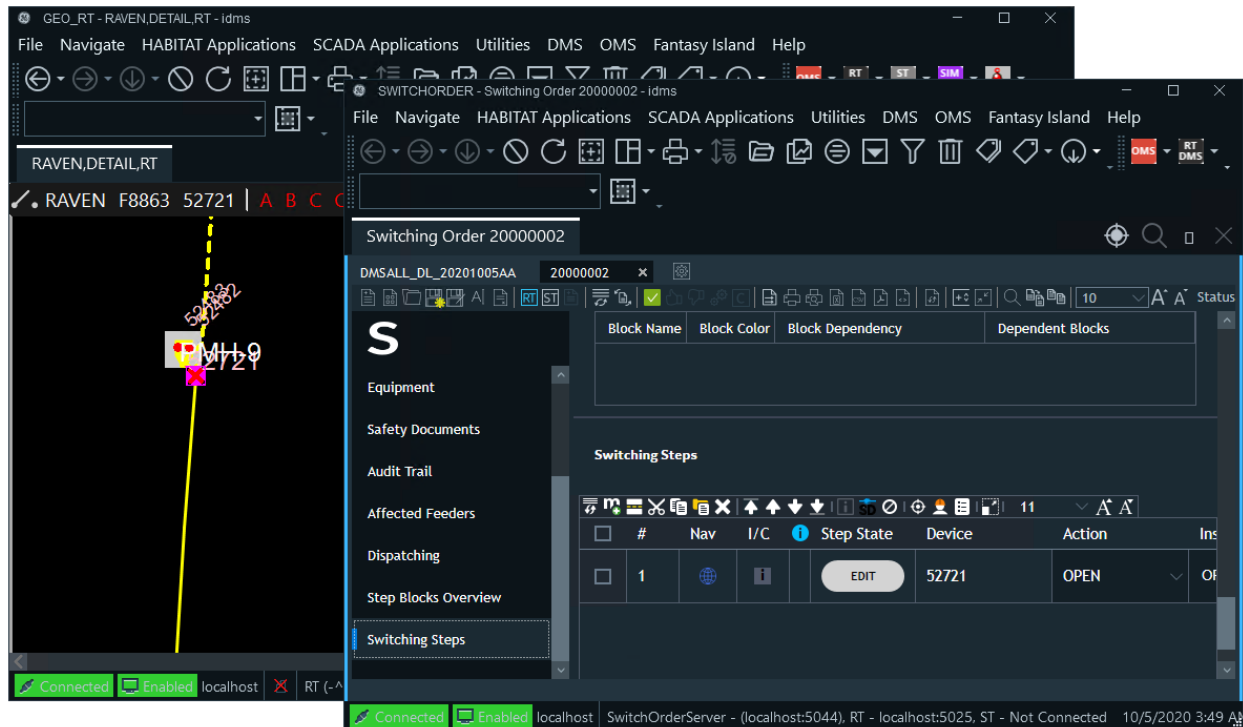


Figure 15. Switching Order Form

Switching orders can be created directly by selecting devices from the user's view of the electrical network provided via the Network View user interface (see above image). In study mode, operator's actions can be recorded as switching steps. Switching orders can also be generated automatically by the Network Optimization functions.

The effect of a switching order can be simulated by applying it to the study network model prior to execution of the order on the live network. Execution of switching orders automatically updates the status of non-telemetered switches in the network model, to ensure that the model always reflects the current status of the system.

While the switching operations act on the DNOM as with all other functions, the operation of the switching orders is managed by the switching order server.

Safety documents are the process by which jurisdiction for a section of the network is passed between the operator and field or maintenance crew. A number of different types of safety documents can be defined. A safety document creates a list of the tags that are to be placed as part of the switching process that isolates the required section of the network. The issuing of a safety document is confirmed by selecting each tag on the network view to verify that the tags have been correctly placed. In this way, the issuance and release of safety documents is interlocked with the execution of the related switching orders.

The screenshot shows a web application titled "SAFETY Document Summary". At the top, there is a "Create new Document" button and a search icon. Below this is a tabbed interface with "All", "Pending/Suspend", "Issued", "Released", and "Template" tabs. The "All" tab is selected, displaying a table with the following columns: Id, State, Name, Issue to, Device ID, Substation, and Document Type. The table contains six rows of data, all with a "Pending" state.

	Id	State	Name	Issue to	Device ID	Substation	Document Type
Edit	492E6F3B	Pending			50713	RAVEN	HOLD OPEN
Edit	492E6F2D	Pending			50713	RAVEN	HOLD OPEN
Edit	492E6F05	Pending			52721	RAVEN	DNO
Edit	492E6ED3	Pending			52721	RAVEN	HOLD OPEN
Edit	492E6E51	Pending					DNO
Edit	492E6D81	Pending					DNO

Figure 16. Safety Document Display

2.7. Network Outage Management

Network Outage Management is a trouble call and outage management system that collects, coordinates, and analyzes outage information from customer trouble calls, automated meter observations, and SCADA information. It assists the dispatcher by determining the most likely point at which customer supply has been lost and coordinating crews and other activities through the repair and restoration process.

Network Outage Management is designed to handle storm situations, provide basic work crew monitoring functionality, and calculate distribution performance indices to meet regulatory reporting requirements. It provides the necessary information to the dispatchers, other utility personnel, and information systems managers involved in restoring unscheduled outages and communicating outage and restoration information to customers.

A complete record of all actions taken in the restoration process is recorded for each incident. Network reliability and performance indices are automatically calculated.

GE's solution for the network outage management system includes the following benefits:

- The Outage Management System uses high level protection devices (feeder breakers, reclosers, station breakers, transmission zero-voltage indicators) and telemetered statuses from customer smart meters (AMI) as well as trouble calls via Interactive Voice-Response (IVR) and direct calls as outage observations.

This ability to merge historical and state-of-the-art outage observations (for example, phone calls and SCADA signals) is a huge performance benefit in large scale events. For example, when a feeder breaker trips there is no need for further bottom-up call-based prediction as the OMS system already knows the feeder is out.

- The OMS utilizes AMI observations to complement phone calls and to leverage AMI for verifying the extent of outages and, more importantly, verifying complete restorations.
- The system treats each customer end-point as an individual entity connected to a dynamic, as-operated wires model. As the OMS system model always reflects actual connectivity, customers are always tied to the correct energization sources (or not, when de-energized). As the GE OMS does not make gross assumptions (for example, assuming a static model) the restoration performance and accuracy of outage reporting metrics (SAIDI and CAIDI) are improved.
- Unlike some other systems, the OMS recognizes and models dynamic partial restorations and abnormal energization paths. Accurately handling partial restorations also improves the accuracy of outage performance indices.

The Outage Management System display is shown in the below image. The display has tabs that show a division overview, details of individual outages, and a geographic view of the affected areas of the network.

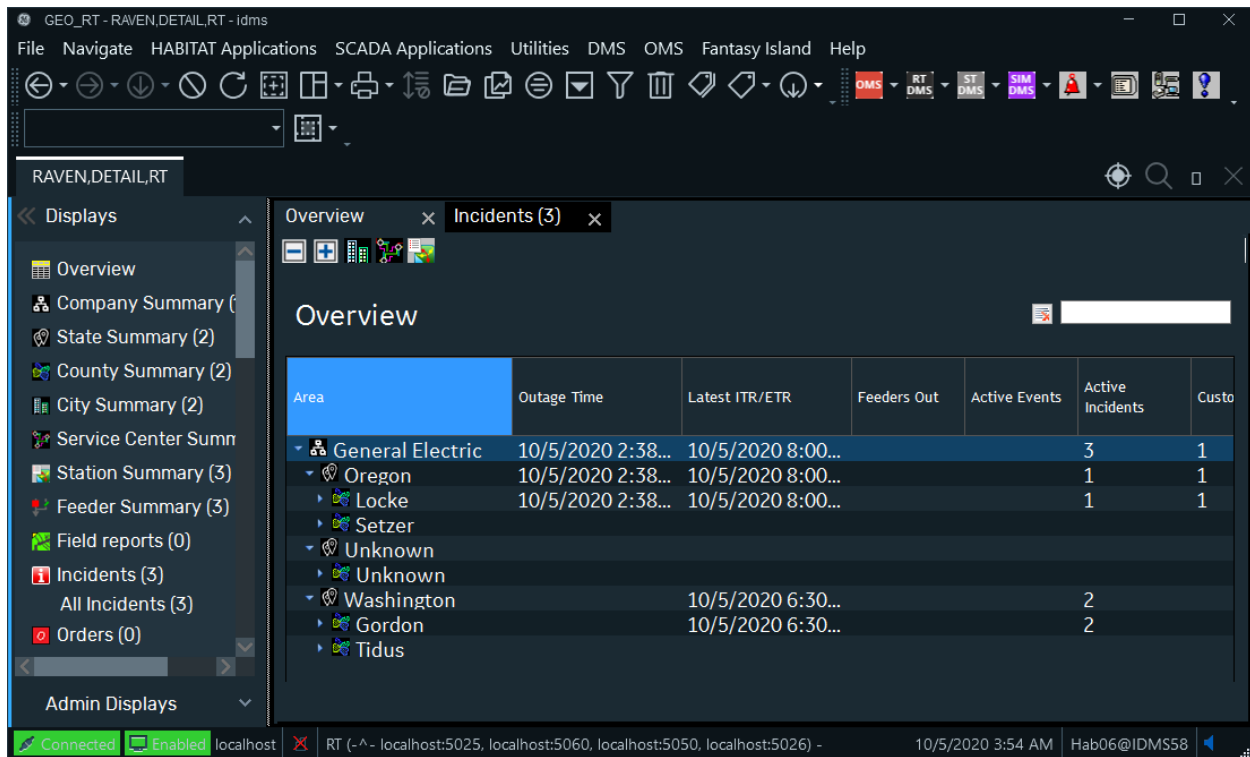


Figure 17. Outage Management System Display

Planned outages can be managed by using the Outage Management System.

The Outage Management System also includes an archive capability where each outage incident is written to a separate database when the incident is closed.

The Network Data server function of OMS can be used to support custom web-based dashboard summary displays of outage activity. This can be shown on all types of web capable devices such as smart phones and tablet computers.

2.8. Distribution Operator Training Simulator

The Distribution Operator Training Simulator (DOTS) provides an integrated system utilizing the same software components, interfaces, and user interfaces as the real-time network management system, as shown in the below image. A simulated power system model generates the required network responses, and the event scripter allows complex training scenarios to be developed. In this way, dispatchers can be trained in both routine and emergency operations that accurately represent the behavior and response of the real system without the risk of its disturbance. In addition, the simulator enables you to safely test new and existing DMS functions.

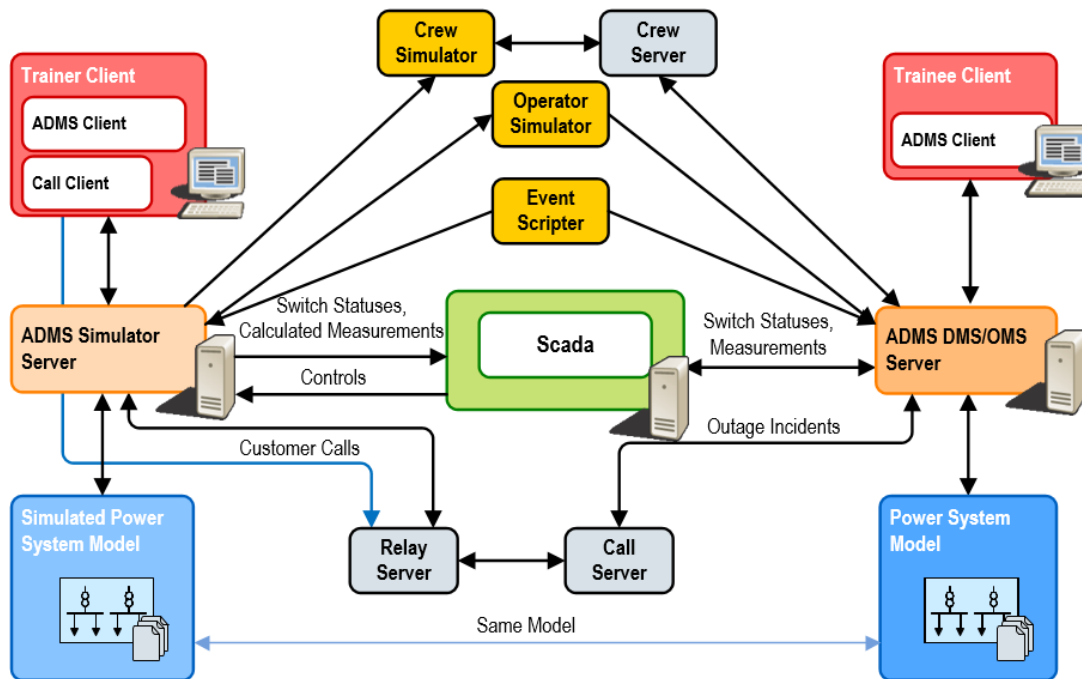


Figure 18. Distribution Operator Training Simulator

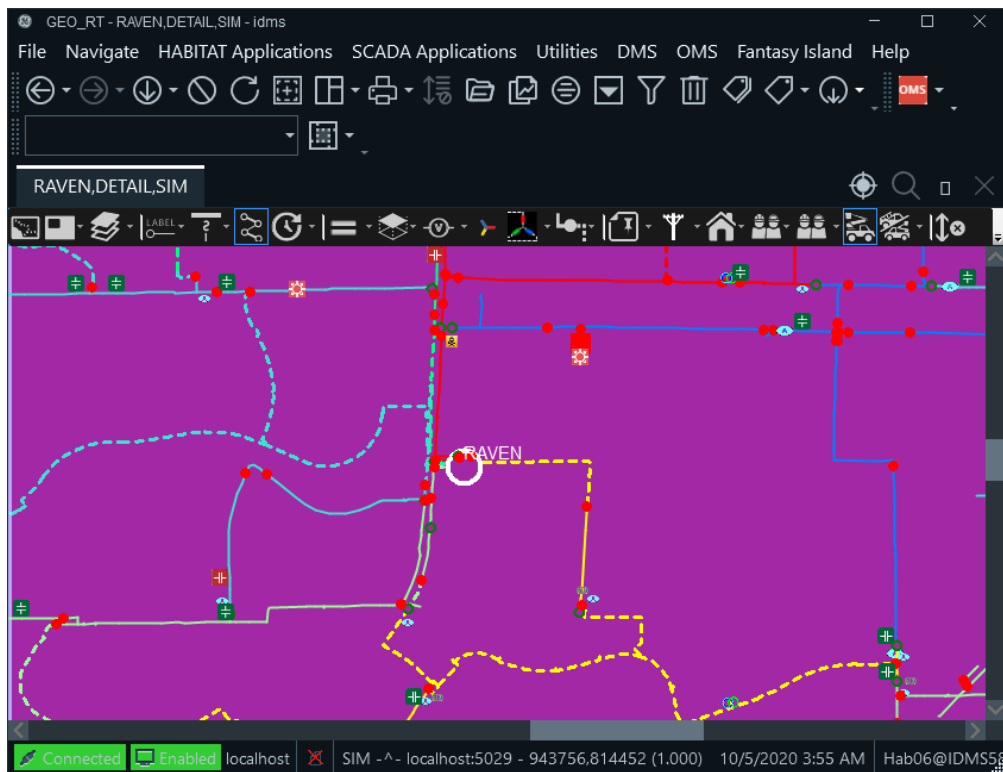


Figure 19. Simulator View of the Network

The training simulator supports the complete user interface of ADMS. The simulation engine is also able to exchange data with a SCADA simulation, providing an environment for the following types of training and testing:

Distribution operators training and testing:

- Study mode and real-time operations
 - User interface presentation and navigation
 - Simulation of actual electric system operation by manual user actions or scripting of events
 - Snapshot of real-time environment
 - Switching operations (manual and supervisory)
 - Annotations, tagging, and topology processing, including feeder coloring and tracing
 - SCADA operations and alarms
 - Network analysis and optimization
 - Crew management training (RoboTools)
- Event forensics
 - Post event analysis
 - Forensic event re-enactments
 - Human Performance Improvement (HPI)

- Training
 - New employee training
 - Employee development (skills, deficiencies, etc.)
 - Scenario training (emergency and day-to-day operations)
- Testing and certification
 - Initial evaluation and testing of potential operators
 - Operator certifications (initial and on-going)
 - Operator testing

Operating environment training and testing:

- Processes, procedures, policies, and configurations
 - Develop, revise, update, and test procedures and processes
 - Demo and study new features and functionality
 - Operational process design simulation and testing
 - Simulation and testing of new smart equipment
- DMS implementation and deployment support
 - Software regression testing
 - Simulation and testing of software functionality
 - Volt/VAR Optimization (LVM), Fault Location (FL), Fault Isolation and System Reconfiguration (FISR), switching, and DMS interfaces to other live, in production systems (such as OMS)
 - Provides a tool for software and device deployment to simulate operation, determine optimal system configurations and settings, and simulate performance
- Emergency, emergency restoration planning, and simulation
 - Storm and weather events, load shed, black start, peak load response, and cold load
- Automated system restoration solutions
 - Simulation and testing of closed loop FISR functionality before implementation of an actual pilot
- Event forensics
 - Post event root cause analysis
 - Forensic switching error simulation and analysis
- Day-to-day operations
 - Distributed generation impact simulations
 - Allows full featured simulations to be run for virtually any operational scenario to evaluate the best course of action
 - Facilitate study and analysis of system operations

- Switching verification

2.9. Historical Data

This section describes the features that are used to archive historical data in ADMS.

2.9.1. Archiving Overview

Archiving ADMS historical data is a collaborative effort of multiple applications. The Historian Sampler application is responsible for archiving the tag and SCADA data, while the ADMS Archive application is responsible for capturing DMS, OMS, switching order, safety document, and reference data.

The Archive application has a separate thread for each data source. Of the five threads, one of the threads:

- archives DMS data from the DMS archive journal
- archives OMS and switching order data from the OMS archive journal
- archives safety documents
- archives reference data (static codes and descriptions)
- archives call and AMI data

The general structures of the four threads are similar but the implementation details differ. This is because the nature of the source data and the triggers for archiving data vary.

The below image shows the data flow of capturing historical data from different data sources.

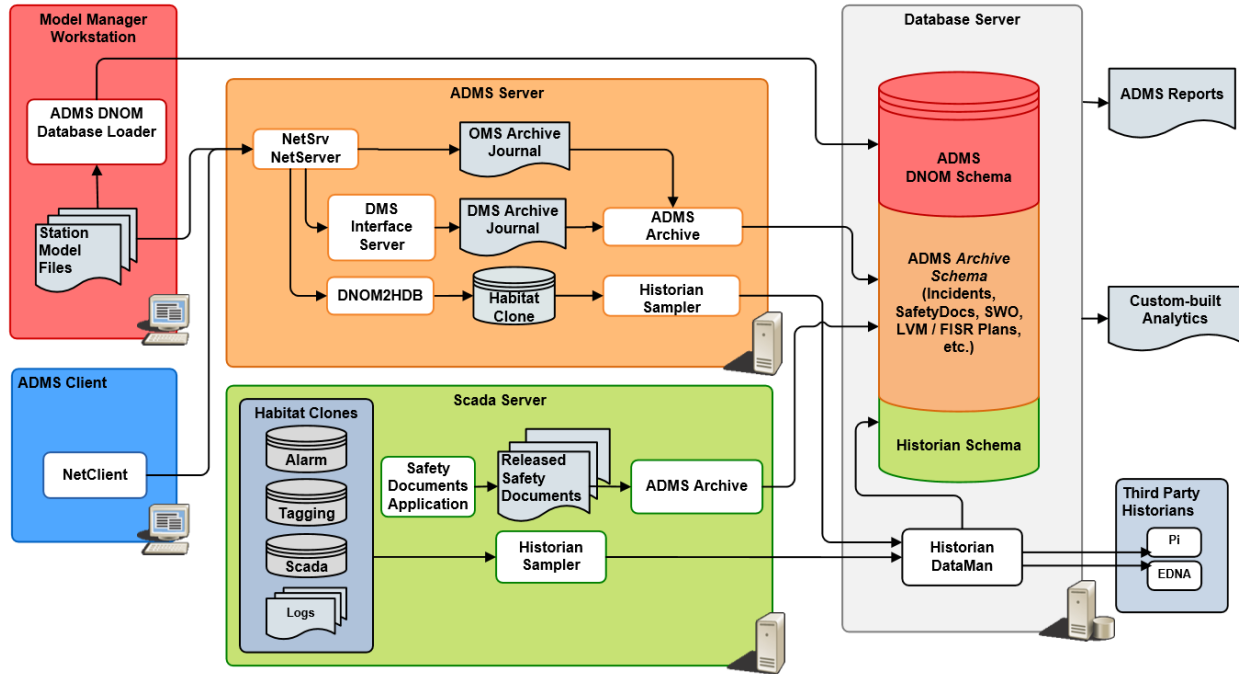


Figure 20. ADMS Archiving Solution

The table below summarizes the archiving components for various types of data in an ADMS installation.

Data Type	Application	Database
Events and alarms	Historian via Scada Alarm	Oracle or SQL Server
Switching operations	Historian via Scada Alarm	Oracle or SQL Server
Switching orders	ADMS Archive	Oracle or SQL Server
DPF solutions (summary only)	ADMS Archive	Oracle or SQL Server
FISR, LVM, AFR solutions	ADMS Archive	Oracle or SQL Server
Fault Location solutions	ADMS Archive	Oracle or SQL Server
Station/feeder exceptions	ADMS Archive	Oracle or SQL Server
Incidents	ADMS Archive	Oracle or SQL Server
Safety docs	ADMS Archive	Oracle or SQL Server
Tags	Historian	Oracle or SQL Server
SCADA measurements	Historian	Oracle or SQL Server
Network model file info	ADMS Archive	Oracle or SQL Server
DNOM nominal state	ADMS DNOM Database Loader	Oracle or SQL Server
DNOM dynamic state	N/A. Refer to the Snapshot Viewer	N/A
DNOM query data	DNOM2HDB	HABITAT

In addition to the components above, Habitat Sampler supports integration with third-party historian products to copy data flowing into Historian. For details, refer to the Scada

documentation.

2.9.2. ADMS Archive

ADMSArchive stores near real time and historical data in a relational database that provides a data source for distribution and outage management reports. Data from other ADMSapplications is automatically stored in the Archive database. Archived data includes:

- Distribution Power Flow (DPF) solution summaries
- Station and feeder exceptions
- Fault Location solutions and results
- Fault Isolation and Service Restoration (FISR) solutions, plans, and steps
- Load and Volt/VAR Management (LVM) solutions, plans, and steps
- Automatic Feeder Reconfiguration (AFR) solutions, plans, and steps
- Active, cleared, and cancelled incidents from the Outage Management System (OMS), including the incident chronology log and restoration steps
- Executed switching orders and switching steps
- Released safety documents
- Devices referenced by other archive records (for example, incidents, switching orders, or safety documents)
- Network model file information
- Reference data such as device types or operating centers
- Customer call, meter events, and statuses

2.10. Read-Only Data Access

In addition to the operation-oriented displays and tools, ADMS provides read-only access to current DMS and OMS data and recent history for users inside and outside of the control room environment, including read-only dashboard displays and reports for analysts and management.

The ADMS read-only system configuration allows replicating a copy of the production ADMS system data so that users who require read-only access to the data can view it using the read-only client without interfacing directly with the production system. Typically, the ADMS read-only system is placed in a "DMZ" located outside the production system's network firewall. For more information about the read-only system, refer to the *ADMS Software Installation and Maintenance Guide*.

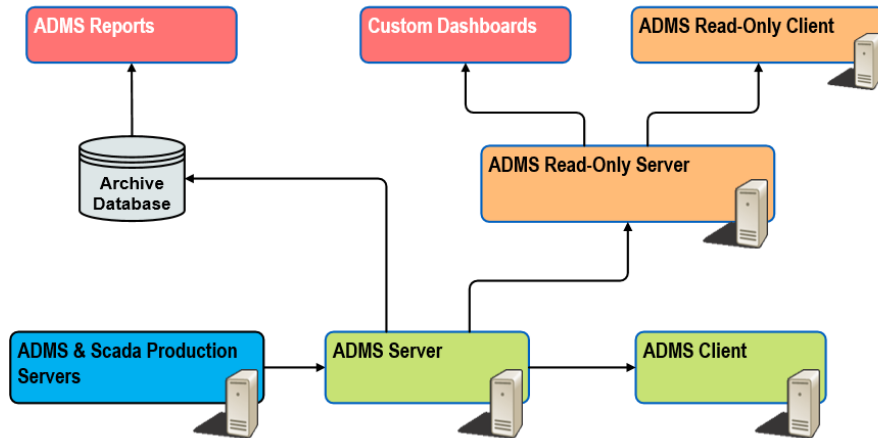


Figure 21. Read-Only Data Access

Custom dashboards can be developed based on the functionality of the DMS Interface server and NetDataServer web services (refer to [ADMS External Interfaces](#)).

The ADMS Reports application provides sample distribution and outage management reports, including reliability performance indices and outage statistics that power companies may use directly, or as templates for their own custom reports. The data source for these reports is the Archive database, which contains historical data such as incidents, switching orders, safety documents, and network analysis results.

2.11. System Architecture

As shown in [ADMS Architecture Overview](#), ADMS is built on a client-server architecture. The smart (thick) client combines information from the ADMS server and the Scada server.

The ADMS hardware configuration, as shown in the below image, is designed to prevent any single point of failure. This ensures the high availability needed for real-time management of an electrical distribution network.

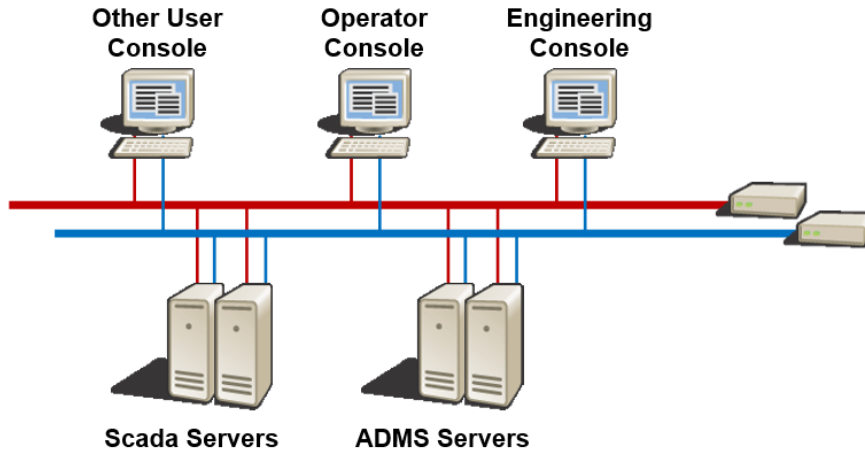


Figure 22. Typical ADMS Hardware Configuration

ADMS uses redundant ADMS servers, redundant network(s), and redundant loggers/printers. Applications run under the control of the Configuration Manager, which monitors the status of both hardware and software components and takes remedial actions when a failure occurs.

A typical dual-redundant distribution management system is made up of two DMS host nodes and several Browser-based operator consoles. Other dedicated computers can be added.

The total processing power of ADMS is distributed among various client consoles and servers that communicate with each other through a dedicated Local Area Network (LAN).

2.11.1. Main ADMS Servers

The main ADMS server applications include the DMS server (NetSrv), OMS server (NetServer), DNAF server (DnafSrv), etc. The main ADMS servers are the core of the GE Distribution Management System solution. The DMS server holds the entire Distribution Network Object Model, including the latest static, dynamic, and state estimation data.

This model represents the current state of the network based on the nominal data, events received from external SCADA subsystems, dynamic changes created by the ADMS clients, and calculated values from Distribution Network Advanced Functions (DNAF).

The ADMS server is a multi-threaded application, separated into several modules. Each module is responsible for providing end users with functions.

A separate, read-only implementation of the server can also be used to support large numbers of clients or other data services (such as executive dashboards) without impacting the performance or security of the main control environment.

For more detailed information about the implementation of the ADMS server, see [Main ADMS Server](#).

2.11.2. ADMS Client

The ADMS client is a smart client, providing operators with a detailed and comprehensive view of the network model and allowing operators to execute the necessary distribution management system functions.

The critical feature of an effective real-time network management system is that it must provide the operator with the most comprehensive and accurate view of the actual situation.

In this regard, the combination of the Distribution Network Object Model and the distribution user interface plays an important role by providing an integrated environment to collect dynamic data from a variety of sources. It also allows advanced applications to use this data in real time and to present consistent information to the operator.

For example, when dealing with unplanned outages, an operator needs immediate and easy access to the following information:

- Actual topology of the network, including both telemetered and manual switches
- Any activity taking place in the same area that might possibly be affecting availability of the equipment, such as maintenance
- Any other available information to help better understand and assess the situation

The ADMS client is fully integrated with GE's standard user interface operating environment, Browser, providing navigation to Supervisor Control and Data Acquisition (SCADA) displays, archive forms, and any Web-enabled applications.

The below image shows an example of how multiple views are presented in ADMS's integrated user interface.

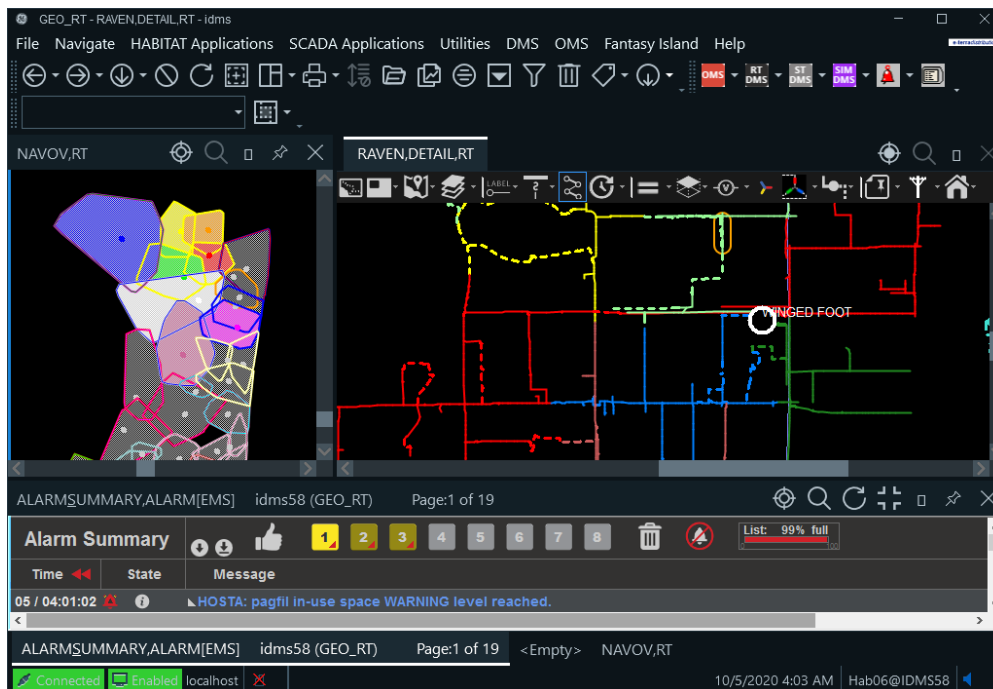


Figure 23. ADMS Integrated User Interface

The client maintains all of the nominal station files and the latest snapshot of the dynamic data available locally. Therefore, clients can run independently of the server itself and can perform offline study analysis.

The client application is highly configurable, and a number of configuration tools are provided for the support engineer to customize the look and operation of the client.

2.12. Scada Server

In a distribution management system, Scada provides Supervisory Control and Data Acquisition (SCADA) capabilities for real-time monitoring and control of power distribution systems.

The Scada subsystem gathers real-time data from remote terminal units and other communication sources in the field, and it enables the control of field devices from consoles. Scada is a distributed, scalable system implemented in a Habitat real-time control system environment. Communication between the Scada server and the ADMS servers is via an InterSite Data (ISD) link.

Scada enhances the functionality of ADMS by performing the following primary functions:

- Real-time status updates of DNOM
- Provides network measurements
- Supervisory control of network devices
- Supports tagging of devices

Scada also relies on Habitat subsystems: Configuration Manager (CFGMAN), Central Alarm

Management (ALARM), and Permission server (PERMIT).

For detailed information about the Scada subsystem, refer to the *Scada Software Maintenance Guide* and the *Scada Operator Guide*.

For detailed information about the Habitat Permission server, and geographical hierarchical permission management in particular, refer to the *PERMIT User Guide*.

3. ADMS Modeling Environment

The Distribution Network Object Model and the tools used to generate and maintain the model form the foundation of ADMS. This single model supports all the operational and analytical functions, and the user interface environment. This single-model concept avoids the problems of data consistency and maintenance that can otherwise arise.

3.1. Distribution Network Object Model

ADMS is built around the Distribution Network Object Model (DNOM). This model combines the static model data with dynamic data and model updates, plus all the other information to support a full suite of advanced network management functions.

DNOM is a single, seamless model purposely built to meet the performance and scalability needs of a real-time distribution management system. Its implementation is based on a modern object-oriented technology, which provides special data structures well-suited to an integrated, operational system.

DNOM supports the following types of data:

- **Static data:** Network equipment electrical parameters, UI display object data
- **Dynamic data:** Temporary modifications, manual entries, abnormal statuses, trouble orders, safety tags, and documents
- **Calculated states:** Power flow results and other output data from network analysis functions

The DNOM is structured on a station basis— i.e., the smallest amount of data that is used for a change or update is the portion of the network that is normally supplied from a single substation.

The DNOM includes a comprehensive list of electrical devices and network components, as well as non-electrical objects commonly used to support functions in the field of outage management, work management, and crew management.

In particular, the definition of the DNOM includes all necessary structures to support:

- Real-time visualization of network operation events on geographic, schematic, and/or tabular displays
- High-performance advanced computation performed by applications, such as network optimization
- Operation organization by area of responsibility

The list of objects supported by the DNOM is based upon the CIM wires and topology packages. These include:

- Substations, which can be either primary stations (HV/MV) or switching stations
- Feeders, which group distribution circuits and components fed from a substation
- Generators, including small customer co-generator plants, such as those used as Distributed

Energy Resources (DER)

- Line segments, which can be single-phase, two-phase, or three-phase and can make up a distribution circuit
- Conductors
- Transformers/regulators, including auto-transformers, and two and three winding transformers with constructions (including star-star, delta-delta, star-delta, and several others)

The tap changing model includes fixed tap, tap changer, and Load Tap Changer (LTC). The model provides the capability to model tap changers on either the HV or LV side of the transformer with the maximum and minimum ratio at either end of the tapping range.

- Capacitors and reactors with various control modes
- Switching devices, including circuit breakers, reclosers, disconnects, sectionalizers, and fuses with a variety of switching schemes depending on the type of the switch and whether it is remotely controllable
- Grounding devices
- Relays, fault detectors
- Temporary connections to represent jumpers and temporary switches made in the field
- Loads and load categories, which refer to different load shapes and characteristics, including load-to-voltage dependency, power quality requirements, etc.
- Operational limits for components such as lines, transformers, and switching devices

DNOM also supports the concept of commissioning and decommissioning equipment, so that new additions and removals to the distribution network can be displayed on a separate layer of Network View until the equipment is placed in service or removed.

3.2. ADMS Modeling Tools

ADMS provides system administrators with a full range of modeling tools, including:

- The ADMS Converter validates and creates the Distribution Network Object Model based on GIS extracted data and/or Substation Editor output files.
- The ADMS Substation Editor is a stand-alone application module. It is a display and network modeling tool that performs the following functions:
 - Creates network models for the internals and externals of substations
 - Creates schematic displays for the internals of substations
 - Creates schematic or geo-schematic views of feeders
 - Creates Scada database import files

The Substation Editor is used to build the network topology for the equipment within a substation. This topology is merged with the model of the equipment outside the substation,

which has been exported from the asset management system (usually a GIS). In addition to the electrical connectivity and properties of the equipment, the Substation Editor generates the one-line display that is used to view the substation internals and/or externals.

- Source is the system modeling environment for Platform. The purpose of Source is to populate and maintain the databases used by Platform applications running in real-time or near-real-time environments and, in particular, the Scada data.

The ADMS Converter creates the DNOM from input files created by the Substation Editor and/or by the customer's asset database. The converter provides an XML file format interface, allowing the customer to extract the necessary data from the utility's asset database (usually a GIS) to create input files that respect the supported converter XML file format interface.

The data extracted from the GIS includes network device information, connectivity, topology, nominal status, and non-electrical data such as roads and buildings.

The extraction process is kept independent of the real-time network management system. Any GIS model can be extracted to build the DNOM regardless of the supplier or internal schema. The only requirement is to extract the data respecting the supported XML file format interface; this is described in the *ADMS Modeling Reference Guide*.

Because changes in the distribution network happen continually, the transfer process has been designed to allow incremental updates with no need to bring the system down or even fail over. The model is built on a per-station basis and is fully scalable from a single-zone substation to the largest distribution network.

With this approach, all of the data maintenance effort can be retained within the GIS or within the Substation Editor, and all parts of the organization are working from the same version of the network information.

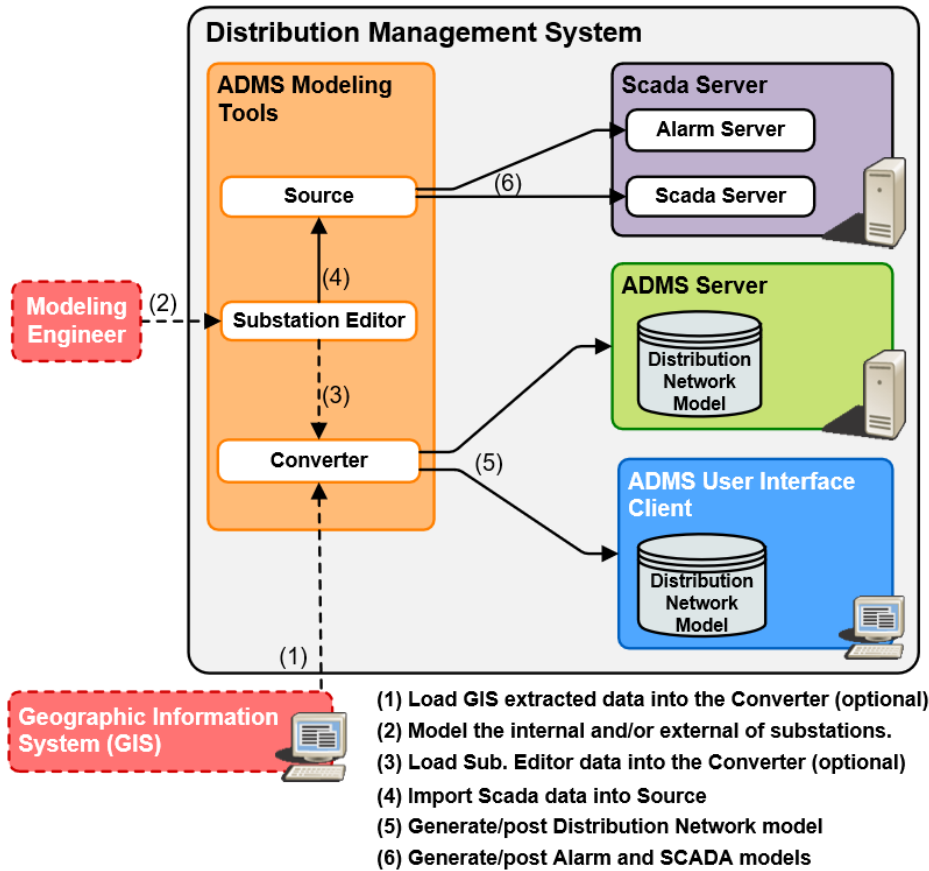


Figure 24. ADMS Modeling Environment

The ADMS client reads this model directly to display geographic and/or schematic views of the network. Special display definitions are not required to show network views. The model supports multiple geographic coordinate sets for each device so that, if available, the network can be displayed in custom geo-schematic views. The network views can also include various levels of detail depending on the zoom level. The geo-spatial information, including roads, property boundaries, building footprints, and associated labels (provided as .dxf or ESRI shape files) can also be displayed as required.

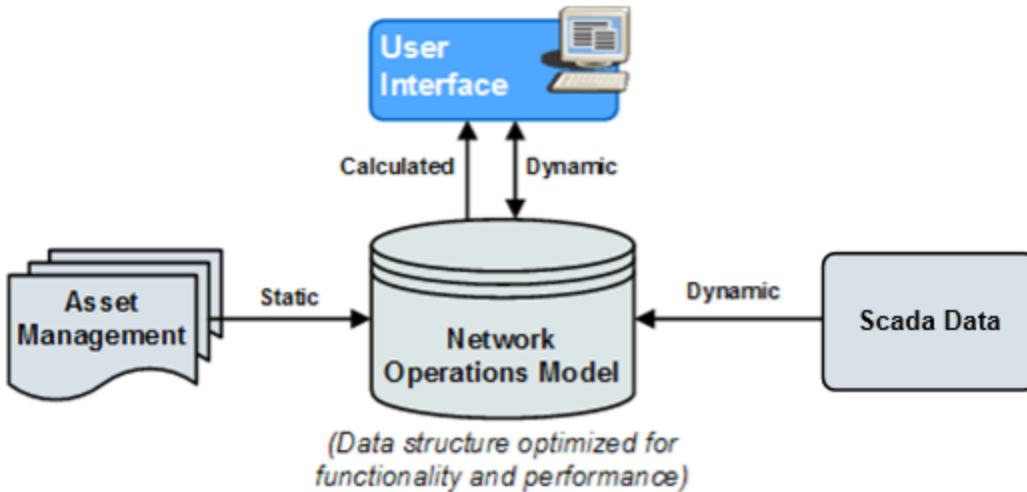


Figure 25. Network Operations Model Data Types

The typical approach to creating and maintaining the databases for a power system and data acquisition system includes four main activities: initially populating the database, maintaining the databases, building one-line displays, and installing new databases and displays.

3.3. Substation Editor

The Substation Editor is a stand-alone application module that is supplied as part of the standard ADMS product suite. It is a display and distribution network modeling tool that performs the following functions:

- Creates distribution network models for the internals of substations
- Creates distribution network models for the externals of substations (if no data extracted from a GIS is available)
- Creates schematic displays for the internals of substations
- Creates schematic or geo-schematic views of feeders
- Creates Scada database import files
- Creates Control database import files

The Substation Editor is used to build the network topology for the equipment within a substation. This topology is merged with the model of the equipment outside the substation, which has been exported from the asset management system (usually a GIS) or created in the Substation Editor if the customer does not leverage a GIS.

In addition to the electrical connectivity and properties of the equipment, the Substation Editor generates the one-line displays that are used to view the substation internals.

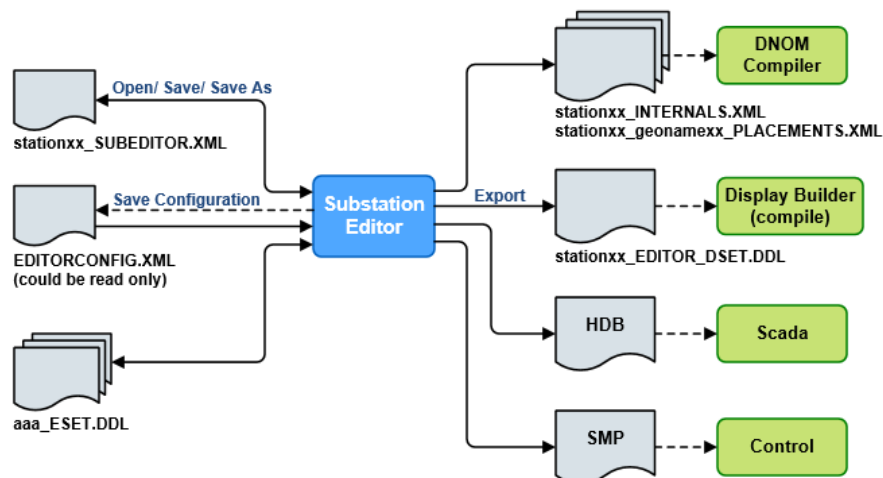


Figure 26. Substation Editor - Conceptual Overview

As shown in the above image, the Substation Editor reads and writes a stationxx_SUBEDITOR.XML file (for example, if the name of the substation is DOUGLAS, then the file name is DOUGLAS_SUBEDITOR.XML). This file contains the information required to export to the DNOM compiler (the converter) and to the Browser display set.

The substation name must exactly match the substation name specified in the GIS. A separate mapping is provided by the converter to map this name to the Habitat substation ID.

The following types of files are exported by the Substation Editor:

- A display set (stationxx_DSET.DDL). Usually, this is the station internals SCADA one-line display.
- The XML files used by the DNOM converter. The converter supports three types of files:
 - Stationxx_INTERNALS.XML: This file contains the station internal equipment and display, and is referred to as the "station internals" file. There can only be one (or none) of these files for each substation.
 - Stationxx.XML: This file contains the equipment and display outside of the station fence. This is referred to as the "station file". Normally, this file is created by exporting the data from the GIS, but it is possible to create it using the Substation Editor. There must be only one (or none) of these files for each substation.
 - Stationxx_PLACEMENTS_geonamexx.XML: This file contains placements for station equipment to be shown in a specific geometry (geonamexx). This is referred to as the "placement file". It is possible that there can be multiple placement files for a single station, and there can be files with the same geometry name for multiple stations. The placement file contains only display information, and it links to equipment identifiers in the station file.
- SCADA database files:
 - Stationxxx_SMP.xml: Contains SCADA record information related to modeled measurements. This file can be imported into the Control database.
 - Stationxxx_HDB.txt: Contains SCADA record information related to modeled measurements. This file can be imported into the Scada database using HDB import.

- Stationxxx.issg: Describes the newly exported station internals and station placements files, used by the Model Manager.
- global_internals.xml: Contains the Associated Models with default properties of network elements.

The Substation Editor uses a configuration file to specify attributes and parameters for the editing session. The attributes and parameters can be modified and saved in the Substation Editor. For users who need to build substation topologies with consistent layout, the configuration file is read-only.

For more information, refer to the *Substation Editor User Guide*.

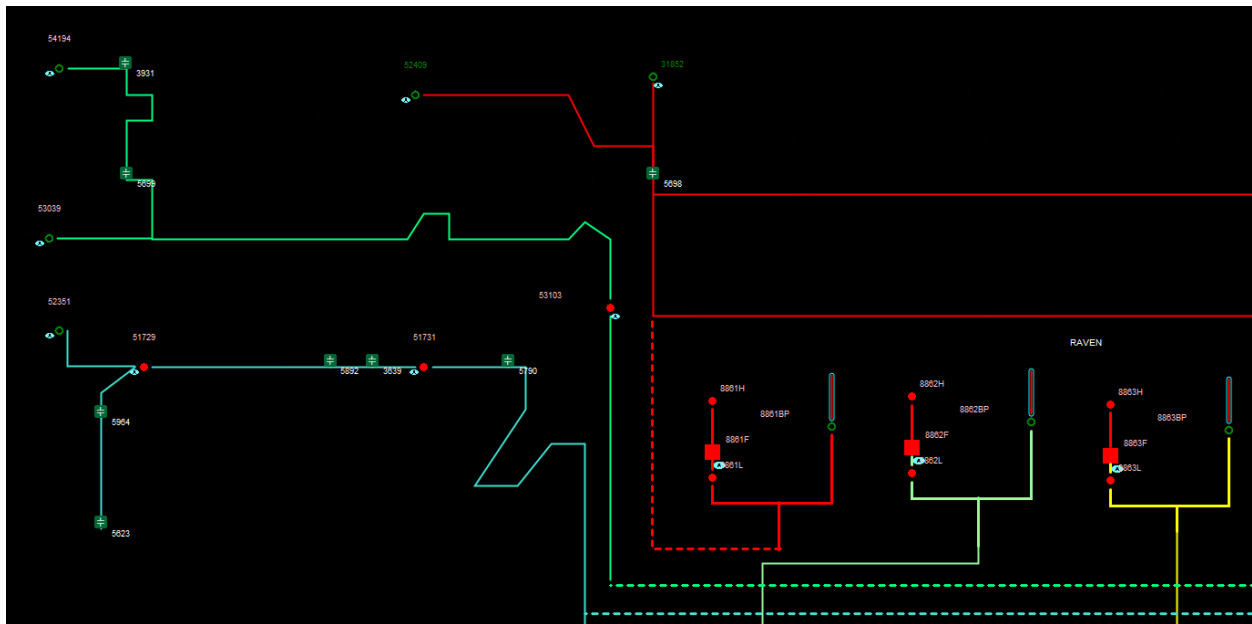


Figure 27. Display Combining Substation and Feeder Elements

3.4. Substation Builder

The Substation Builder tool is a modeling tool that produces <Station>_subeditor.xml files that are used as an input to the Substation Editor tool, which produces Scada displays and portions of the Scada model and the ADMS network topology model (DNOM model).

For more information, refer to the *Substation Builder User Guide*.

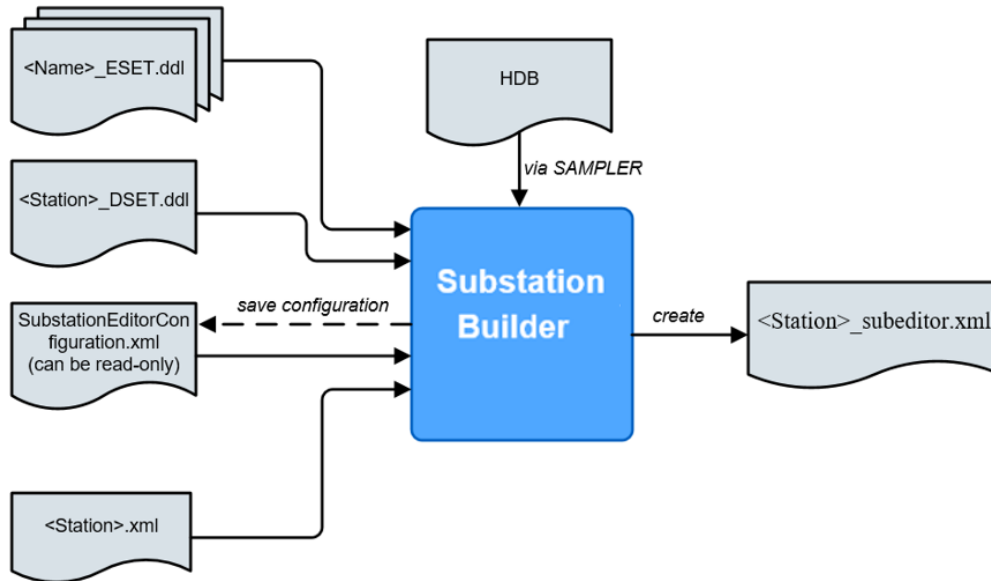


Figure 28. Substation Builder

3.5. Substation File Builder Lite

The Substation File Builder Lite (SFBLite) tool is used to update station externals files <Station>.xml files with the measurement and value placement information from the Scada database. SFBLite produces updated <Station>.xml files that could be used for building station models using the Converter or Model Manager.

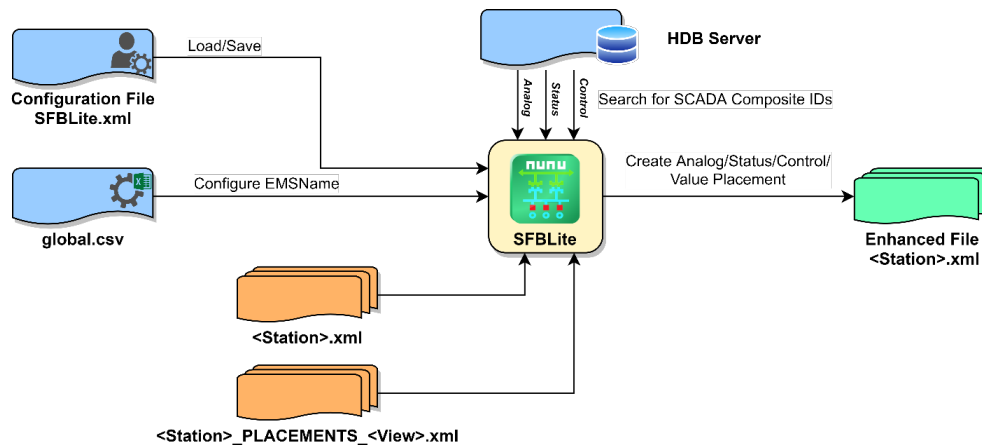


Figure 29. Substation File Builder Lite

3.6. Converter

The network operations model is the core of the DMS. It maintains both the static and the dynamic data describing the distribution network. The data is structured to support all the required functionality of the DMS: real-time operations, network analysis functions, and user interface

displays for the distribution network. This functionality, combined with the performance requirements for the efficient operation of very large distribution networks, means that a highly specialized and optimized data structure is required.

The Common Information Model (CIM) is an abstract model that represents all the major objects in an electric utility enterprise typically involved in utility operations. The CIM provides a standard way of representing power system resources as object classes and attributes along with their relationships. This is accomplished by defining a common language (i.e., semantics and syntax). It is a schema, not a database or a specific data file format.

For GE's DMS solution, the information for the distribution facilities (lines, switching devices, transformers, energy consumers, and others) is stored in a CIM-based extensible markup language (XML) file for each distribution substation. The data in this station model file typically represents utility assets outside the substation fence, downstream of the station getaway point or feeder head. The source of the facility data can be a distribution utility asset management system (for example, a GIS) or an output file created through the Substation Editor application.

In addition, a global resources file is used to efficiently store modeling data that is common to many station file components. There is also a default data file, which contains default data for selected fields that have not been provided by the asset management system.

The data in the model files is combined to create a station binary file that is the single data source for both the network analysis applications and the network view. This binary file is created using the DMS converter application.

The converter validates and merges the data from the station and EMS model files, and outputs the binary operations model files. The validation process consists of data integrity checks that verify that the data can be processed by the DMS. If data errors exist, the system administrator is notified with messages indicating the extent of the problem.

When the modeling errors are corrected and new station model files have been created by the converter, the model files can be utilized in the DMS by placing them into the appropriate directories.

For more information about bringing new model files into the production system, refer to the *ADMS Software Installation and Maintenance Guide*.

The process for populating the operations model is shown in the below image.

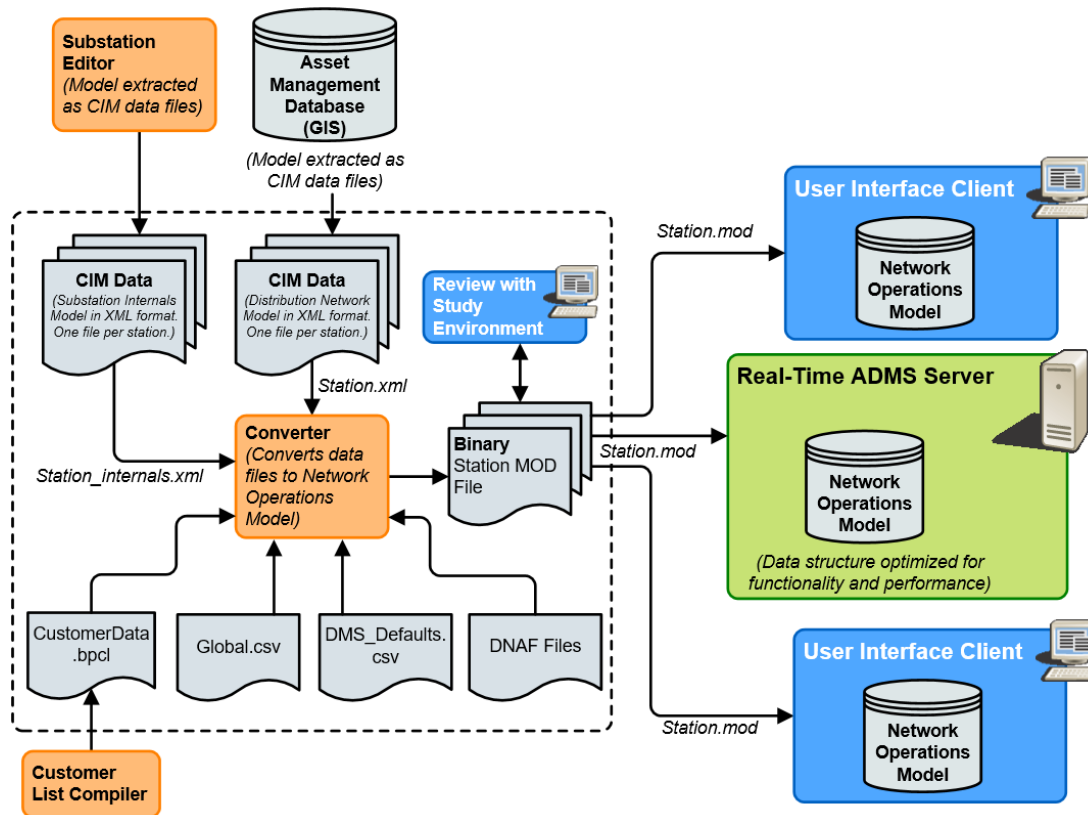


Figure 30. Process Flow for Populating the Operations Model

The DMS converter is a Microsoft Windows-only application that reads in the model files, validates the data, and writes out the data into a binary file. This resulting binary file is called a "station model file". Station model files have the extension ".mod". From the station model file, the topology processing application can be executed and, if equipment placement information is modeled, UI displays can be automatically generated.

In conjunction with the station CIM data files, two additional files are used to provide global network data and default data values.

For more information, refer to the *ADMS Modeling Reference Guide*.

3.7. Model Manager

The ADMS Model Manager tool is responsible for converting station data files. The resulting station model files are then stored in the appropriate folder for further review or transferred into the online ADMS servers.

The Model Manager directly calls the Converter and the Navigation Overview Builder through the command-line interface. Although the Model Manager wraps the Converter and the Navigation Overview Builder, they can still function as standalone applications.

The Model Manager tool can run in Automatic mode, where it constantly scans for new files, or in

Manual mode.

The Model Manager tools consist of the following:

- **Model Manager Service:** A Windows service that runs the business logic of the Model Manager application separately from the client and connects to the Model Manager client.
- **Model Manager Service Client:** Connects to the model manager service and provides the user interface to process and review files manually.
- **Model Manager Configuration Editor:** Provides options to configure the Model Manager client.
- **Model Manager Application:** Provides a user interface similar to the Model Manager Service client user interface and allows the user to manage all of the processes manually. Unlike the service client, the stand-alone client does not run automatically or automate any processes. The Configuration Editor is available to the stand-alone client via the corresponding menu option.

The Model Manager tool performs the following functions:

- Reads the substation group (SSG) or internal substation group (ISSG) file and determines the stations to be converted.
- Passes the necessary attributes to the Converter in a command-line call, including station name, input file paths, and output file paths.
- Receives a completion and validation status from the Converter through the Converter's logs. If the Converter fails to complete or crashes, the Model Manager application detects it and displays this information and the validation status in the Model Manager's display tabs, including the Converter's log tab, and also logs it in the audit log.
- Copies the station model files and resulting Converter log/analyst files to the appropriate folders based on the individual station validation status, the Auto-Process flag (from the SSG file), and the list of stations that must be processed together.
- If a new station file has been detected (based on a master station list stored in a file), converts the new station. After processing a pre-configured number of stations, calls the Navigation Overview Builder application. Copies the resulting navoverview.nvw file to the File Updater folder for deployment to the ADMS client consoles.
- Displays the station files that are currently being processed and their conversion status. Displays the change summary provided in the SSG file.

The Model Manager processes data from various sources, as shown in the below image.

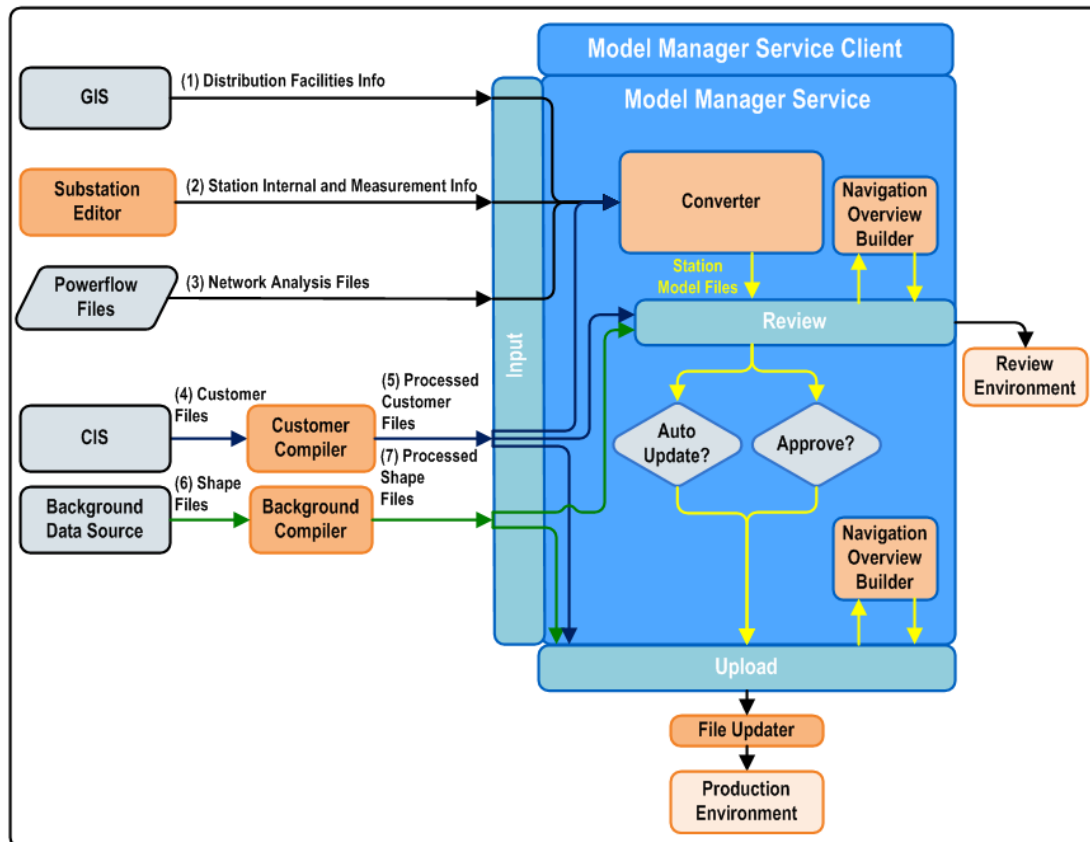


Figure 31. Model Management Process Overview

4. Main ADMS Server

The main ADMS server applications are the core of the GE Distribution Management System solution. The main servers include the following:

- **DMS Main Server (NetSrv):** Provides core system functionality and maintains interaction among ADMS system components.
- **OMS Server (NetServer):** Provides functionality for the Outage Management System.
- **DNAF Server (DnafSrv):** Provides functionality for the Distribution Network Analysis Functions (DNAF).
- **Simulator Server (SimSrv and NetServerSIM):** Provides functionality for the Distribution Operator Training Simulator environment.
- **Read-Only Server (NetServerRO):** Provides functionality for the Read-Only mode.
- **Customer-Only Mode Server (NetServerCM):** Provides functionality for the Customer-Only mode, which operates with the Advanced Metering Infrastructure (AMI) system.
- **DMS Server in Follower Mode:** A DMS server can run in follower mode to the enabled DMS server. In Independent Follower mode, follower servers receive changes from the enabled server and independently implement DMS commands from the follower server clients. In Dependent Follower mode, follower servers receive changes from the enabled server and send requests to the enabled server to implement DMS commands from the follower server clients. An OMS server can receive permissions from a DMS server in follower mode.
- **OMS Server in Follower Mode:** An OMS server can run in follower mode to the enabled OMS server. In Independent Follower mode, follower servers receive changes from the enabled server and independently implement OMS commands from the follower server clients. In Dependent Follower mode, follower servers receive changes from the enabled server and send requests to the enabled server to implement OMS commands from the follower server clients.

The DMS server application (NetSrv) is designed to run on Microsoft Windows Server operating systems.

The DMS server holds the entire Distribution Network Object Model (DNOM) in memory. This includes the latest static, dynamic, and state estimation data. This model represents the current state of the distribution network based on nominal data, events received from the integrated Scada system, dynamic changes created by the ADMS clients, and calculated values from Distribution Network Analysis Functions (DNAF) execution.

The ADMS servers respond to requests sent by the ADMS clients, updating the dynamic model or executing analysis functions as appropriate.

The communication between server and clients leverages a proprietary protocol developed by GE for the Distribution Management System solution. Specifically, ADMS defines a set of message types that are used for communication. Messages are constructed and serialized into binary format and transferred over the Internet protocol suite (commonly known as "TCP/IP").

The ADMS servers can be integrated with the Habitat Permission Management server (PERMIT), the

Habitat Configuration Manager (CFGMAN), and Scada. PERMIT is used to authenticate and authorize user access to the system, while Scada provides data regarding the current state of the network. CFGMAN monitors the integrated applications, assigning them a role (ENABLED, DISABLED, or STANDBY).

The ADMS servers are composed of a number of modules, each of which is a multi-threaded component, as highlighted in the below image.

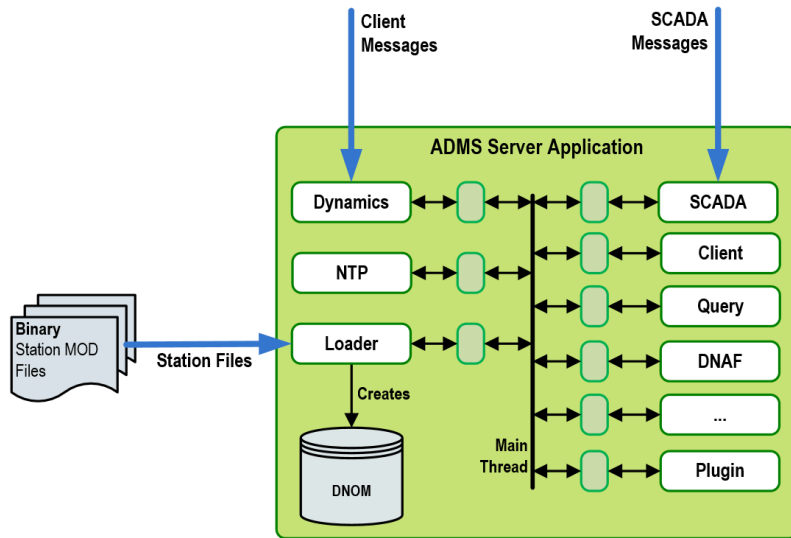


Figure 32. ADMS Server Overview

Message objects are passed between modules by the application's main thread. The main thread is responsible for delivering intra-module messages, as illustrated below.

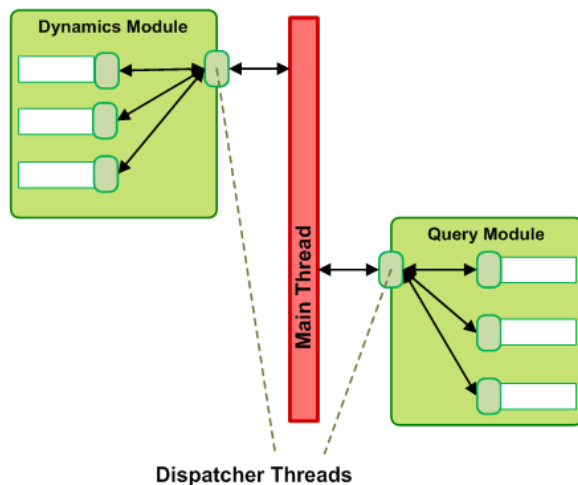


Figure 33. ADMS Server Messages Concepts

This architectural pattern is repeated inside each module. Every module has a single "dispatch thread", which schedules worker threads and exchanges intra-module messages, as well as sending and receiving messages from the main thread.

4.1. Loader Module

The loader module monitors changes to station files and DNAF parameter files placed on the server. The location for these files is set in the ADMS server applications' configuration files.

When updating a station model:

1. The existing station file is replaced with a new file.
2. The loader thread polls the directory and identifies the new file.
3. The loader deserializes the binary station data from the file into memory.
4. The loader unloads the station whose model file is removed.

4.2. Dynamics Module

The dynamics module handles dynamic or topology changes from the nominal position submitted by ADMS clients or by SCADA. A special thread maintains the TCP/IP connection with the clients, sending and receiving the DMS binary messages.

There is a worker thread pool that is used for dynamic processing.

The dynamics module also maintains a special thread that manages a connection with PERMIT in EMS. The permission structure from PERMIT is loaded and stored on the server. This structure defines permissions against a GUID that is submitted by the DMS client with each request. Communication uses PERMIT's proprietary protocol.

4.3. SCADA Module

The SCADA module maintains a TCP/IP connection with the remote Scada; therefore, it uses GE's proprietary ISD protocol for communication. The SCADA module processes the SCADA messages and creates a DMS message to notify the dynamics module. The dynamics module applies the change to the dynamic data.

The SCADA module also can communicate with Control for SCADA data. It uses the same internal DMS message, so the rest of DMS is insulated from the SCADA communication protocol.

4.4. Network Topology Processor Module

The Network Topology Processor (NTP) module is notified of changes to the topology of the network, and it makes a network selection based on the changes. This includes the station that contained the changes and all of its connected stations. Each station is processed one thread at a time to prevent conflicts.

The NTP module implements its own scheduling. If it is receiving a lot of changes, it can schedule

the processing to run once following the receipt of the entire group of changes.

The processor determines the energization/de-energization of the network, including identifying any loop states (where a section is receiving power from more than one source) within the network. The results of the processing are written to the ADMS server's DNOM. These results are not pushed to the clients; they execute their own topology processing on the local model.

This client processor is built on the same code that the server uses; however, the processing is simpler. Topology processing occurs quite often, so executing it on the client greatly reduces the amount of information that must be sent from the server and makes the system more efficient.

Results of this processing are used by the DNAFs to perform network analysis.

4.5. Query Module

The query module receives requests for network analysis information from the ADMS clients. These requests access data within the DNOM created by a particular analysis function. The client's request is received by the dynamics module, which recognizes it as a query, so it passes the message through to the query module.

Queries are submitted using a simplified SQL-like syntax. Every class within the DNOM has a serializable name, and each attribute has a name. Every object has an associated ID and name. These are used within the SQL syntax to identify the data to retrieve. Dynamic data also has record and field names, so these can also be included in a query.

For example, the following statement returns the stated data from all switches with station DOUGLAS:

```
SELECT ID, Name, State, NormalState
FROM rt.sub.swt
WHERE Name = 'DOUGLAS'
```

Results from the query can be provided in XML, or as tab-separated or comma-separated formats.

This module also provides the ability to write helper functions, which can be included in queries. These helper functions are created within code; therefore, they can provide a more sophisticated query. The query engine holds a pointer to this function, allowing it to execute it on request.

4.6. Client Module

The client module provides communications among the Habitat Configuration Manager (CFGMAN), the active server, and the standby server (or servers). The standby machine appears as a client to the active server; therefore, it receives the updated station files and dynamic data as any normal client does.

When standby starts up, it loads all station files into memory so that they are immediately available, if required. The dynamics that it receives are not applied to the model but are written to its

dynamics file. If the standby machine becomes active, it stops listening for dynamics and loads the current snapshot of the dynamics.

4.7. Distribution Network Analysis Functions Module

The Distribution Network Analysis Functions (DNAF) module manages the execution of the various analysis functions written for distribution management systems. These analysis functions execute against the current state of the network, providing information to the operator as required. Different events can trigger an analysis function, including a user request, a network change, or a periodic execution.

These analysis functions are built to work with a three-phase unbalanced network. However, these analysis functions work with either.

The Distribution Power Flow (DPF) provides the base analysis for the network. It uses information provided by the integrated SCADA and the distribution network model to determine power details across the different elements within the network. This information is used by the other analysis functions during execution.

In addition to the basic distribution power flow, optimization functions can be added to provide a more comprehensive suite of network applications. It is also possible to turn functions on and off in the DNAF server configuration file or using the client DNAF System Configuration display.

4.8. OMS Modules

This section provides a brief description of the modules that are related to the Outage Management System.

- OMS Message Processor Module

The dynamics module (unmanaged code) exchanges messages with the ADMS clients. For OMS-related messages, it needs to pass to the OMS modules (managed code). The OMS message processor module is responsible for the transformation to handle exchanges of intra-module messages among the managed and unmanaged code modules.

- OMS Loader Module

The OMS loader module monitors and loads data files related to the OMS functions. The static file of interest is the compiled customer Crew file. Upon server application startups or model updates, the OMS loader module is also responsible for reading and applying dynamics from the snapshot and journal files to restore the application to the current system state.

- OMS Query Module

The OMS query module receives requests for OMS information from the ADMS clients. These requests access data within the in-memory OMSDB or information stored in files. The client's request is received by the dynamics module, which recognizes it as an OMS query. It passes the

message to the OMS message processor module, which performs the transformation and then passes to the OMS query module.

- OMS Engine Module

The OMS engine module is the core analytical module of the OMS. It performs the business logic of incident management, outage extent analysis, outage life-cycle management, calculation of outage indices, monitoring and assigning crews, etc. This module is also responsible for writing the OMS journal files and call journal files. It also creates and maintains snapshot files. In the read-only data server, this module is also responsible for maintaining the archive journal files.

- Call Interface Module

The call interface module manages communication among the NetServer application and the call server application. It maintains the TCP/IP connection with the call server application, sending and receiving call messages. The call messages are then queued or relayed to the OMS engine module.

- Crew Interface Module

The crew interface module manages communication among the NetServer application and the crew server application. It maintains the TCP/IP connection with the crew server application, sending and receiving crew messages. The crew messages are then queued or relayed to the OMS engine module.

- OMS Customer Engine Module

The OMS customer engine module is the core module for the DMS running in customer-only mode.

4.9. Plugin Module

The plugin module provides a mechanism for quickly extending the functionality of the main server by loading pre-registered Dynamically Linked application Libraries (DLLs). These libraries can contain additional functionality specific to a customer or entirely new applications that can be optionally used. By using separate libraries, updates to the system functionality can be made without having to update the main server. Functionality can be adjusted for specific clients by using different libraries or library combinations.

The plugin module is event driven. When system changes occur, such as NTP processing completion, DNAF processing completion, and configuration file changes, the main server notifies the plugin module. The plugin module forwards these notifications to all registered application libraries so that they may do specific processing.

The application libraries have access to the DNOM object model and can use any data that the main server applies to perform its calculations.

The plugin module includes worker threads, configuration parameters, and triggering messages.

4.10. Plan Manager Module

The plugin module provides a mechanism for managing reconfiguration and optimization plans generated by real-time advanced DNAF applications (AFR, FISR, and LVM). The module supports the Advanced Application Plan Status System View display. The main functionality of the module is to:

- Implement, remove, cancel, and fail plans.
- Implement plans automatically or step-by-step.
- Check SCADA status measurements after implementing a step.

5. ADMS Client

The GE Distribution Management System (DMS) uses an application client for its user interface, called the ADMS client.

The ADMS client is designed and implemented in a combination of C/C++ and .NET (C#), thus leveraging all the advantages of the latest object-oriented technologies.

The ADMS client runs within the Browser environment as a plug-in component. As such, the ADMS client can leverage the existing authentication capability and viewport structure provided by Browser.

The ADMS client provides the operators with a view of the distribution network model, and it allows the execution and control of the DMS functions. The ADMS client has all of the nominal station files and the latest snapshot of the dynamic data available locally; therefore, clients can run independently of the server itself, providing topology processing, tracing functionality, etc.

For instance, in the worst-case scenario, if the ADMS client loses its connection with the real-time server, the operator cannot submit dynamic changes to the network— but the operator can view the network as of the last time the client was connected to the server.

This "smart client" approach also makes use of the processing power on the operator's workstation, thereby optimizing the load on the server and on the ITnetwork.

To start the ADMS client, the operator uses the appropriate startup shortcut, which launches Browser. Browser reads the network dynamics configuration XML file (netconf.xml) and loads the ADMS plug-in components, as shown in the below image.

The network display (NetDisplay) provides the dynamics information. A separate instance is created for each server connection. For example, if running both study mode and real-time mode, Browser creates two separate instances for each server. The server connections are defined in the network dynamics configuration file with the dynamics.

From a software standpoint, when the operator creates a new Browser viewer port and calls up an ADMS display, Browser creates a NetDisplay instance, which makes use of a particular dynamics instance and other components from the NetComponents library.

The ADMS user interface displayed to the operator is rendered based on the DNOM stored in memory. The DNOM is populated from the static station files delivered by the File Updater client, and the dynamic data is delivered by the ADMS server.

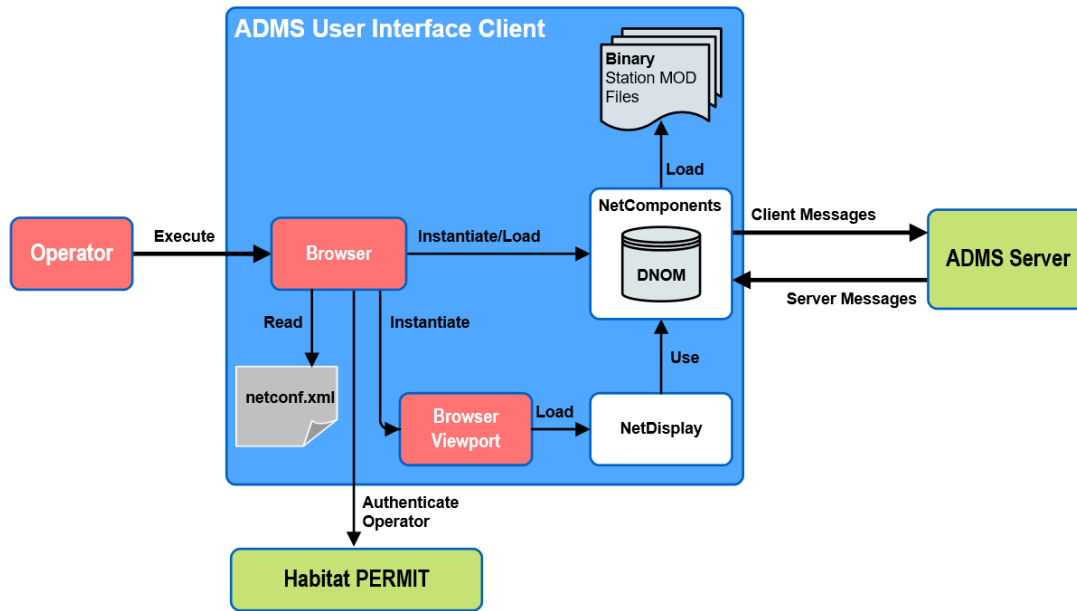


Figure 34. ADMS Client Overview

Due to the physical constraints of the client environment, it usually loads a partial (configurable) model view. This means that the client holds only the current station model and the immediately surrounding stations in memory. This creates some latency when moving around stations; however, it allows the ADMS client to run within the constraints of a typical client machine.

The concept of the partial model view means that, when the user clicks on the station overview (or specifies to center the main geographic view), the proper subset of stations is loaded into memory to represent a partial view of the system. The "proper" stations mean all of the stations whose geographical boundaries intersect with the selected location, plus all of the stations that directly tie electrically with the station(s) from this initial selection. To "pan" outside of this partial view, the user can re-position on the navigational overview or use the Reload View command on the geographic view. When a different subset or neighborhood of stations is selected for viewing, the stations that are no longer in the neighborhood are unloaded from memory and the new stations are brought in. In addition to the reload mechanism, other navigational means such as the station overview can be used to load different stations into the partial model view.

5.1. Network Topology Processor Module

The Network Topology Processor (NTP) module is included in both the ADMS client and server. In this way, topology changes are processed locally on each client workstation, greatly reducing the amount of information that must be sent from the server.

When the NTP module is notified of a change in the network state, the processor determines the energization/de-energization of the network, including identifying any loop states (where a section is receiving power from more than one source) within the network. This includes the station that contained the changes and all of its connected stations. Each station is processed one thread at a

time to prevent conflicts. The results of the processing are written to the ADMS local DNOM.

The NTP module implements its own scheduling. If it is receiving a lot of changes, it can schedule the processing to run once following the receipt of the entire group of changes.

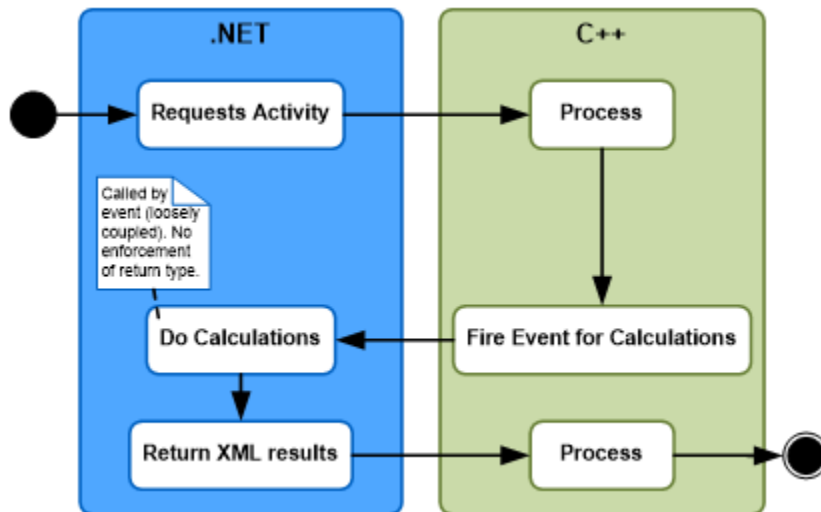


Figure 35. ADMS Client Activity

5.2. Navigation

Navigation between views is implemented as a URL-style protocol, and it is encapsulated in the NetDisplay class, which uses the UrlParser class to break down the navigation URLs. This approach can be used consistently for navigating and/or triggering functionality (such as server requests); therefore, it is extensible by design.

5.3. Configuration

The configuration mechanism is based on XML and is designed to be extensible. There are already quite a few settings to accommodate different installations, and a GUI tool is provided to manage those settings - for example, the Viewer Configuration Editor. For detailed information, refer to the *Viewer Configuration Editor User Guide*.

The settings are read and parsed at startup, and no performance degradation is expected with the current design. Parsing is done by the means of a generic XML serialization class (included in the .NET framework), which can parse any valid XML into an object model. This makes it quite easy to extend the configuration files in the future without breaking existing functionality; however, code making use of any additional configuration settings needs to be added.

5.4. Browser

Browser is a graphical user interface product developed by GE for its suite of real-time products. As discussed in [ADMS Client](#), the ADMS client runs within the Browser environment, leveraging its existing authentication and viewport structure.

Browser is an interface that provides state-of-the-art display and navigation, such as windows, menus, mouse clicks, as well as specific functions such as "room management", which makes it much easier for the operator to deal with multiple displays on multiple screens.

Browser supports multiple types of displays, including schematic and tabular displays. It features context-sensitive online help as well as dynamic toolbars and menus.

Menu items are driven by the user's permissions. When the user is working in a particular subsystem or application, special drop-down menus of related displays become available.

Browser is designed with the latest technology, and it provides the best integration with the Microsoft Windows environment. Browser provides all the features that control room operators expect, such as multiple viewports, pop-ups, pull-down menus, multiple fonts, efficient panning and zooming, support for ActiveX controls, scripting, etc.

Browser provides operators with complete and easy access to the SCADA/EMS system's operation. Workstations can be dedicated to specific tasks, or they can be dynamically and transparently reconfigured as users log in. In any case, multiple displays can be viewed in windows, and a wide variety of techniques can be used to view and control the operation of the network and computer system as well.

Browser's design provides for:

- A consistent "look and feel" across related GE products
- Flexible console operations and permissions
- Rapid notification of state and device changes
- The ability to easily customize and extend the user interface
- Isolating operators and application programmers from low-level details, so they can concentrate on the application-specific factors of the user interface

6. ADMS Configuration Tools

ADMS is supplied with a full range of configuration and support tools to allow system administrators to perform both the initial set-up and the ongoing maintenance of the system.

It is an important feature of the ADMS product that emphasis has been placed on broad-ranging configurability to reduce the need for customization and to allow utilities to more easily maintain and modify the operation of the system without the need for vendor support.

The relationship between the configuration tools and the ADMS functional modules is shown in the below image.

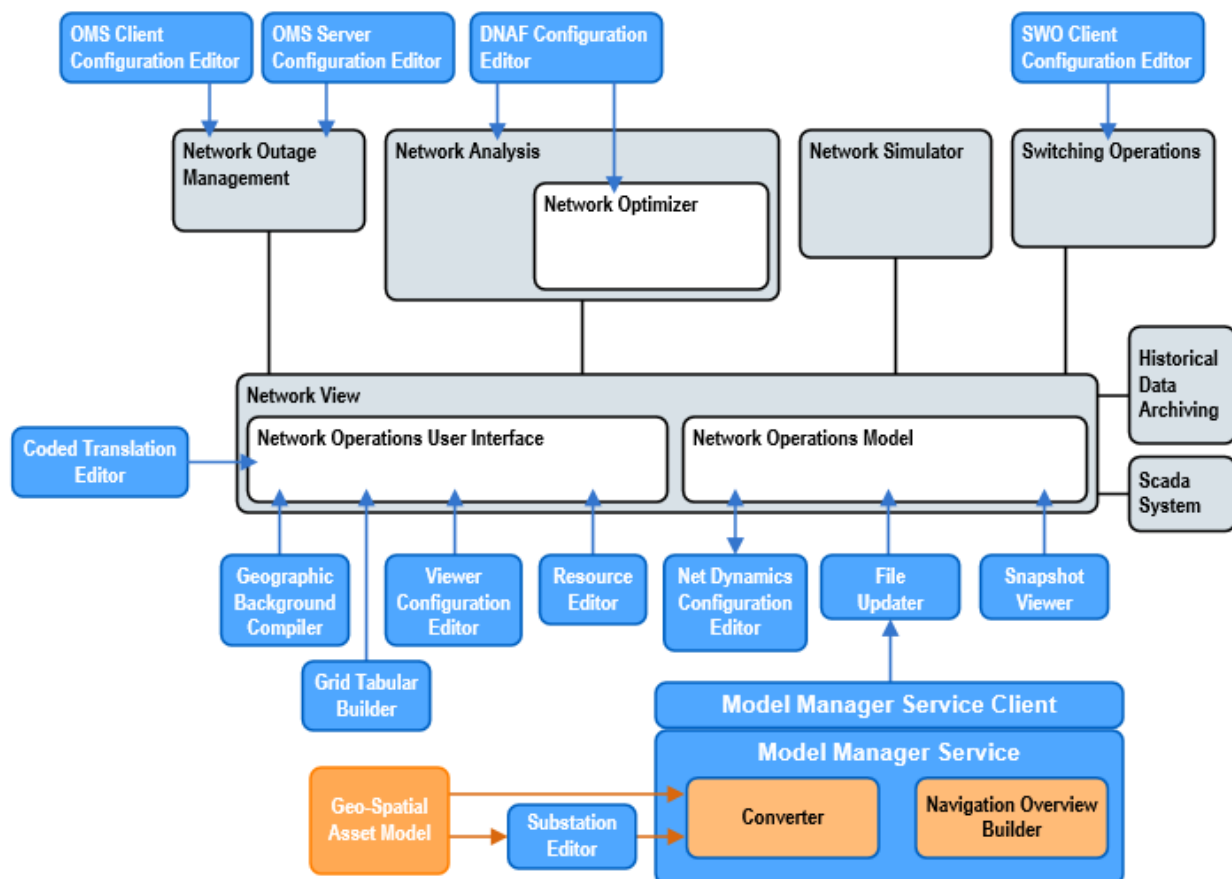


Figure 36. Modeling, Configuration, and Support Tools

6.1. Geographic Background Compiler

This tool is used to create the land-based background layers for use by the DMS client: roads, water, building footprints, etc.

The geographic background data can be supplied as a DXF or ESRI shape file. The compiler generates a background file that is read in by the Network View. The display of the background

Using ESRI shape files provides a greater degree of intelligence to be incorporated into the geographic background layer. Object layers can be combined as required, allowing users to search for a specific street address.

For more information about the Geographic Background Compiler, refer to the *Geographic Background Compiler User Guide*.

This tool analyzes all of the station model files to create a Station Overview file (overview.nvw), which is used by the ADMS clients. The overview.nvw file is the basis for the Station Overview display, which is used by operators to open a new viewport for any part of the network (see the below image). It also provides electrical and geographic adjacency information used by Network View in the creation of partial model views.

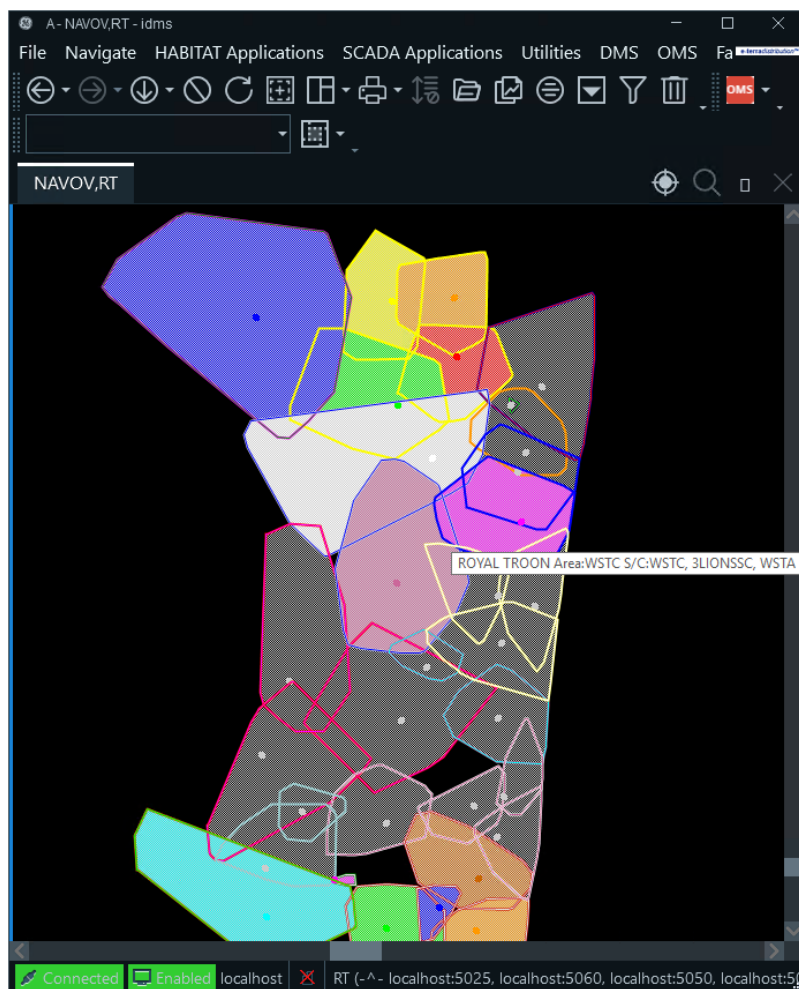


Figure 38. Station Overview Display

6.3. Net Dynamics Configuration Editor

The Net Dynamics Configuration Editor provides a graphic method to safely edit the `netconf.xml` file. This tool is used to define and modify the instances of the dynamic data that are used by ADMS. While ADMS uses a standard set of static data, a separate instance of dynamic data is defined for each mode of operation (i.e., real-time, study, and simulation).

The editor includes tools to create new dynamic instances, modify the paths for the location of dynamics files, set display background colors for each instance, configure files for saving annotation dynamic data, and define web maps and background land-based data files.

For more information about the Net Dynamics Configuration Editor, refer to the *Net Dynamics Configuration Editor User Guide*.

6.4. Viewer Configuration Editor

The Viewer Configuration Editor provides a graphical user interface for modifying the look and feel of the ADMS client, the display of network objects (symbols, zoom, and de-clutter behavior), and the definition of toolbars, right-click menus, function keys, and mouse operations.

An example of the editor page for the definition of network line symbology is shown in the below image. The tabs at the bottom of the screen allow selection of other viewer configuration functions. The output of the editor is an XML file that is read by the ADMS client.

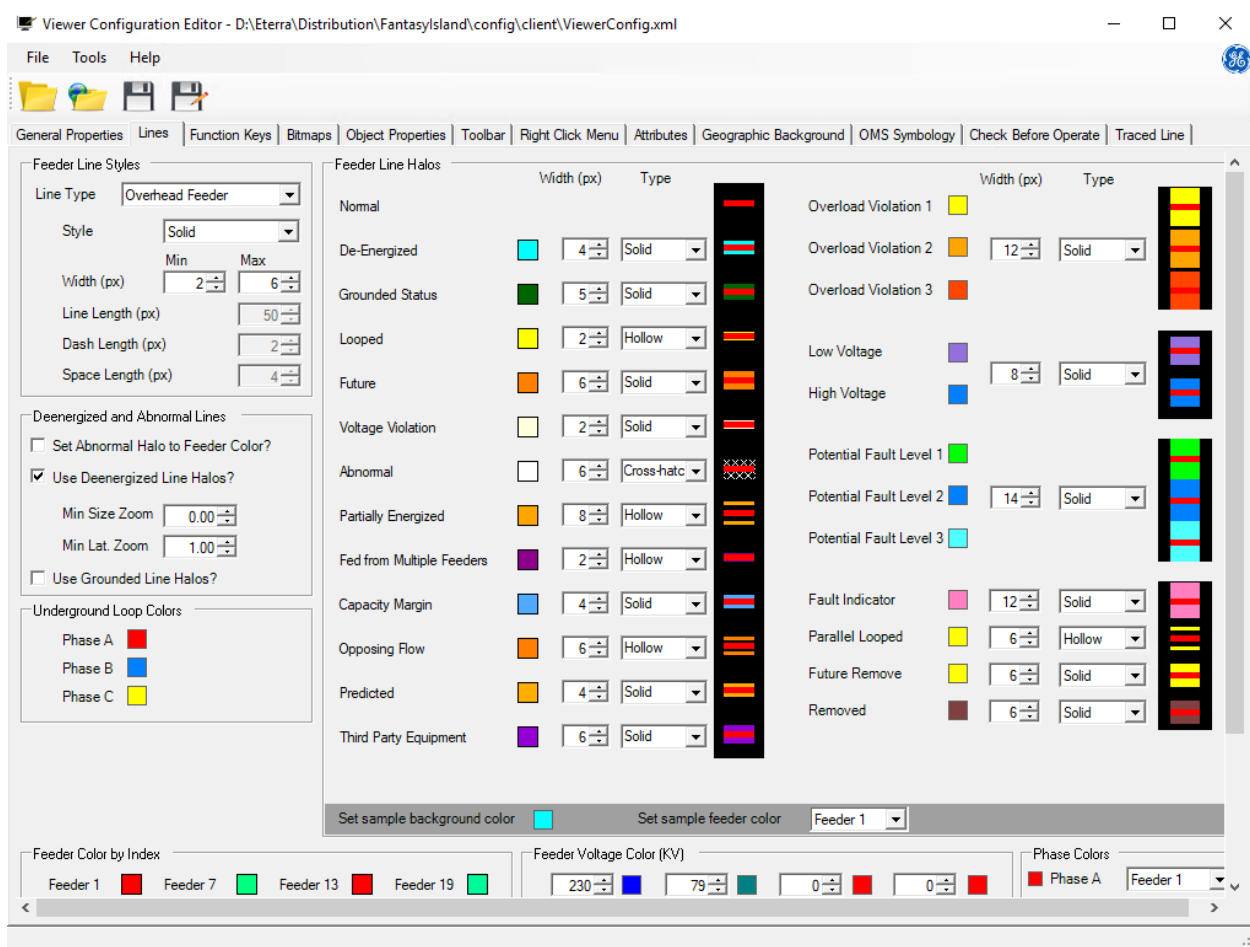


Figure 39. Line Symbology Definition in the Viewer Configuration Editor

For more information about the Viewer Configuration Editor, refer to the *Viewer Configuration Editor User Guide*.

6.5. Resource Editor

This tool enables the project/support engineers to localize the ADMS display components to a language other than English.

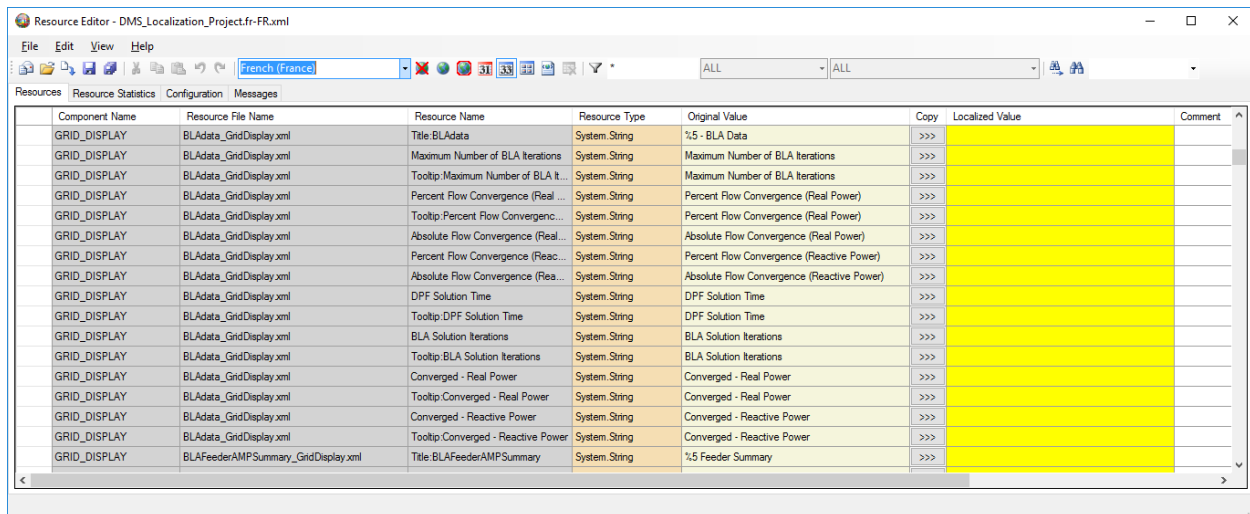


Figure 40. Resource Editor

For more information about the Resource Editor, refer to the *Resource Editor User Guide*.

6.6. File Updater Utility

In addition to the offline configuration tools, ADMS includes a FileUpdater utility to keep clients and the server up to date with the latest model and configuration files.

The ADMS clients and server each maintain a complete set of station model files. This approach facilitates online changes. For instance, when a new switch is added to a particular feeder, updating the model used by the real-time system consists of loading the updated model of the substation of interest into the appropriate folder in the server and each client machine.

The ADMS File Updater software is responsible for keeping the ADMS servers and clients up to date in terms of data files. In addition to the station model files (substationname.mod), there are XML-based toolbar files, XML-based tabular displays, and bitmap files that also need to be updated as changes arise.

The below image shows the different components and the data flow between nodes and components.

For more information about the File Updater components, refer to the *File Updater Software Installation and Maintenance Guide*.

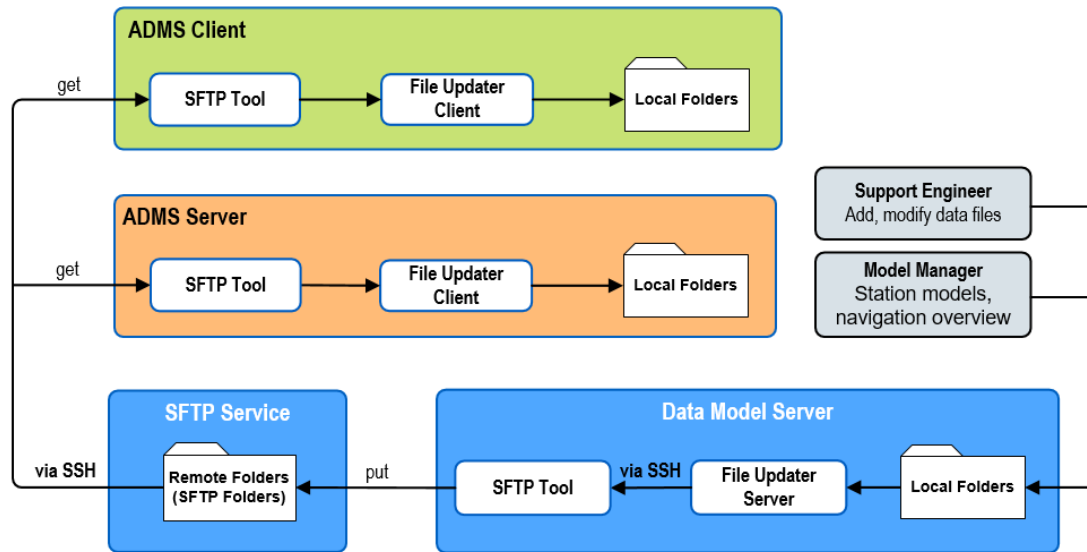


Figure 41. File Updater Components

6.7. Grid Tabular Builder

Tabular displays are used throughout ADMS to display the results of analysis or measured values. Tabular displays are also used to input power flow analysis parameters.

This tool is used to define tabular displays and the data extracted from DNOM. The builder provides tools for logical tests and conditional displays of information using colors or symbols. The display definition is generated as an XML file, which is displayed by the ADMS Network View.

The below image shows an example of the grid tabular builder forms and the resulting display.

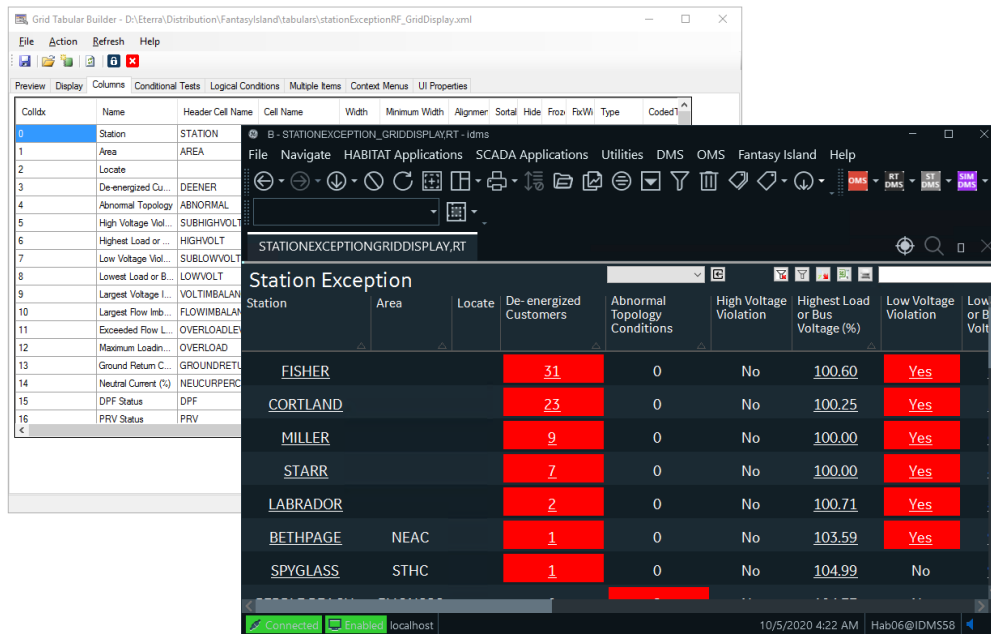


Figure 42. Example of Grid Tabular Definition and Resultant Display

For more information about the Grid Tabular Builder, refer to the *Grid Tabular Builder User Guide*.

6.8. DNAF Configuration Editor

There are a large number of parameters that are used in the configuration of the Distribution Network Analysis and Network Optimization functions. These parameters are all maintained in the `DNAF_PARAMS.xml` file, which is read by the ADMS DNAF application. This parameter file is edited by the DNAF Configuration Editor.

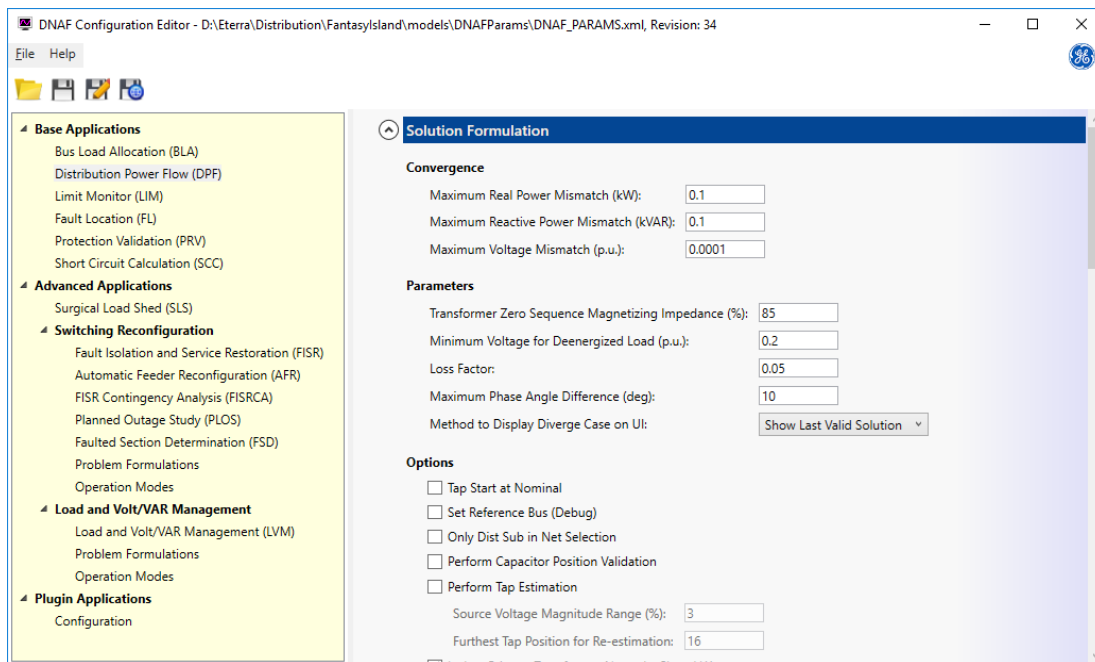


Figure 43. DNAF Configuration Editor

For more information about the configuration of the DNAF parameters, refer to the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide* and *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

6.9. Switching Order Client Configuration Editor

This tool is used to configure the look and behavior of the Switching Order application used for switching management.

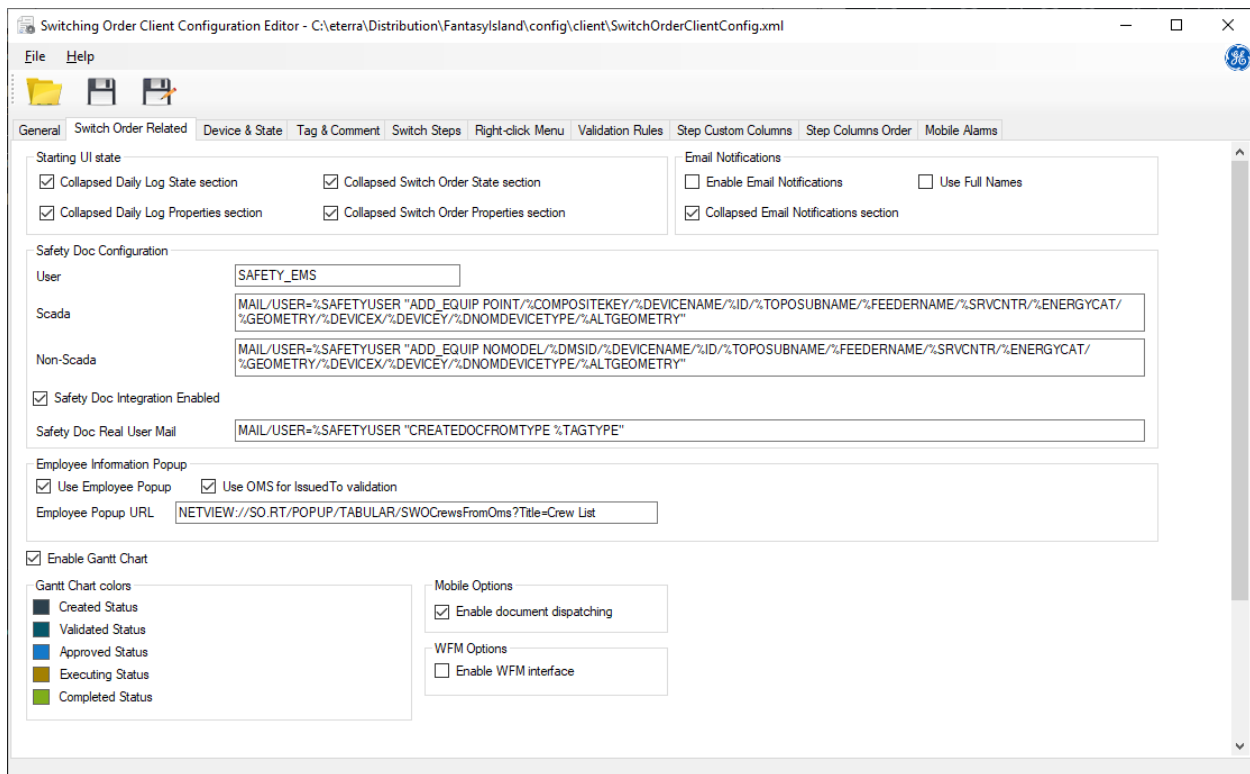


Figure 44. Switching Order Client Configuration Editor

For more information about the Switching Order Client Configuration Editor, refer to the *Switching Order Client Configuration Editor User Guide*.

6.10. Switching Order Server Configuration Editor

This tool is used to configure the workflows of the Switching Order application used for switching management.

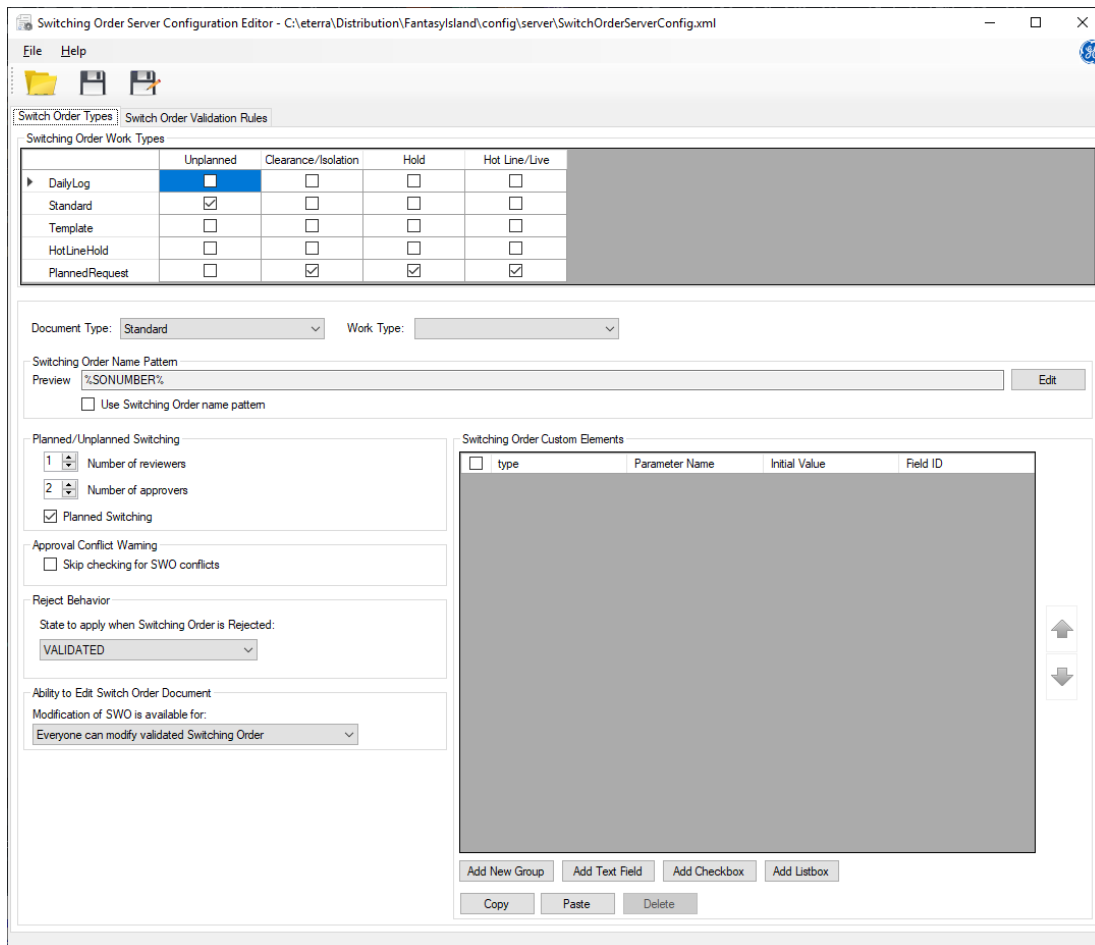


Figure 45. Switching Order Server Configuration Editor

For more information about the Switching Order Server Configuration Editor, refer to the *Switching Order Server Configuration Editor User Guide*.

6.11. OMS Client Configuration Editor

This tool is used to configure the look and behavior of the Outage Management System (OMS) displays.

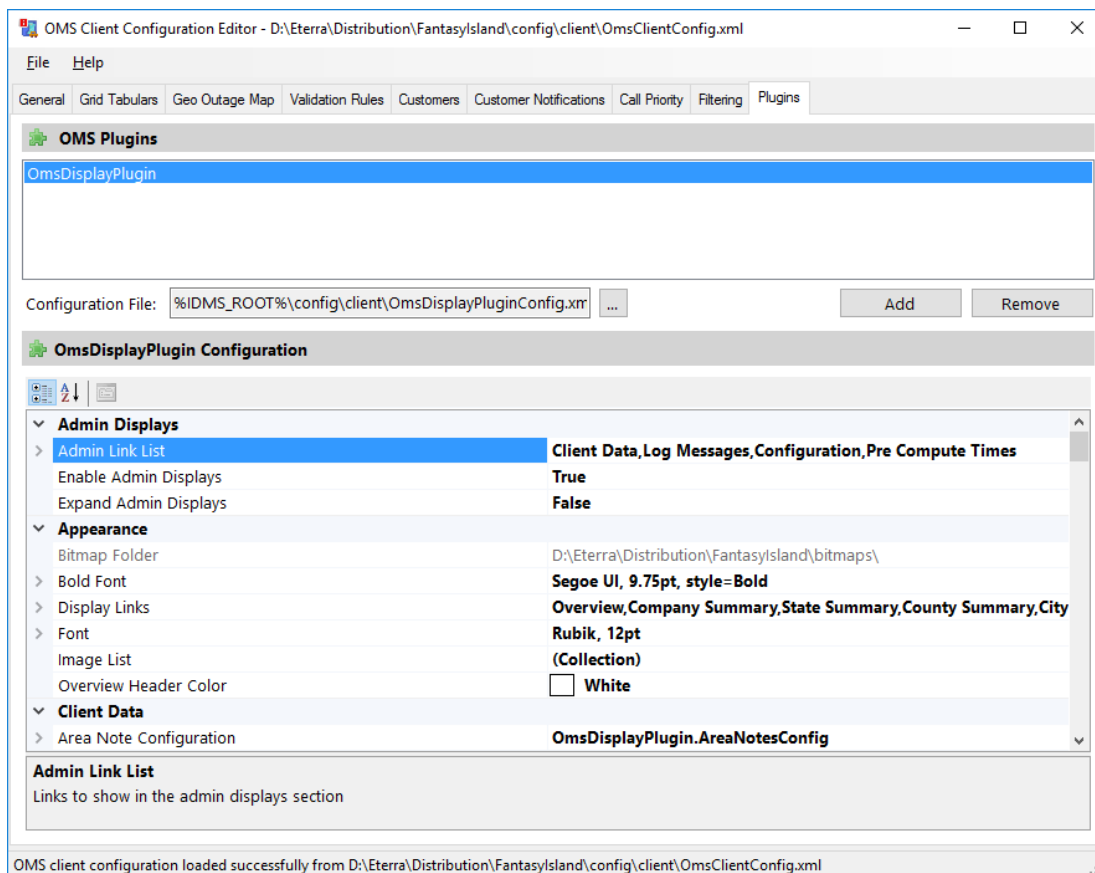


Figure 46. OMS Client Configuration Editor

For more information about the ADMS Outage Management Client Configuration Editor, refer to the *Outage Management System Client Configuration Editor User Guide*.

6.12. OMS Server Configuration Editor

The main purpose of the OMS Server Configuration Editor tool is to configure the Outage Management System (OMS) server. It also allows an analyst to set up rules that the system follows when certain events occur, and to configure conditions to meet the needs of individual utilities.

Some of the aspects that can be configured using the OMS Server Configuration Editor include:

- Definition of default ETR values for different outage reasons
- Creation of several event types
- Configuration of chronology messages
- Configuration of case notes

The output of the OMS Server Configuration Editor is an XML file, *OmsServerConfig.xml*.

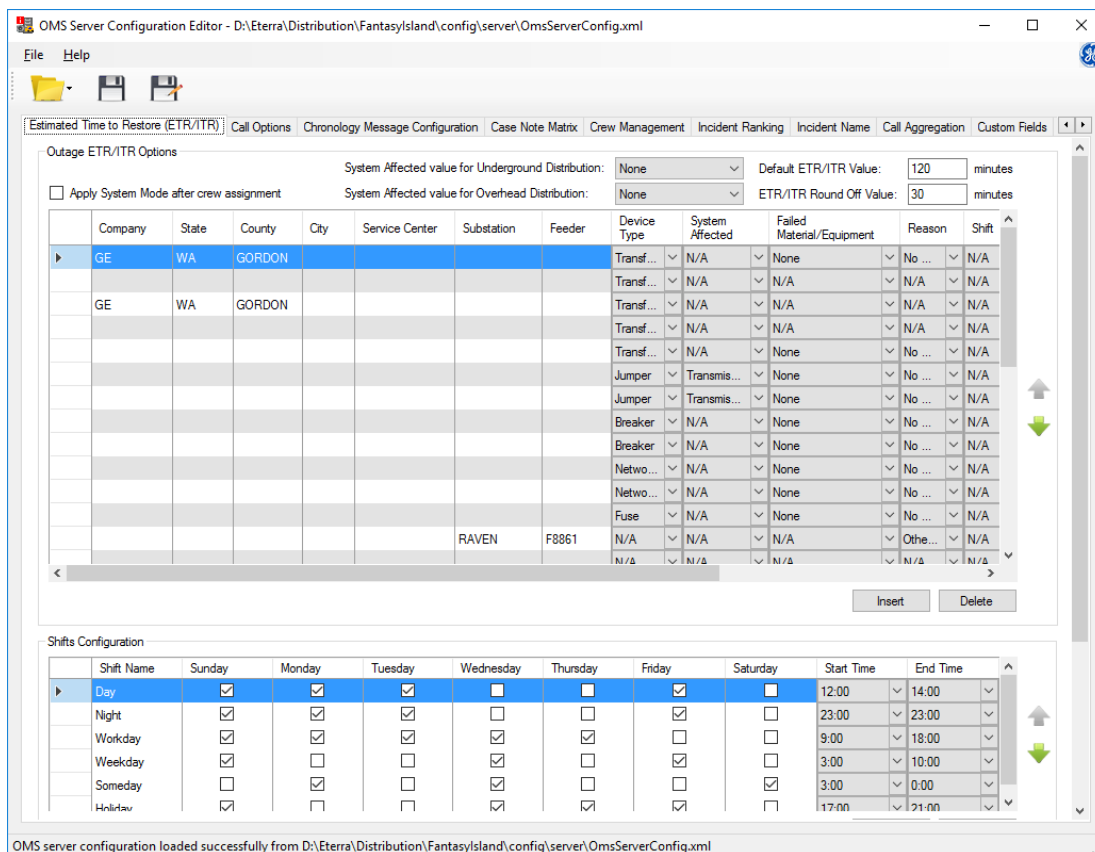


Figure 47. OMS Server Configuration Editor, Estimated Time to Restore Tab

For more information about the ADMS Outage Management Server Configuration Editor, refer to the *Outage Management System Server Configuration Editor User Guide*.

6.13. Model Manager Configuration Editor

The Model Manager Configuration Editor allows you to modify the locations of the Model Manager's folders, components, and log files.

The changes are saved in the Model Manager configuration file, *ModelManager.xml*. The default location of the configuration file is indicated in the "Config File" field at the bottom of each tab of the Model Manager Configuration Editor.

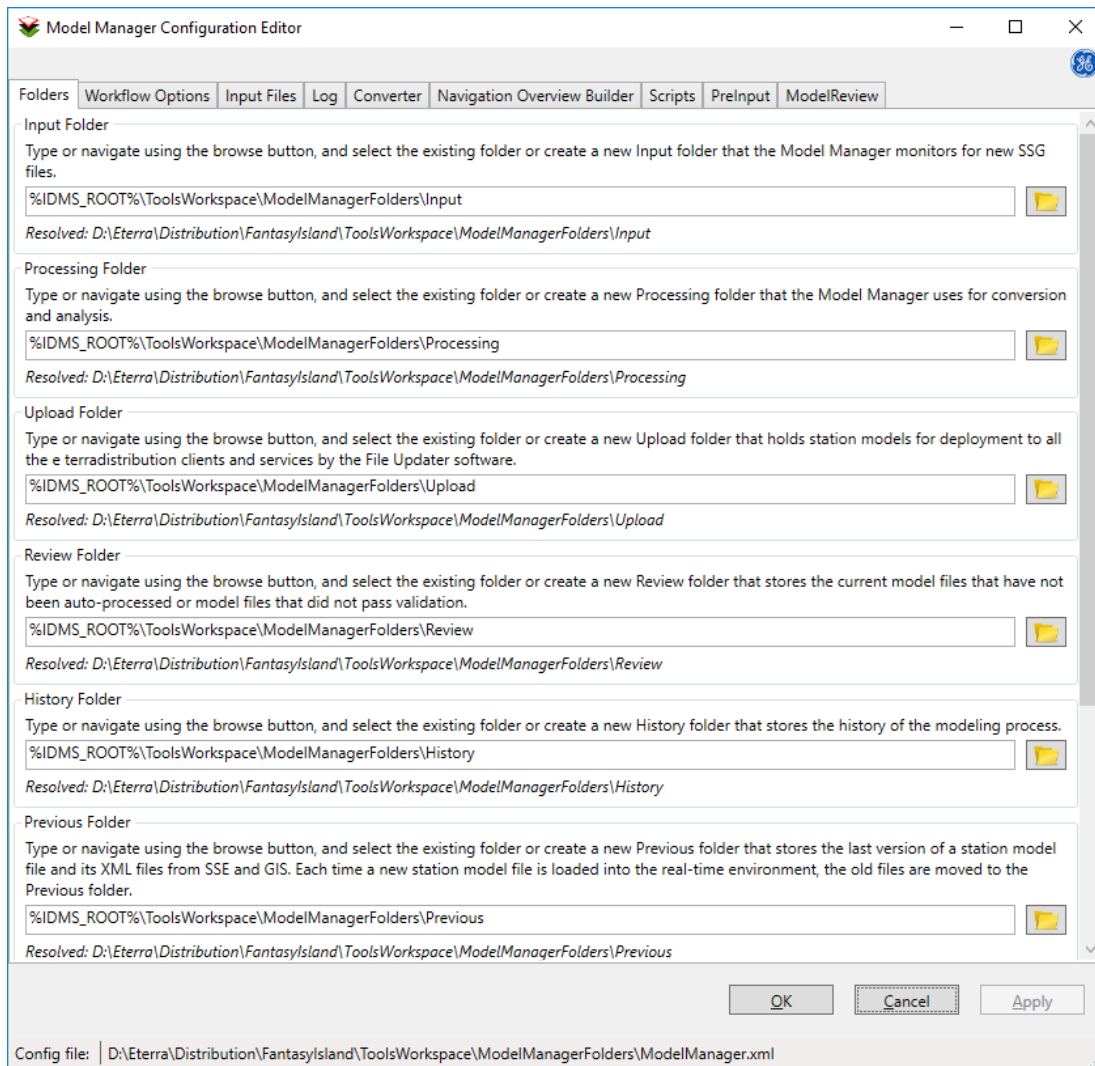


Figure 48. Model Manager Configuration Editor Folders Tab

For more information about the ADMS Model Manager Configuration Editor, refer to the *Model Manager User Guide*.

6.14. Coded Translation Editor

The Coded Translation Editor is a support tool provided with ADMS.

The main purpose of this tool is to define translations (human-readable names, descriptions, and images) for values from the source code, used in ADMS grid tabular displays.

The output of the Coded Translation Editor is an XML file, `CodedTranslationsConfig.xml`.

The Coded Translation Editor output is used for the OMS displays and for the ADMS network view Find tool. However, when you use the Grid Tabular Builder tool, you can configure any ADMS tabular display to use the coded translation types.

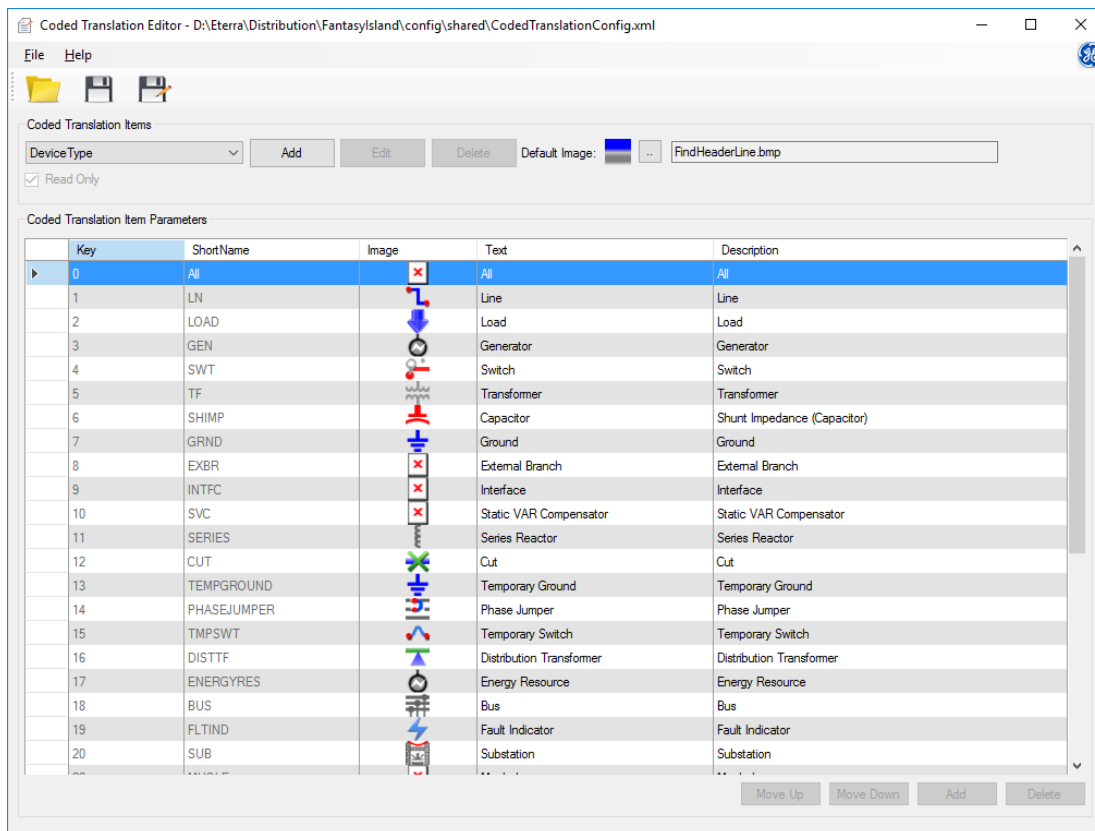


Figure 49. Coded Translation Editor Tool

For more information about the Coded Translation Editor, refer to the *Coded Translation Editor User Guide*.

6.15. Snapshot Viewer

The Snapshot Viewer is an ADMS Outage Management System (OMS) debugging and disaster recovery tool used when all other options of fixing a problem have been exhausted.

The snapshot viewer application allows you to view the entries of snapshot, dynamics, and journal files in a grid view, modify them, and save the changes. For NetServer and NetDataServer, it is possible to apply journal messages up until a specified point in time and then create a new snapshot file with the saved results.

The snapshot viewer also allows an administrator to view and edit dynamics files. Dynamics files are standalone, such that a dynamics file can be modified in isolation without requiring modifications in other files before the change is valid. The snapshot viewer allows an administrator to view any dynamics file, modify any of the contents, and save the file.

For system restoration, it is necessary for the administrator to either copy the files to a separate location, or coordinate to stop the run-time application manually before saving the file.

It is possible for "orphan" dynamics entries (those that no longer link to a valid device) to be deleted

from an administrative display. These do not require the snapshot viewer.

An example of when the display is used is for a cut that is placed on a line, but a GIS import occurs where the original line no longer exists in the system. In this case, DMS server is unable to apply the cut. The cut can be deleted and potentially re-applied to the system using the normal display technology rather than the Snapshot Viewer.

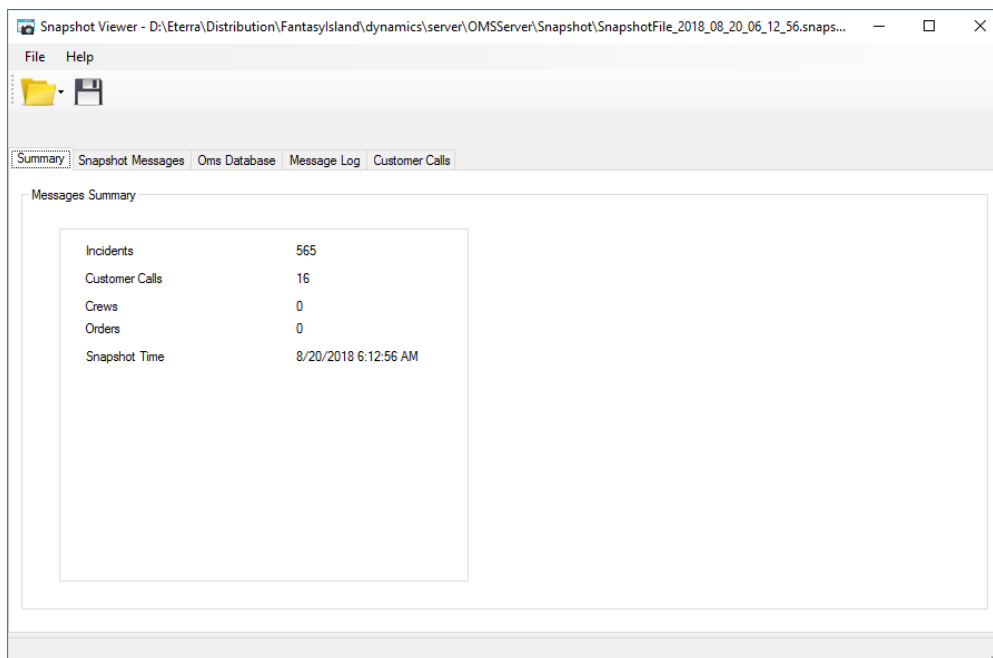


Figure 50. Summary Tab of the Snapshot Viewer Tool

Note

The Snapshot Viewer is designed for system administrators as a debugging tool. For operation staff to view and make corrections to the cleared incidents, use the OMS History Correction display, which is part of the ADMS client.

For more information about the Snapshot Viewer, refer to the *Snapshot Viewer User Guide*.

6.16. File Updater Configuration Editor

The File Updater Configuration Editor allows the analyst to configure the File Updater utility's settings, such as user name, directory entries, and validation log options. The can tool also be used to start and stop File Updater services.

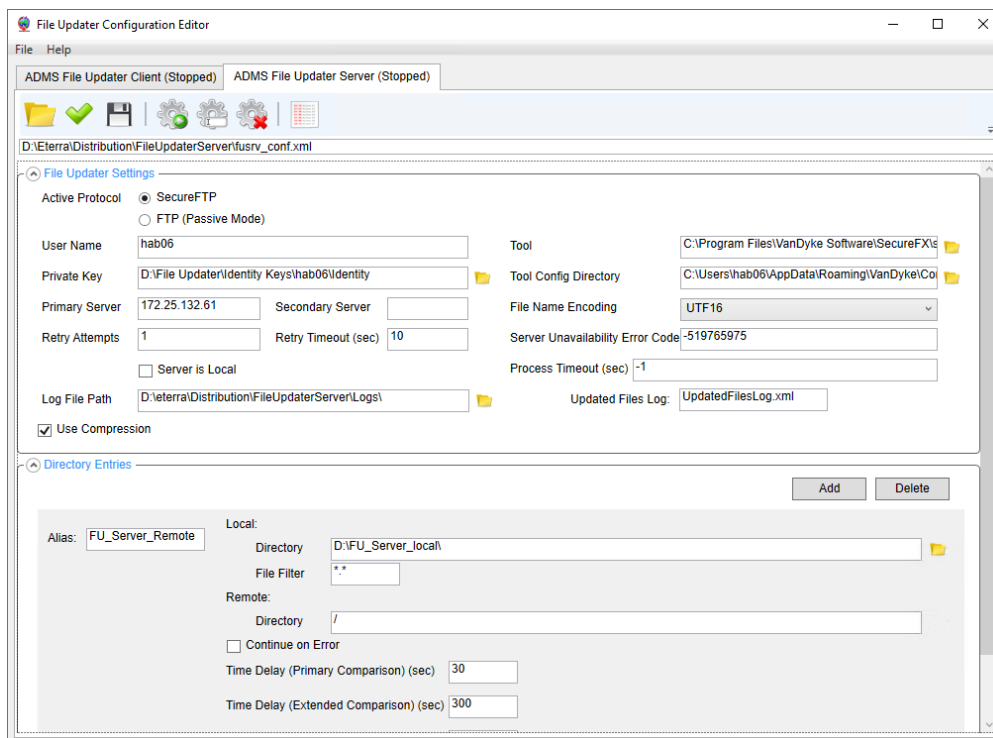


Figure 51. File Updater Configuration Editor

For more information about the File Updater Configuration Editor, refer to the *File Updater Software Installation and Maintenance Guide*.

7. ADMS External Interfaces

ADMS provides a number of programmatic interfaces that use a RESTful web service and TCP-based protocols to enable integration with third-party products and enterprise applications. The interfaces also support read-only access to ADMS data.

Customers can use the existing solutions or develop custom applications and service requests using the code samples supplied with the Software Development Kit (SDK). Access to the interfaces is subject to an additional license.

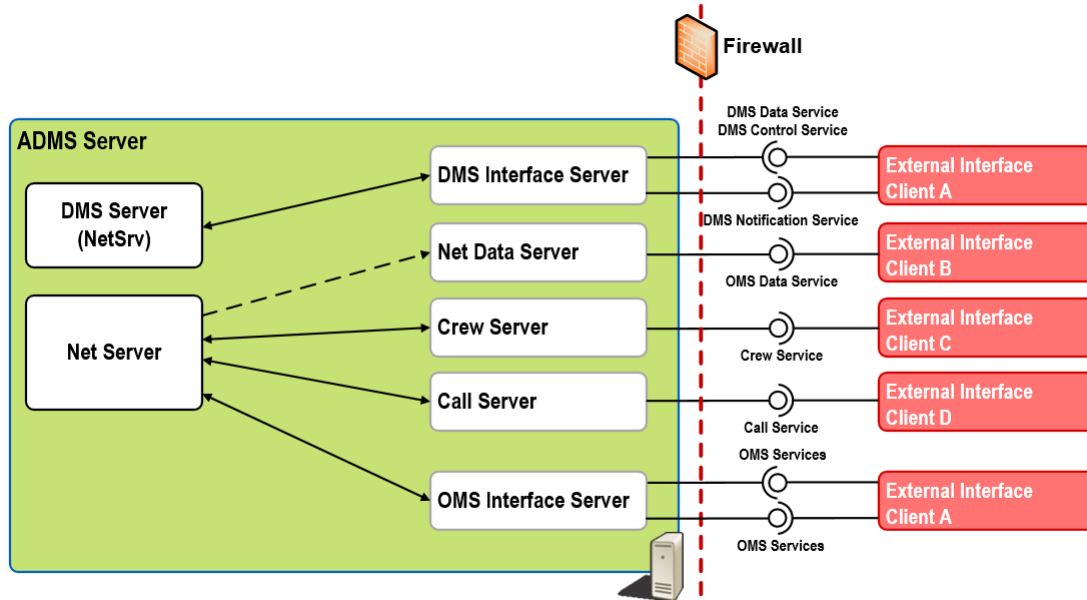


Figure 52. External Interfaces

7.1. Distribution Management Interface

The distribution management interface is supported by the DMS Interface server that communicates with the DMS server (NetSrv). This web service interface allows external applications to read and update DMS data such as switch status, device data, DNAF results, switching order data, and annotations. The interface also allows applications to send requests to a study server. The DMS Interface server also supports custom integrations based on its functionality.

The DMS Interface server acts as a client to the DMS server, using the same communication protocol that the Browser client uses to communicate with the DMS server. Externally, it provides a standard web server interface (REST) to allow third-party applications to communicate with the server.

The web interface is implemented using Windows Communication Foundation (WCF). DMS Data Collections specifically are provided through the Open Data Protocol (OData). Queried data can be returned in AtomPub or JSON formats.

Security aspects include using shared key authentication, trusted clients, firewalls, and HTTPS connections.

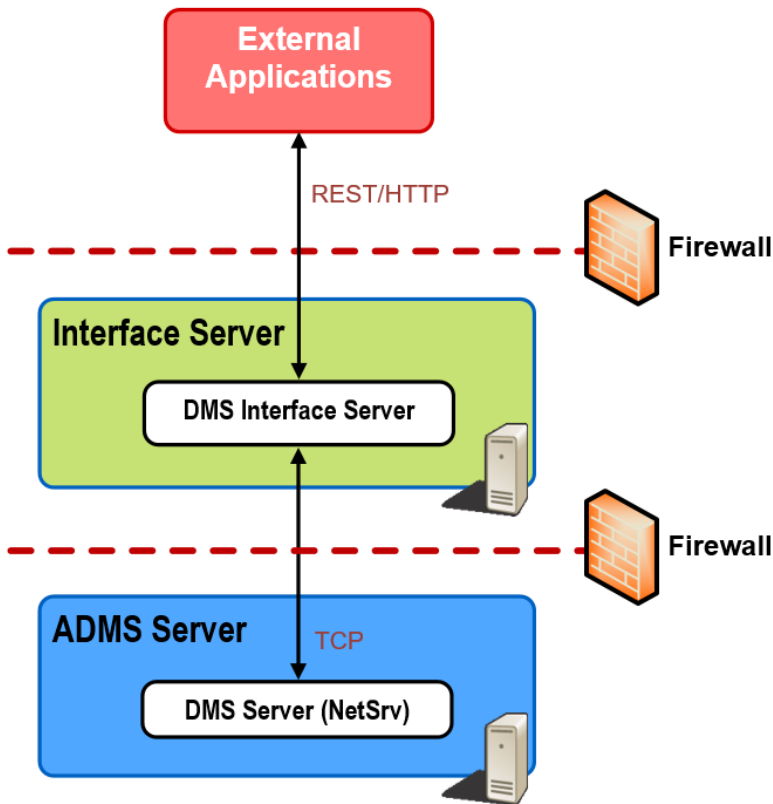


Figure 53. Distribution Management Interface

For more information, refer to the *Distribution Management Interface Server Programmer Guide*.

7.2. Outage Management Interfaces

The OMS Interface server provides a standard web server interface to allow third-party applications to communicate with the OMS server (NetServer).

The OMS Adapter provides a web services portal for messages among the OMS Interface server and external applications, including the Outage Assist solution.

The Customer Experience Adapter, Automated Notification Paging Services Adapter, and View Outage Portal Adapter use the same approach that the OMS Adapter does to send data to external applications.

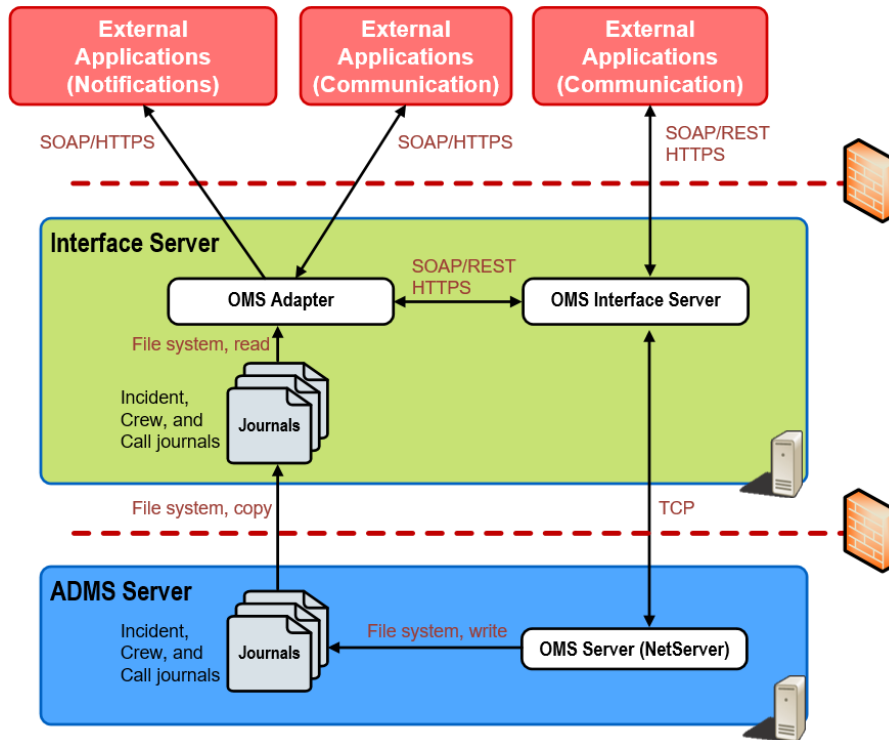


Figure 54. OMS Interface Server and OMS Adapter

The OMS Interface server and OMS Adapter can be run using standard HTTP or HTTPS. To provide additional security for web service messages, an "authorization" header can be used for requests. In addition, OMS Adapter can include trusted clients with encrypted passwords.

The OMS Interface server also supports UAA authentication and using SSL certificates for the HTTPS mode.

For more information, refer to the *Outage Management Interface Server Programmer Guide*.

7.3. Outage Management Read-Only Interface

The outage management interface is supported by NetDataServer, which is a read-only web service interface. It reads snapshots, journals, crew files, and customer models to provide real-time and historical outage management data.

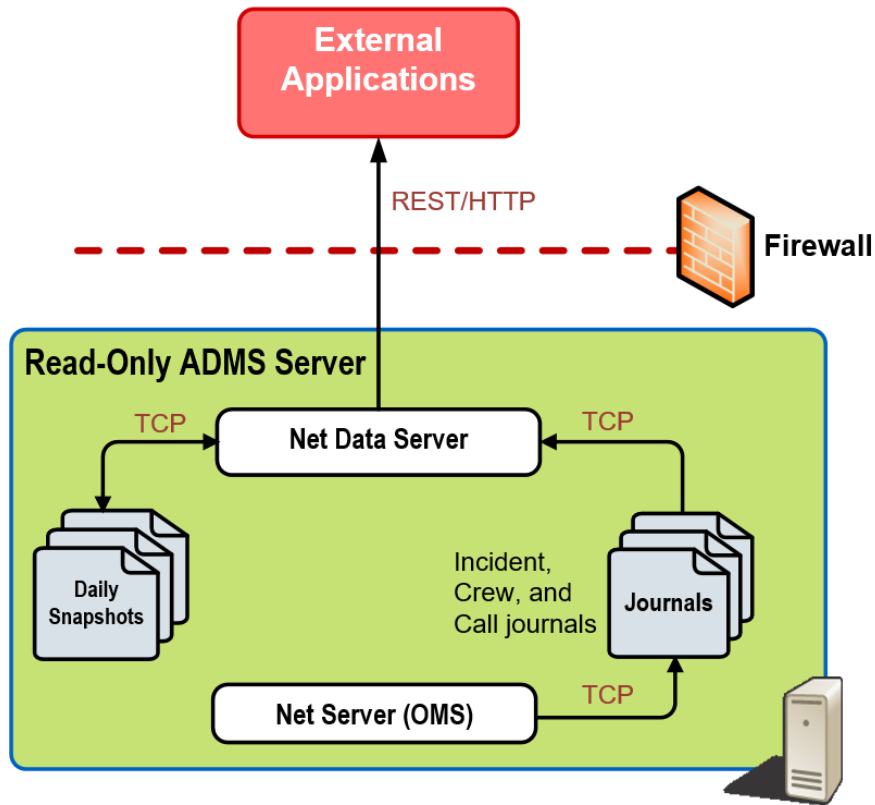


Figure 55. Outage Management Read-Only Interface

Interface resources are exposed based on conventions defined in the Open Data Protocol (OData), which is based on a Representational State Transfer (REST) architecture. Queried data can be returned in AtomPub or JSON format.

Security aspects include using firewalls and HTTPS connections.

For more information, refer to the *Outage Management Read-Only Interface Programmer Guide*.

7.4. Call and Crew Interfaces

ADMS Call and Crew servers support communication with external systems, such as a Workforce Management System (WFM), Customer Information System (CIS), Interactive Voice Response System (IVR), or Advanced Metering Infrastructure (AMI). The integrations can be created point-to-point or via an integration bus (for example, Enterprise Service Bus).

The Call server and Crew server can also run as web services. In web service mode, they support communicating with external applications (WFM and IVR) via HTTP instead of legacy TCP. They can post notifications and send data to external interfaces and allow external applications to update OMS data.

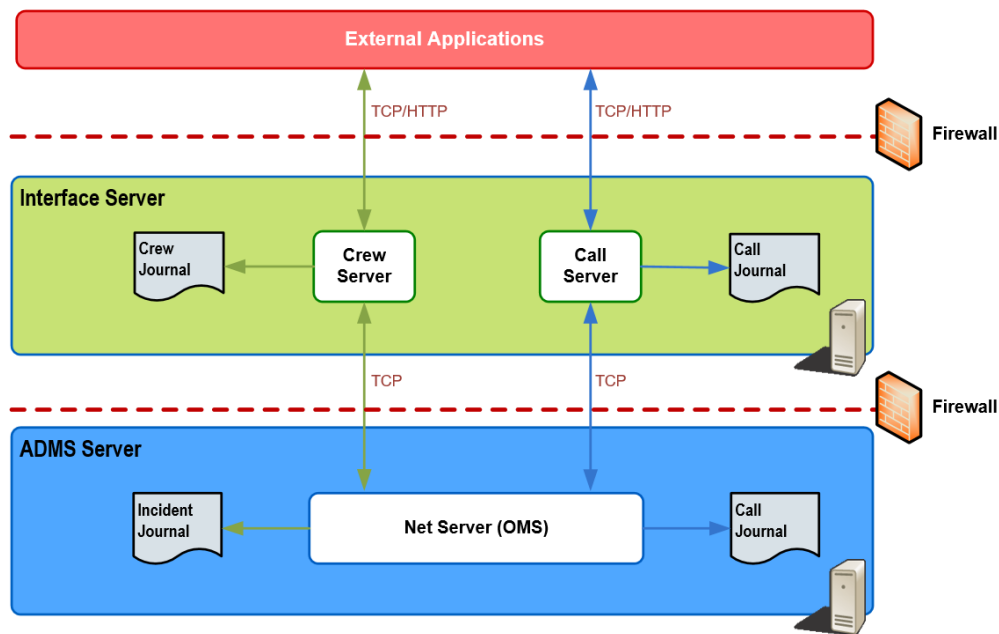


Figure 56. Call and Crew Server Interfaces

The Call and Crew interfaces have been implemented using .NET-based technology. They use XML messages for communication and perform XML serialization and de-serialization using .NET-provided classes. Interfaces communicate via TCP/IP or HTML.

Security aspects include using shared key authentication, trusted clients, firewalls, and message encoding.

For more details, refer to the *OMS Integration Programmer Guide*.

8. Other System Functions

The ADMS servers rely on the standard real-time software management functions that are used by all GE products.

In an integrated ADMS Scada implementation, Habitat is utilized to provide the following functions:

- High availability
- Authentication and authorization
- Centralized alarm management

8.1. High Availability

Habitat includes software to monitor the health and well-being of the hardware and software in the system. It uses knowledge of which components are redundant and their current states in order to maximize the availability of the overall control system.

Two components contribute to the high-availability design of ADMS:

- **PROCMAN:** The central Process Manager for all SCADA and DMS applications
- **CFGMAN:** The Configuration Manager that monitors the integrated applications

PROCMAN is the Habitat process management utility responsible for managing the entire collection of processes making up an integrated SCADA/DMS solution.

In this context, the term "management" means deciding the order in which tasks are to be started, monitoring the status of those tasks, and deciding when they should be restarted, as well as sequencing the execution of internal functions within the tasks themselves.

A typical SCADA/DMS solution consists of 40 to 60 tasks running on the same host computer. PROCMAN manages the SCADA and DMS tasks while addressing the following requirements:

- Many tasks require that a collection of tasks all be up and running in a cooperative manner.
- Failure detection and processing is complex, since some tasks can be restarted when they fail, whereas others require several other tasks to be stopped and the collection started up together in a prescribed order.
- Changing the internal sequencing logic of a program without changing the code requires having this logic modeled in a database.

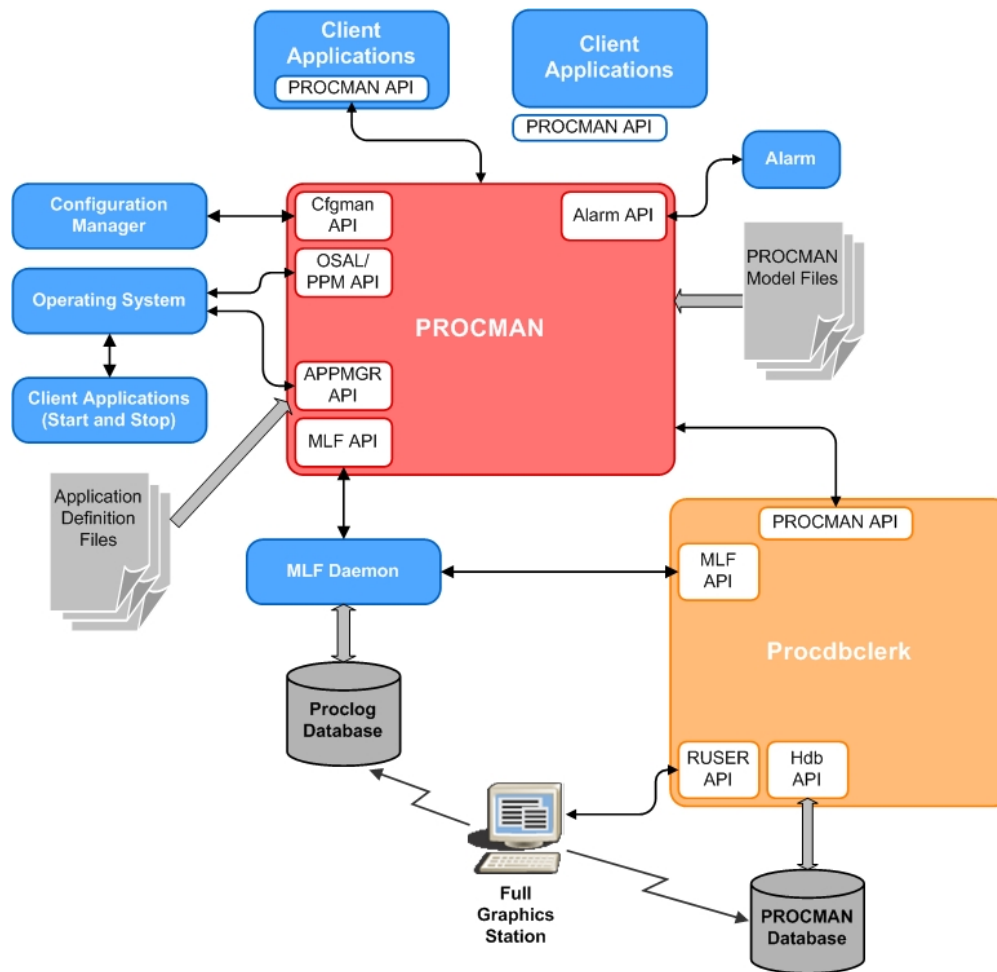


Figure 57. Process Manager Data Flow

CFGMAN is the configuration management software that monitors hardware and software failure, takes appropriate remedial actions, and provides overall system status to maintenance personnel. CFGMAN keeps track of errors and, if counts exceed predefined thresholds, it issues alarms and possibly activates the backup configuration for the device.

CFGMAN gathers error information as detected by the operating system, such as that related to hard drives, network interfaces, and memory. It also polls devices such as front-end processors, application servers, and workstations to determine the connectivity and availability of those devices.

CFGMAN contributes to a highly reliable and manageable system configuration by performing the following functions:

- Monitors software and hardware
- Graphically presents the status of all devices and processes
- Assigns roles to software components
- Recovers systems from failed software components
- Recovers systems from failed critical hardware components

- Alarms and reports significant events detected and manual requests
- Provides the operator with the ability to monitor and control the configuration

The SCADA and DMS applications and hardware devices are modeled in the Configuration Manager database. The CFGMAN's model defines many different objects (systems, application sets, processes, disks, equipment, etc.), as well as their relationships to one another and how their availability affects the availability of the integrated DMS solution.

8.2. Authentication and Authorization

Authentication

Authentication is the process of determining whether someone or something is, in fact, who or what it is declared to be.

There are three layers of user authentication: the operating system level, the Citrix level (for Citrix-based clients), and the ADMS level.

- Users located in the production control system networks (who have the ADMS client installed) log in to the workstation at the operating system level and then log in to ADMS.
- Users who use the Citrix client to access the real-time ADMS servers log in to the operating system, the Citrix server, and to ADMS.

Note

Web-based users log in to the operating system and use web-based authentication.

ADMS authentication is performed through Habitat's centralized permission management application, called PERMIT. The PERMIT application provides both authentication capability (who is attempting to log on) and authorization capability (what an identified user is allowed to do).

By using the PERMIT application for both ADMS and Scada, a single login can be provided for users and the defined Areas of Responsibility (AORs) can be coordinated. In this way, an operator who is designated a particular section of the distribution network is automatically authorized to perform manual and SCADA switching for all devices within the area and to receive relevant alarms or outage messages for that area. Changing an operator's area of responsibility is a quick and simple task. For more information about defining permissions for ADMS users, refer to the *ADMS Software Installation and Maintenance Guide* and the *PERMIT User Guide*.

The login procedure checks the PERMIT user model to authenticate the user. Users are notified if invalid user names or passwords are entered. Messages and alarms are issued to mark the authentication event and record the results.

Browser supports a variety of authentication options to control access to the system. Any time an operator console opens a new connection to a given server, the server authenticates the connection. Once a connection from client to server is established, the operator can open any number of additional client windows without having to repeat the login sequence.

As an alternative to PERMIT, the Kerberos network authentication protocol can be used for authentication. Kerberos provides strong authentication for client/server applications by using secret-key cryptography.

Authorization

Authorization is the process of giving someone permission to do or have something. Role-based authorization grants access rights to service operations or resources based on the functional role(s) of a user.

ADMS integrates with the PERMIT server to provide role-based authorization. User roles (also known as "user types") are defined in the PERMIT database. For details, refer to the *PERMIT User Guide*.

The ADMS real-time server interacts with PERMIT through the ADMS permission manager proxy server (netpermsrv.exe), which is a Habitat add-on component.

The below image illustrates permission checking principles.

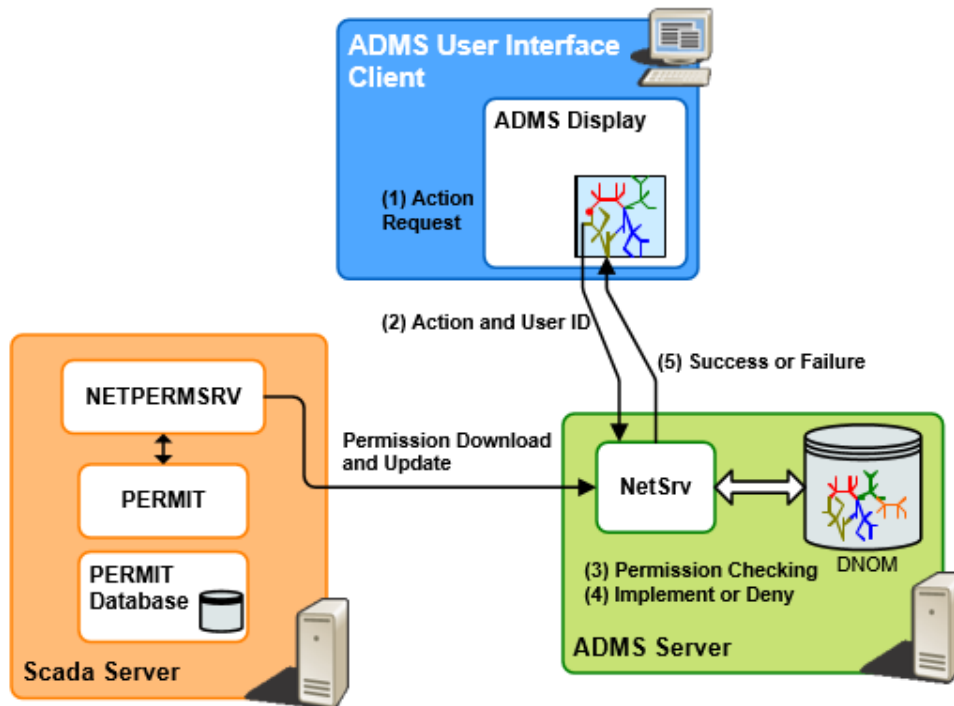


Figure 58. Permission Checking

An additional plug-in, the Permission Management Plug-In, allows permissions to be managed from within the ADMS display environment, as shown in the below image. It supports the more complex arrangement of Areas of Responsibility (AORs) that are frequently configured for the distribution network and provides the operators with an easier method of managing these AORs.

The Permission Management Plug-In provides a hierarchical tree view of the permission structure and the current states. Operators can then change, take, or drop access to AORs directly from this view.

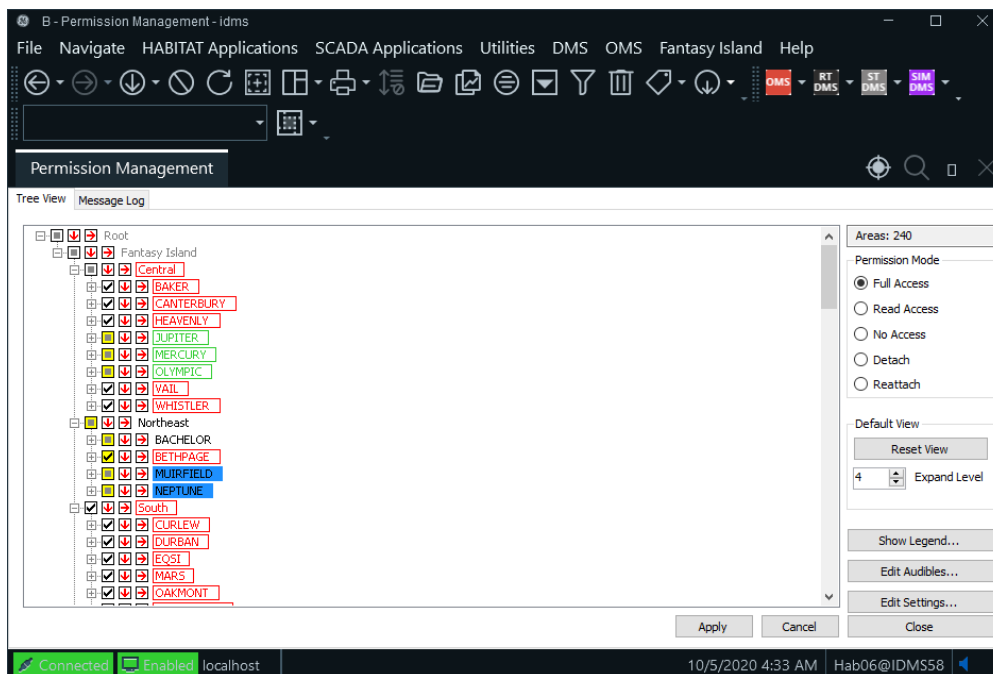


Figure 59. Permission Management Plug-In

For more information about defining AOR access modes, refer to the *Permission Management Plug-In User Guide*.

8.3. Centralized Alarm Management

The ALARM application, delivered as part of Habitat, provides all the means required to notify operators in real time about events occurring on the SCADA and DMS solution.

A critical part of an Integrated Distribution Management System is a single coordinated alarm and event logging system. Alarms generated by the SCADA system and by distribution analysis functions, such as limit violations, must appear in the same system to present a consistent display to operators.

System events, switching operations, annotations, temporary modifications, etc. similarly must be managed in a coordinated fashion so that operators can easily analyze and review the operation of the entire ADMS.

Events are changes to the system's state that are detected by application software. An event can be printed and archived by the ALARM application. It can be processed as an alarm depending on the attributes modeled in the alarm database and attached to the event. An audible and/or visual indicator is used to report the alarm to the operator. An alarm requires operator acknowledgment and/or deletion.

The ALARM application follows the traditional client-server model, where clients, such as Scada for SCADA-related alarms, are sending messages and requests to a server that processes them. Each event is typically associated with a database object (for example, a SCADA point representing a

circuit breaker, a PROCMAN task). An event is posted by the client application with a set of attributes. ALARM processes the event according to these attributes and the modeling information stored in the alarm database.

Time	State	Message
05 / 02:50:17	▲	PROCMAN FLAG ISD_SCANNING ON CPU IDMS58 SET TRUE BY TASK SCADA
05 / 02:49:14	▲	HOSTA: Application set EMSHOST database set is now VALID.
05 / 02:49:07	▲	HOSTA: Application set EMSHOST transitioned from DISABLED to ENABLED.
05 / 02:49:07	▲	HOSTA: Application set EMSHOST is now AVAILABLE.
05 / 02:49:07	▲	HOSTA: System has transitioned to MASTER from IDLE.
05 / 02:49:07	▲	HOSTA: System HOSTA is now AVAILABLE.
05 / 02:45:24	▲	PROCMAN FLAG ISD_SCANNING ON CPU IDMS58 SET TRUE BY TASK SCADA
05 / 02:44:02	▲	HOSTA: Application set EMSHOST database set is now VALID.
05 / 02:43:59	▲	HOSTA: Application set EMSHOST transitioned from DISABLED to ENABLED.
05 / 02:43:59	▲	HOSTA: Application set EMSHOST is now AVAILABLE.
05 / 02:43:59	▲	HOSTA: System has transitioned to MASTER from IDLE.
05 / 02:43:59	▲	HOSTA: System HOSTA is now AVAILABLE.
05 / 02:38:48	▲	22225199 OPEN changed to BETHPAGE
05 / 02:38:48	▲	22225199_C OPEN changed to BETHPAGE
05 / 02:38:48	▲	22225199_A OPEN changed to BETHPAGE
05 / 02:38:48	▲	22225199_B OPEN changed to BETHPAGE
05 / 02:38:48	▲	22225042 OPEN changed to BETHPAGE
05 / 02:38:48	▲	22225042_C OPEN changed to BETHPAGE
05 / 02:38:48	▲	22225042_A OPEN changed to BETHPAGE
05 / 02:38:48	▲	22225042_B OPEN changed to BETHPAGE
05 / 02:38:08	▲	PROCMAN FLAG ISD_SCANNING ON CPU IDMS58 SET TRUE BY TASK SCADA
05 / 02:37:20	▲	HOSTA: Application set EMSHOST database set is now VALID.
05 / 02:37:17	▲	HOSTA: Application set EMSHOST transitioned from DISABLED to ENABLED.
05 / 02:37:17	▲	HOSTA: Application set EMSHOST is now AVAILABLE.

Figure 60. Alarm Summary Display

The below image depicts the overall architecture of ALARM.

The ADMS user interface system, Browser, manages the presentation of alarms across all client workstations, so that a given alarm is shown on every operator display that contains the object that is subject to the alarm condition.

When tracking down the source of an event, the operator and analyst can use displays such as the system activity log. This log presents views showing all system activity, acknowledgement activity, and events. Users can annotate items on the logs.

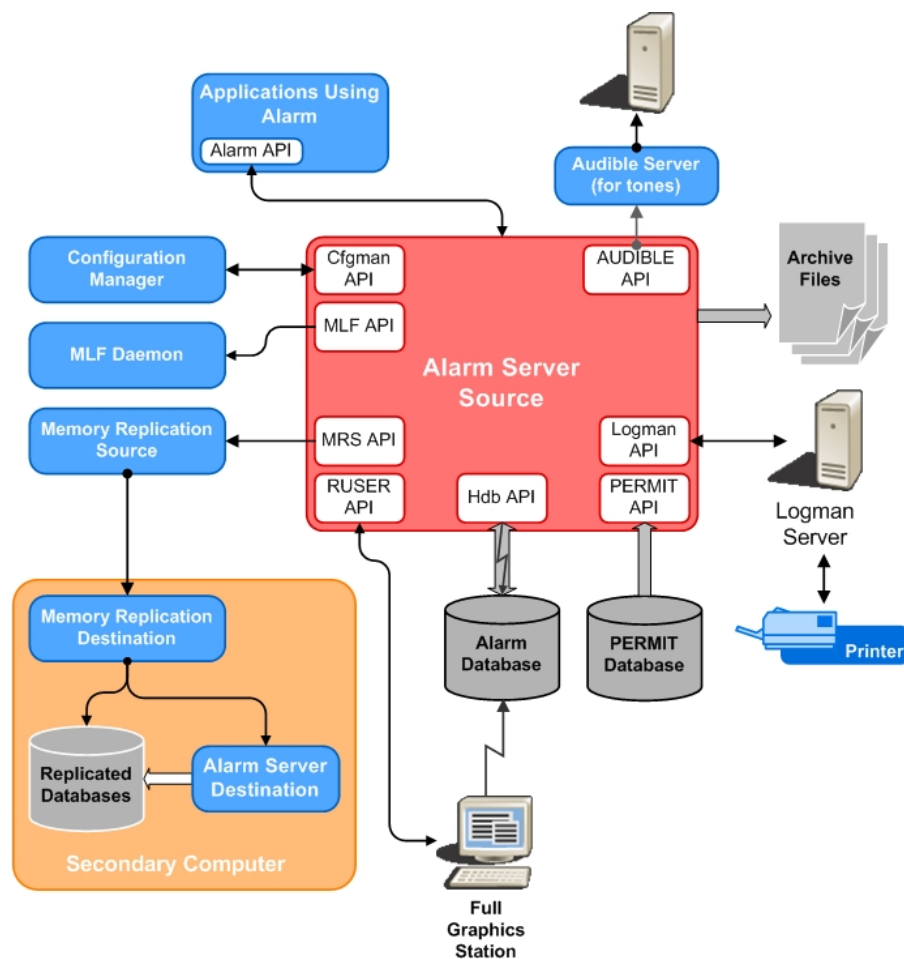


Figure 61. Alarm Application Architecture